

Solution to Stein&Shakarchi *Complex Analysis**

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2022.7

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1 前言

2022年上半年,我为泰山学堂2020级本科生讲授《复分析》(在更多的情况下,这门课的课程名称是复变函数,讲授内容区别于研究生课程复分析)。Stein和Shakarchi的复分析教材是我们的参考书,但并不是按照这本书授课。开学初,了解到班上两位同学合作,利用暑假时间完成了Stein和Shakarchi的复分析教材的前十章的习题解答,并整理出了电子版。为他们的学习热情点赞。

复变函数是大学数学专业的基础课程,其在数学专业的重要性不必多言。这里仅强调,这门课的很多内容和微积分一样,也是理工科需要学习和掌握的内容。此外,复变函数也是本人的研究方向解析数论的基础课程。事实上,Stein和Shakarchi的复分析(以及傅里叶分析教材)揭示了该门课与解析数论之间的深刻联系。

这里更多想就如何参考习题解答聊几句。我们大多情况下需要做适量的习题,包括适量有难度的习题。但是需要强调,做习题这个过程是为了辅助我们更深刻的理解、掌握和应用所学的知识。不能简单地把“刷习题”(尤其是刷套路重复的习题)当成我们学习的主要任务。当然,似乎在微积分和线性代数这样的课程里,更容易陷入刷习题的旋涡里。对于高年级数学课程,套路相同的习题过多重复的现象也未必多见。同时,一些好的习题也会有助于兴趣的培养。我们可能常常遇见一些证明冗长且难度较大的定理证明,这弄不好会打击或者磨灭兴趣。但是,面对一个好的习题,在经历各种尝试且一步步将其解开过程中,往往能培养和激发我们的学习兴趣。

手上有一个好的习题解答,使用恰当会对我们的学习有一定帮助。然而,完成习题的过程,切忌轻易参考习题解答!一个完美的理想状态当然是咱们并不需要参考这个解答(就完成了自己想做的习题)。如果这样,相信两位作者也会为学弟学妹们感到高兴。我们在大多数情况下,总会或多或少遇到一些自己无法解答,或者至少短时间内无法解答的习题。面对这样一些有一定困难的习题,我们还是要努力尝试自己解决,不要轻易放弃。至少在形式上,应该一次花上几十分钟甚至更长时间去努力攻克,而且这样的努力尝试应该有好多次。当然,在若干次努力尝试后仍然无法解决,再来参考两位学长的解答,这个时候这份解答才会是真正宝贵的资源!

最后,两位提供的解答,将会作为我讲授相关课程的重要参考资料,在此表示感谢!

赵立璐

2022年9月于济南

2 致谢

这份答案是我与我的室友写就，计划开始于2022年7月，因为疫情等因素的影响被迫中断了一个月，前前后后大概花了三周，以至于9月才最终完稿。其中我负责1.2.3.8.9章，1.2.3章是复变函数的基础理论，一般来说本科生能够掌握这三章及共形映射（第八章）的部分内容就已经足够。室友负责4.5.6.7.10章，这一部分可以分为两块，第一块是整函数零点估计（第五章）和复变函数的傅里叶分析（第四章），这一部分能够让我们对复分析的认识得以健全，是比较推荐去了解的；第二块是 ζ 函数的解析延拓（第六，七章）和 Θ 函数的应用（第十章），它们和第九章（ $\wp$ 函数）一样研究的是具体的函数，且和解析数论相关度很大，这四章可以根据个人口味去学习。

Stein&Shakarchi复分析的缺陷是，它的正文以理论为主，具体的计算和证明方面鲜有手把手的“baby”式教学，这和国内的课本是大相径庭的。初步阅读了此书的读者马上就能发现诸如洛朗级数的概念，计算积分非常有用的Jordan引理等均被放置或者隐藏在了习题中。换言之仅看正文不做习题，又不结合其他书来看的话（尤其是以Stein复分析为课本的教学班级）很难对这一块的内容有系统的全面的认知，导致学习的效果不佳。另一方面，它的习题非常具有代表性，质量很高，比较困难的问题也采取给Hint循循善诱的态度，如果积极去接触这里的习题(exercise)乃至问题(problem)，必然对于这门课程以及对于分析的理解有所提升。这两个方面的因素是我们写这份答案的动力之一。

但其实说是写就，不如说是编纂，事实上很多的习题可以找到现成的答案，这也使得我们的工作量减少了很多，且在此基础上能够更进一步。我们并非仅仅是答案的搬运工和校稿员，这部答案与市面上流传的版本有两个区别，一个是全面，全十章都有涉及，一个是完整，不仅有绝大部分习题的解答，也有几乎所有问题的解答。那么问题的解答均是我们自己独立写就，正确与否有待检验。我们期盼着有读者能够提出疑问，并将其发送到预留的邮箱中，这是对我们的工作的最高敬意。

对于这份工作，我们对以下人员表示感谢，首先是我们的老师赵立璐教授，他在复分析课程的教学上认真敬业，教学水平很高，讲解了本书大部分内容，这是我们得以在课程结束后得以完成这份答案的坚实基础。然后是来自浙江大学的B站up主CaserLius，他在很早之前分享的本书前三章的习题解答，不仅成为了我们工作的重要参考，也帮助我度过了复分析期中考试，而这项工作也得到了他的支持。然后是一位外国友人Robert C.Rhoades，他在2006年写的5.6.7.8.9.10章的部分习题答案对本工作也是一份很好的材料。最后是我们的同学们，山东大学数学系的氛围良好，求知与交流的风气盛行，如果没有这样的氛围，我们不会产生这样的想法并付诸行动。

以上。

墨蝦

9.7于浙江省金华市

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3 Chapter 1

3.1 Exercise

1.2. Simple exercise. We leave it to reader. □

3. $\sqrt[n]{se^{\frac{i}{n}(\varphi+2k\pi)}} (k = 0, \dots, n-1)$. □

4. Without loss of generality, suppose $i \succ 0 \succ -i$, then $i \cdot i \succ i \cdot (-i)$ i.e. $-1 \succ 1 \Rightarrow -i \succ i$. □

5.(a) Since Ω_1 is an open set, $z(t^*) \notin \Omega_1$, or a open ball $B \subset \Omega_1$ will contain $z(t^*) \Rightarrow B \cap \gamma \neq \emptyset$ and obviously there exists $t' > t^*$ s.t. $z(t) \in \Omega_1$. Contradiction! So $z(t^*) \in \Omega_2$, similarly we will find $t'' < t^*$ s.t. $z(t'') \in \Omega_2$ but by the definition of t^* we have $z(t'') \in \Omega_1$. Contradiction! So Ω cannot be written as the union of two disjoint open set. □

(b) Let $p \in \Omega_1 \Rightarrow$ there is a curve γ from p to ω . Choose an open ball B s.t. $p \in B \subset \Omega$. Any point $q \in B$ we can choose a curve γ' from q to p , the union of γ and γ' is the desired curve from q to $\omega \Rightarrow q \in \Omega_1$. Hence Ω_1 is open set. Similarly we can prove Ω_2 is open. Obviously they are disjoint and $\Omega = \Omega_1 \cup \Omega_2$ so by (a) we get the contradiction. Since open ball is pathwise connected, we choose $\omega \in B' \subset \Omega \Rightarrow B' \subset \Omega_1$. Hence $\Omega_1 \neq \emptyset \Rightarrow \Omega = \Omega_1$. □

6.(a) By ex 5 clearly C_z is open. The check of equivalent relation is trivial and we leave it to reader. □

(b) Suppose Ω have uncountable connected component then by (a) we can choose uncountable ball belong to these component respectively and they are disjoint. Since every ball contains a rational point, so we get an uncountable set which elements are rational points. Contradiction! □

(c) Choose a large disc O containing the compact set $\Rightarrow O^c$ is pathwise connected. If Ω have two unbounded components A, B , assume $A \cap O^c \neq \emptyset \Rightarrow O^c \subset A$. Since $A \cap B = \emptyset$, we have $B \subset O$. Contradiction! □

7.(a) We choose a proper angle θ s.t. $e^{i\theta}z \in \mathbb{R}$, then we have

$$\left| \frac{\omega - z}{1 - \bar{\omega}z} \right| < 1 \Leftrightarrow \left| \frac{e^{i\theta}(\omega - z)}{e^{i\theta}(1 - \bar{\omega}z)} \right| < 1 \quad (1)$$

Without loss of generality, let $z \in \mathbb{R}$. Then it suffices to prove $(r - \omega)(r - \bar{\omega}) \leq (1 - r\omega)(1 - r\bar{\omega}) \Leftrightarrow r^2 + |\omega|^2 - |r\omega|^2 \leq 1(r \in \mathbb{R})$. Observe that $r^2 + |\omega|^2 - |r\omega|^2 = (1 - |\omega|^2)r^2 + |\omega|^2 \leq 1 - |\omega|^2 + |\omega|^2 = 1$. The equality occurs iff $|\omega| = 1$ or $r = 1$. So we draw the conclusion. □

(b) Actually these are basic propositions from conformal mappings, please turn to 8.2.1 “The aut of the disc” (page 219) □

8. Just calculate it directly. □

9. We use the same method as the proof of the C.-R. equation in rectangular coordinate. If $f = u + iv$ is differentiable at the point p , then we will get the same result in the formula

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{u(z + \Delta z) - u(z)}{\Delta z} + \lim_{\Delta z \rightarrow 0} i \frac{v(z + \Delta z) - v(z)}{\Delta z} \quad (2)$$

as the $z + \Delta z$ approximating z in the tangential and normal direction. Let $z = re^{i\theta}$, we have

$$\begin{aligned} & \lim_{\Delta r \rightarrow 0} \frac{u(z + \Delta r e^{i\theta}) - u(z)}{\Delta r e^{i\theta}} + \lim_{\Delta r \rightarrow 0} i \frac{v(z + \Delta r e^{i\theta}) - v(z)}{\Delta r e^{i\theta}} = \\ & \lim_{\Delta \theta \rightarrow 0} \frac{u(z + r e^{i\theta}(1 - e^{i\Delta\theta})) - u(z)}{r e^{i\theta}(1 - e^{i\Delta\theta})} + \lim_{\Delta \theta \rightarrow 0} i \frac{v(z + r e^{i\theta}(1 - e^{i\Delta\theta})) - v(z)}{r e^{i\theta}(1 - e^{i\Delta\theta})}. \end{aligned}$$

Notice that $1 - e^{i\Delta\theta} \rightarrow -i\Delta\theta$ when $\theta \rightarrow 0$, so compare the real and imagine part we have

$$\begin{aligned} \frac{\partial u}{\partial r} &= \frac{1}{r} \frac{\partial v}{\partial \theta} \\ \frac{1}{r} \frac{\partial u}{\partial \theta} &= -\frac{\partial v}{\partial r} \end{aligned}$$

So it is easy to check that $\log z$ is holomorphic, we leave it to reader. Besides, we point out that

$$f'(z) = \frac{r}{z} \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right) \quad (3)$$

So $(\log z)' = \frac{1}{z}$. □

10. Check it by calculating it directly. We leave it to reader. □

11. $f = u + iv$ is holomorphic, then $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$. Then

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial x} \left(\frac{\partial v}{\partial y} \right) + \frac{\partial}{\partial y} \left(-\frac{\partial v}{\partial x} \right) = 0$$

$$\Delta v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{\partial}{\partial x} \left(-\frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right) = 0$$

12. This is a classical example. We Check it directly. Since $u = \sqrt{|xy|}$, $v = 0$

$$\begin{aligned} u_x(0, 0) &= \lim_{\Delta x \rightarrow 0} \frac{u(\Delta x, 0) - u(0, 0)}{\Delta x} = 0 = v_y(0, 0) \\ u_y(0, 0) &= \lim_{\Delta y \rightarrow 0} \frac{u(0, \Delta y) - u(0, 0)}{\Delta y} = 0 = -v_x(0, 0) \end{aligned}$$

However, we observe that

$$\frac{f(\Delta z) - f(0)}{\Delta z} = \frac{\sqrt{|\Delta x \Delta y|}}{\Delta x + i \Delta y} \quad (4)$$

Let Δz approximate 0 along the path $\Delta y = k\Delta x (\Delta x \rightarrow 0)$, we get

$$\frac{f(\Delta z) - f(0)}{\Delta z} = \frac{\sqrt{|k|}}{1 + ki} \quad (5)$$

So f is not differentiable at $(0, 0)$. $\square$

13.(a) If $|f| = c$, then $\bar{f} = \frac{c^2}{f}$ is analytic in Ω . However, if both $u + iv$ and $u - iv$ satisfy the C.-R. equation it is easy to get $u_x = u_y = v_x = v_y = 0 \Rightarrow f$ is constant. $\square$

(b)(c) According to C.-R. equation, we can also get $u_x = u_y = v_x = v_y = 0 \Rightarrow f$ is constant. $\square$

14. This is the classical result we have learnt it in mathematical analysis. We leave it to reader. $\square$

15. Let $s_N = a_1 + \dots + a_N$. Use ex 14 we have $\sum_{n=1}^{N+1} a_n r^n = (1-r) \sum_{n=1}^N s_n r^n + s_{N+1} r^{N+1}$. First assume that $s = \sum_{n=1}^{\infty} a_n = 0$. Let $N \rightarrow \infty$ we have

$$\left| \sum_{n=1}^{\infty} a_n r^n \right| = \left| (1-r) \sum_{n=1}^{\infty} s_n r^n \right| \quad (6)$$

Choose K s.t. $\forall k > K, |s_k| < \epsilon$, then we have

$$\left| \sum_{n=1}^{\infty} a_n r^n \right| \leq (1-r) \sum_{k=1}^{K-1} M r^k + (1-r) \epsilon \frac{r^K}{1-r} \quad (7)$$

So let $r \rightarrow 1$ we have

$$\lim_{r \rightarrow 1} \sup \left| \sum_{n=1}^{\infty} a_n r^n \right| = 0 \quad (8)$$

Then if $s \neq 0$, we define $d_n = a_n - \frac{s}{2^n} \Rightarrow \sum_{n=1}^{\infty} d_n = 0$. Repeat the process above and we draw the conclusion. $\square$

16. This is an exercise for mathematical analysis and we leave it to reader. $\square$

17. We can choose N s.t. $\forall n > N, \frac{|a_{n+1}|}{|a_n|} \in (L - \epsilon, L + \epsilon)$. We multiply them and get $|a_n| \in (|a_N|(L - \epsilon)^{n-N+1}, |a_N|(L + \epsilon)^{n-N+1}) \Rightarrow \lim_{n \rightarrow +\infty} |a_n|^{\frac{1}{n}} = L$. $\square$

18. Let

$$f(z) = \sum_{n=0}^{\infty} a_n z^n \quad (9)$$

and for any $z_0 \in \Omega$ we write $z = z_0 + (z - z_0)$ then we have

$$f(z) = \sum_{n=0}^{\infty} a_n \left(\sum_{i=0}^n C_n^i z_0^{n-i} (z - z_0)^i \right) = \sum_{n=0}^{\infty} b_n (z - z_0)^n \quad (10)$$

Observe that $b_n = \sum_{i=0}^{\infty} C_{n+i}^n a_{n+i} z_0^i$. By Stirling's approximation, we have

$$\frac{1}{\lim_{i \rightarrow +\infty} \sqrt[i]{|C_{n+i}^n a_{n+i}|}} = \frac{1}{\lim_{i \rightarrow +\infty} \sqrt[i]{|a_{n+i}|}}$$

Hence b_n are convergent. $\square$

19. (i) On the unit circle, we suppose $z = \cos \theta + i \sin \theta, \theta \in [0, 2\pi)$, then $a_n = nz^n = n \cos \theta + in \sin \theta$. We know that $\lim_{n \rightarrow \infty} n \cos n\theta$ doesn't exist $\forall \theta \in (0, 2\pi)$. But when $\theta = 0$, we have $\lim_{n \rightarrow \infty} n \cos n\theta = +\infty$, so $\lim_{n \rightarrow +\infty} a_n$ doesn't exist too. Hence $\sum_{n=1}^{+\infty} a_n$ diverges. $\square$

(ii) On the unit circle, we suppose $z = \cos \theta + i \sin \theta, \theta \in [0, 2\pi)$, then $\alpha_n = z^n/n^2 = \cos n\theta/n^2 + i \sin n\theta/n^2 = a_n + ib_n$

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{|\cos n\theta|}{n^2} \leq \sum_{n=1}^{\infty} \frac{1}{n^2}$$

As $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, so $\sum_{n=1}^{\infty} a_n$ converges. Similarly,

$$\sum_{n=1}^{\infty} |b_n| = \sum_{n=1}^{\infty} \frac{|\sin n\theta|}{n^2} \leq \sum_{n=1}^{\infty} \frac{1}{n^2}$$

As $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, so $\sum_{n=1}^{\infty} b_n$ converges. In summary, $\sum_{n=1}^{\infty} \alpha_n$ also converges as to be proved above. $\square$

(iii) On the unit circle, we suppose $z = \cos \theta + i \sin \theta, \theta \in [0, 2\pi)$, then $\alpha_n = z^n/n = \frac{\cos n\theta}{n} + i \frac{\sin n\theta}{n} = a_n + ib_n$. When $\theta = 0$, that is, when $z = 1$, $\sum_{n=1}^{+\infty} \frac{z^n}{n} = \sum_{n=1}^{\infty} \frac{1}{n}$ obviously diverges. When $\theta \in (0, 2\pi)$, $\sum \frac{\cos n\theta}{n}$ and $\sum \frac{\sin n\theta}{n}$ both converges by Dirichlet discriminance. Hence the power series $\sum z^n/n$ converges at every point of the unit circle except $z = 1$. $\square$

20. Directly use Taylor's formula at zero, we get

$$(1 - z)^{-m} = \sum_{n=0}^{\infty} a_n z^n$$

where

$$a_n = \frac{(m + n - 1)!}{(m - 1)!n!}$$

To prove

$$a_n \sim \frac{1}{(m-1)!} n^{m-1} \quad \text{as } n \rightarrow \infty$$

we just need to prove

$$(n+1) \cdots (m+n-1) \sim n^{m-1}$$

The number of items in the left formula is actually $m-1$, thus we just need to prove

$$\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \cdots \left(1 + \frac{m-1}{n}\right) \sim 1$$

We already learn that m is a fixed positive integer and $n \rightarrow \infty$, thus the number of items in the left formula is limited, we can directly prove it. $\square$

21. For the first formula, a calculation can solve it.

$$\begin{aligned} & -\frac{z}{1-z} + \frac{z}{1-z^2} + \frac{z^2}{1-z^4} + \cdots + \frac{z^{2^n}}{1-z^{2^{n+1}}} + \cdots \\ &= \frac{-z^2}{1-z^2} + \frac{z^2}{1-z^4} + \cdots + \frac{z^{2^n}}{1-z^{2^{n+1}}} + \cdots \\ &= \frac{-z^4}{1-z^4} + \cdots + \frac{z^{2^n}}{1-z^{2^{n+1}}} + \cdots \\ & \quad \dots \\ &= \lim_{n \rightarrow \infty} \frac{z^{2^n}}{1-z^{2^n}} \end{aligned}$$

for $|z| < 1$, the formula equals 0, which means

$$\frac{z}{1-z^2} + \frac{z^2}{1-z^4} + \cdots + \frac{z^{2^n}}{1-z^{2^{n+1}}} + \cdots = \frac{z}{1-z}$$

As for the second formula, similarly,

$$\begin{aligned} & -\frac{z}{1-z} + \frac{z}{1+z} + \frac{2z^2}{1+z^2} + \cdots + \frac{2^k z^{2^k}}{1+z^{2^k}} + \cdots \\ &= \frac{-2z^2}{1-z^2} + \frac{2z^2}{1+z^2} + \cdots + \frac{2^k z^{2^k}}{1+z^{2^k}} + \cdots \\ &= \frac{-2^2 z^4}{1-z^4} + \cdots + \frac{2^k z^{2^k}}{1+z^{2^k}} + \cdots \\ & \quad \dots \\ &= \lim_{n \rightarrow \infty} \frac{-2^n z^{2^n}}{1+z^{2^n}} \end{aligned}$$

for $|z| < 1$, the formula equals 0, which means

$$\frac{z}{1+z} + \frac{2z^2}{1+z^2} + \cdots + \frac{2^k z^{2^k}}{1+z^{2^k}} + \cdots = \frac{z}{1-z}$$

□

22. Otherwise, suppose there are finite arithmetic progression $S_i = a_i + nd_i$ with $1 \leq i \leq k$. Then

$$\sum_{t=0}^{\infty} z^t = \sum_{i=1}^k \sum_{n=0}^{\infty} z^{a_i + nd_i} \Rightarrow \frac{1}{1-z} = \sum_{i=1}^k \frac{z^{a_i}}{1-z^{d_i}}$$

For example, take $z = e^{\frac{2\pi i}{d_1}}$, we get contradiction. □

23. When $x \in (-\infty, 0]$, $f^{(0)}(x) = 0$ is obvious. So it's derivable of any order and

$$\lim_{x \rightarrow 0^-} f^{(n)}(x) = 0$$

When $x \in (0, +\infty)$, $n = 1$, $f'(x) = \frac{-2}{x^3} e^{-\frac{1}{x^2}} \Rightarrow f'(0) = 0$. Use mathematical methods of induction, then suppose when $n = k$,

$$f^{(k)}(x) = e^{-\frac{1}{x^2}} \cdot P(x), P(x) = \sum C_i \frac{1}{x^i}$$

then when $n = k + 1$, we can deduce that

$$\begin{aligned} f^{(k+1)}(x) &= e^{-\frac{1}{x^2}} \cdot P'(x) - \frac{2}{x^3} \\ &= e^{-\frac{1}{x^2}} \cdot \left[P'(x) - \frac{2}{x^3} P(x) \right] \\ &= e^{-\frac{1}{x^2}} \cdot Q(x) \end{aligned}$$

where

$$Q(x) = \sum C'_i \frac{1}{x^i}$$

and we have

$$\lim_{x \rightarrow 0^+} \frac{e^{-\frac{1}{x^2}}}{x^n} = \lim_{x \rightarrow +\infty} e^{-x^2} x^n = 0$$

Hence

$$f^{(k+1)}(0) = \lim_{x \rightarrow 0^+} \frac{f^{(k)}(0)}{x} = 0$$

□

24. Trivial verification from the definition.

25. (a) Suppose $z = re^{i\theta}$, $\theta \in [0, 2\pi)$, then we have

$$\int_{\gamma} z^n dz = \int_{\gamma} r^n e^{i\theta n} dr e^{i\theta} = r^{n+1} \int_0^{2\pi} i e^{i\theta(n+1)} d\theta = \frac{r^{n+1}}{n+1} \cdot \int_0^{2\pi} d e^{i(n+1)\theta} = \frac{r^{n+1}}{n+1} \cdot [e^{i(n+1)2\pi} - 1] = 0$$

□

(b) Suppose $z = re^{i\theta} + ke^{i\varphi}$, $\theta, \varphi \in [0, 2\pi)$. φ, r, k here are all constant. Then

$$\begin{aligned}\int_{\gamma} z^n dz &= \left. \frac{z^{n+1}}{n+1} \right|_{\gamma} = \left. \frac{(re^{i\theta} + ke^{i\varphi})^{n+1}}{n+1} \right|_0^{2\pi} = \frac{(re^{i2\pi} + ke^{i\varphi})^{n+1}}{n+1} - \frac{(r + ke^{i\varphi})^{n+1}}{n+1} \\ &= \frac{(r + ke^{i\varphi})^{n+1}}{n+1} - \frac{(r + ke^{i\varphi})^{n+1}}{n+1} \\ &= 0\end{aligned}$$

(c) It may be complex if you try to calculate it directly, but this is trivial by Cauchy Theorem in next chapter. $\square$

26. We suppose F, G are any two primitive functions of f . Suppose $H = F - G$, then

$$[H]' = [F - G]' = F' - G' = f - f = 0$$

we may suppose $H = u + iv$, because f is continuous, so F, G is analytic, therefore H is also analytic. So we have

$$\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = 0$$

by Cauchy-Riemann equations,

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} = 0, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = 0 \Rightarrow \\ u &= c_1, v = c_2, \quad c_1, c_2 \text{ are constants.} \Rightarrow F - G = c_1 + ic_2\end{aligned}$$

$\square$

4 Chapter 2

4.1 Exercise

1. Let $f(z) = e^{iz^2}$. We integrate $f(z)$ around a circular sector of radius R running from $\theta = 0$ to $\frac{\pi}{4}$ just as the figure showed. Along the x axis the integral is

$$\int_0^R e^{ix^2} dx$$

Along the curved part we have $z = Re^{i\theta}$ and the integral is

$$\int_0^{\frac{\pi}{4}} e^{iR^2 e^{2i\theta}} iRe^{i\theta} d\theta = iR \int_0^{\frac{\pi}{4}} e^{-R^2 \sin 2\theta} \cdot e^{i(\theta + R^2 \cos 2\theta)} d\theta$$

Finally, along the segment at angle $\frac{\pi}{4}$ we have $z = re^{i\pi/4}$ and the integral is

$$\int_R^0 e^{ir^2 e^{i\pi/2}} e^{i\pi/4} dr$$

The total integral is zero since f is analytic everywhere. As $R \rightarrow \infty$, the integral over the third piece approaches

$$-e^{i\pi/4} \int_0^\infty e^{-x^2} dx = -e^{i\pi/4} \cdot \frac{\sqrt{\pi}}{2} = -\frac{\sqrt{2\pi}}{4} - \frac{\sqrt{2\pi}}{4}i$$

To estimate the integral over the curved piece, we shall take its absolute value, and we use that when $0 \leq \varphi \leq \pi/2$, we have $\sin \varphi \geq \frac{\varphi}{\pi/2}$ and

$$\begin{aligned} \left| iR \int_0^{\frac{\pi}{4}} e^{-R^2 \sin 2\theta} \cdot e^{i(\theta + R^2 \cos 2\theta)} d\theta \right| &\leq R \int_0^{\frac{\pi}{4}} e^{-R^2 \sin 2\theta} d\theta \leq \\ R \int_0^{\frac{\pi}{4}} e^{-4R^2 \theta / \pi} d\theta &= -\frac{\pi}{4R} e^{-4R^2 \theta / \pi} \Big|_0^{\frac{\pi}{4}} = \frac{\pi(1 - e^{-R^2})}{4R} \end{aligned}$$

As $R \rightarrow \infty$, this result approaches zero and we have

$$\int_0^\infty e^{ix^2} dx - \frac{\sqrt{2\pi}}{4} - \frac{\sqrt{2\pi}}{4}i = 0$$

Take real and imaginary parts, we have

$$\int_0^\infty \sin(x^2) dx = \int_0^\infty \cos(x^2) dx = \frac{\sqrt{2\pi}}{4}$$

2. Here we consider the function $f(z) = \frac{e^{iz}}{z}$, around an indented semicircular contour

bounded by circles of radius ϵ and R in the upper half plane. The integrals along the two portions of the real axis add up to

$$\int_{-R}^{-\epsilon} \frac{\cos x + i \sin x}{x} dx + \int_{\epsilon}^R \frac{\cos x + i \sin x}{x} dx = 2i \int_{\epsilon}^R \frac{\sin x}{x} dx$$

The integral around the arc of radius R tends to zero as $R \rightarrow \infty$, by the Jordan lemma; since this lemma isn't mentioned in the book, here's a proof for this specific case: On the arc, $z = Re^{i\theta}$ so the integral is

$$\int_0^{\pi} \frac{e^{iRe^{i\theta}}}{Re^{i\theta}} iRe^{i\theta} d\theta = i \int_0^{\pi} e^{-R \sin \theta} e^{iR \cos \theta} d\theta$$

we take its absolute value

$$\begin{aligned} \left| i \int_0^{\pi} e^{-R \sin \theta} e^{iR \cos \theta} d\theta \right| &\leq \int_0^{\pi} e^{-R \sin \theta} d\theta = 2 \int_0^{\pi/2} e^{-R \sin \theta} d\theta \\ &\leq 2 \int_0^{\pi/2} e^{-2R\theta/\pi} d\theta = - \frac{\pi e^{-2R\theta/\pi}}{R} \Big|_0^{\pi/2} = \frac{\pi(1 - e^{-R})}{R} \end{aligned}$$

which tends to 0 as $R \rightarrow \infty$. Finally, the integral over the inner semicircle tends to $-\pi i$, which can be easily proved and I'll show you behind. As we know, $\frac{e^{iz}}{z} = \frac{1}{z} + O(1)$ as $z \rightarrow 0$, and since the length of the semicircle is tending to zero, the integral over it approaches the integral of $\frac{1}{z}$ over it, which is

$$\int_{\pi}^0 \frac{1}{\epsilon e^{i\theta}} i e^{i\theta} d\theta = - \int_0^{\pi} i d\theta = -\pi i$$

Putting the pieces together and letting $R \rightarrow \infty$ and $\epsilon \rightarrow 0$, we have

$$2i \int_0^{\infty} \frac{\sin x}{x} dx - \pi i = 0 \Rightarrow \int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}$$

3. We consider $e^{(ib-a)x}$. Let $-u = (ib-a)x$, since

$$\int_0^{\infty} e^{(ib-a)x} dx = \lim_{t \rightarrow +\infty} \frac{a+ib}{\sqrt{a^2+b^2}} \int_0^{(ib-a)t} e^{-u} du$$

Like ex 1, we consider the circular sector of radius $R = \sqrt{a^2+b^2}t$. The path of integral is along $0 \rightarrow (ib-a)t \rightarrow \sqrt{a^2+b^2}t \rightarrow 0$. Similarly we know the integral of arc tends to 0 as $R \rightarrow \infty$. Hence

$$\int_0^{\infty} e^{(ib-a)x} dx = \lim_{t \rightarrow +\infty} \frac{a+ib}{\sqrt{a^2+b^2}} \int_0^{(ib-a)t} e^{-u} du = \lim_{t \rightarrow +\infty} \frac{a+ib}{\sqrt{a^2+b^2}} \int_0^{\sqrt{a^2+b^2}t} e^{-u} du = \frac{a+ib}{\sqrt{a^2+b^2}}$$

So

$$\begin{aligned}\int_0^\infty e^{-ax} \cos bxdx &= \frac{a}{\sqrt{a^2 + b^2}} \\ \int_0^\infty e^{-ax} \sin bxdx &= \frac{b}{\sqrt{a^2 + b^2}}\end{aligned}$$

4. In page 42 we know the proof of condition $\xi \in \mathbb{R}$. Let $\xi = a + bi$, we have

$$\begin{aligned}\int_{-\infty}^{+\infty} e^{-\pi x^2} e^{-2\pi i x \xi} dx &= e^{\pi b^2} \int_{-\infty}^{+\infty} e^{-\pi(x-b)^2} e^{-2\pi i a(x-b)} dx = \\ e^{\pi b^2} e^{-2\pi i ab} \int_{-\infty}^{+\infty} e^{-\pi u^2} e^{-2\pi i a u} du &= e^{\pi b^2} e^{-2\pi i ab} e^{-\pi a^2} = e^{-\pi \xi^2}\end{aligned}$$

Where $u = x - b$.

5. Write $f(z)$ as $f(x, y) = u(x, y) + iv(x, y)$ where u, v are real-valued and $z = x + iy$. Then $dz = dx + idy$ so

$$\begin{aligned}\oint_T f(z)dz &= \oint_T [u(x, y) + iv(x, y)](dx + idy) \\ &= \oint_T udx - vdy + i \oint_T vdx + udy \\ &= - \iint \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) dx dy + i \iint \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) dx dy \\ &= 0\end{aligned}$$

by the Cauchy-Riemann equations.

6. Let γ_ϵ be a circle of radius ϵ centered at ω , where ϵ is sufficiently small that γ_ϵ lies within the interior of T . Since f is holomorphic in the region R between T and γ_ϵ ,

$$\int_{\partial R} f(z)dz = \int_T f(z)dz - \int_{\gamma_\epsilon} f(z)dz = 0$$

which means that

$$\int_T f(z)dz = \int_{\gamma_\epsilon} f(z)dz$$

But f is bounded near ω and the length of γ_ϵ goes to 0 as $\epsilon \rightarrow 0$, so $\int_{\gamma_\epsilon} f(z)dz \rightarrow 0$ and therefore $\int_T f(z)dz = 0$.

7. By the Cauchy derivative formula, we have

$$f'(0) = \frac{1}{2\pi i} \oint_{C_r} \frac{f(\zeta)}{\zeta^2} d\zeta$$

where C_r is the circle of radius r centered at zero with $0 < r < 1$. Substituting $-\zeta$ for ζ and

adding the two equations yields

$$\begin{aligned}
|2f'(0)| &= \left| \frac{1}{2\pi i} \oint_{C_r} \frac{f(\zeta) - f(-\zeta)}{\zeta^2} d\zeta \right| \\
&\leq \frac{1}{2\pi} \oint_{C_r} \left| \frac{f(\zeta) - f(-\zeta)}{\zeta^2} \right| d\zeta \\
&\leq \frac{1}{2\pi} \oint \frac{d}{r^2} d\zeta \\
&= \frac{d}{r}
\end{aligned}$$

Letting $r \rightarrow 1$ yields the desired result.

8. For any $x \in \mathbb{R}$, consider a circle C centered at x of radius $\frac{1}{2}$, for $z \in C$,

$$1 + |z| \leq 1 + |x| + |z - x| \leq \frac{3}{2} + |x| < 2(1 + |x|)$$

So

$$|f(z)| \leq A(1 + |z|)^\eta \leq A2^\eta(1 + |x|)^\eta$$

When we suppose $\|f\|_C = \sup_{z \in C} |f(z)|$, we have $\|f\|_C \leq A2^\eta(1 + |x|)^\eta$, thus by Cauchy inequalities,

$$f^{(n)}(x) \leq \frac{n! \|f\|_C}{\left(\frac{1}{2}\right)^n} \leq n! 2^n A2^\eta(1 + |x|)^\eta = A_n(1 + |x|)^\eta$$

with $A_n = n! 2^n A2^\eta$.

9. If we let $f(z) = \varphi(z + z_0) - z_0$ for $z \in \Omega - z_0$, we have $f(z) \in \Omega - z_0$ and f is linear if φ is. Thus, we may assume that $z_0 = 0$. Expanding in a power series around 0, we have $\varphi(z) = z + a_2 z^2 + \dots$

Suppose a_n is the first nonzero coefficient with $n > 1$. Then $\varphi(z) = z + a_n z^n + O(z^{n+1})$, if $\varphi_k(z) = \varphi \circ \dots \circ \varphi$, then as we have proved the case that $k = 1$, we use the mathematical induction to prove that $\varphi_k(z) = z + k a_n z^n + O(z^{n+1})$. If it's true for k , it follows that

$$\begin{aligned}
\varphi_{k+1}(z) &= (z + k a_n z^n + O(z^{n+1})) + a_n (z + k a_n z^n + O(z^{n+1}))^n \\
&\quad + O\left((z + k a_n z^n + O(z^{n+1}))^{n+1}\right) \\
&= z + (k + 1) a_n z^n + O(z^{n+1})
\end{aligned}$$

Now we shall let $r > 0$ and $z \in \Omega$ for $|z| \leq r$. By the Cauchy inequalities,

$$\left| \varphi_k^{(n)}(0) \right| \leq \frac{n! \|\varphi_k\|_r}{r^n}$$

where

$$\|\varphi_k\|_r = \sup_{|z|=r} |\varphi_k(z)|$$

But $\varphi_k(z) \in \Omega$ is bounded, so there must have a constant independent of n and k , which we call it M . And we simply calculate $\varphi_k^{(n)}, \varphi_k(z) = z + ka_n z^n + O(z^{n+1})$, we have $\varphi_k^{(n)}(0) = kn!a_n$. Then we plug them in the pervious formula, we have

$$kn!|a_n| \leq \frac{n!M}{r^n} \Rightarrow |a_n| \leq \frac{M}{kr^n}$$

for all k . Letting $k \rightarrow \infty$, we have $a_n = 0$. Thus, there cannot be no nonzero terms of order $n > 1$ in the power series expansion of φ , which means that φ is linear.

10. Assume it is right. Let $\{P_n\}_{n \in \mathbb{N}_+}$ approximate uniformly to continous function f . By Cauchy theorem, for any closed curve γ in $\mathbb{D}$, we have

$$\int_{\gamma} f(z) dz = \int_{\gamma} \lim_{n \rightarrow \infty} P_n(z) dz = \lim_{n \rightarrow \infty} \int_{\gamma} P_n(z) dz = 0$$

Hence by Morera's theorem we have f is analytic in $\mathbb{D}$. However, choose $f = \bar{z}$ we get the contradiction. $\square$

11.(a) We may start with RHS,

$$\begin{aligned} & \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) Re \left(\frac{Re^{i\varphi} + z}{Re^{i\varphi} - z} \right) d\varphi \\ &= \frac{1}{4\pi} \int_0^{2\pi} f(Re^{i\varphi}) \left(\frac{Re^{i\varphi} + z}{Re^{i\varphi} - z} + \frac{Re^{-i\varphi} + \bar{z}}{Re^{-i\varphi} - \bar{z}} \right) d\varphi \\ &= \frac{1}{4\pi} \int_0^{2\pi} f(Re^{i\varphi}) \left(2 \frac{Re^{i\varphi}}{Re^{i\varphi} - z} - 1 + 1 - 2 \frac{\bar{z}}{\bar{z} - Re^{-i\varphi}} \right) d\varphi \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \frac{Re^{i\varphi} d\varphi}{Re^{i\varphi} - z} - \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \frac{\bar{z} d\varphi}{\bar{z} - Re^{-i\varphi}} \\ &= \frac{1}{2\pi i} \int_0^{2\pi} f(Re^{i\varphi}) \frac{d(Re^{i\varphi})}{Re^{i\varphi} - z} - \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \frac{Re^{i\varphi} d\varphi}{Re^{i\varphi} - R^2/\bar{z}} \\ &= \frac{1}{2\pi i} \int_0^{2\pi} f(Re^{i\varphi}) \frac{d(Re^{i\varphi})}{Re^{i\varphi} - z} - \frac{1}{2\pi i} \int_0^{2\pi} f(Re^{i\varphi}) \frac{d(Re^{i\varphi})}{Re^{i\varphi} - R^2/\bar{z}} \end{aligned}$$

By the Cauchy integral formula, the first part is equal to $f(z)$ and the latter is equal to zero.

(b) A straightforward calculation can solve it.

$$\begin{aligned}
\operatorname{Re} \left(\frac{Re^{i\gamma} + r}{Re^{i\gamma} - r} \right) &= \operatorname{Re} \left(\frac{R \cos \gamma + r + iR \sin \gamma}{R \cos \gamma - r + iR \sin \gamma} \right) \\
&= \operatorname{Re} \left(\frac{[R \cos \gamma + r + iR \sin \gamma][R \cos \gamma - r - iR \sin \gamma]}{[R \cos \gamma - r]^2 + R^2 \sin^2 \gamma} \right) \\
&= \operatorname{Re} \left(\frac{R^2 \cos^2 \gamma - r^2 + R^2 \sin^2 \gamma}{R^2 - 2Rr \cos \gamma + r^2} \right) = \frac{R^2 - r^2}{R^2 - 2Rr \cos \gamma + r^2}
\end{aligned}$$

12. (a) Define the imaginary part v as following:

$$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x}$$

Then we solve v :

$$v(x, y) = v_0 + \int_{y_0}^y \frac{\partial u}{\partial x}(x, y') dy'$$

Then we can find $u + iv$ is exactly holomorphic by C-R equation. $\square$

(b) u is harmonic then there exists holomorphic f s.t. $\Re(f) = u$. By Exercise 11, we have:

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \Re \left(\frac{Re^{i\varphi} + z}{Re^{i\varphi} - z} \right) d\varphi$$

Then take the real part we derive:

$$\begin{aligned}
u(z) &= \frac{1}{2\pi} \int_0^{2\pi} u(Re^{i\varphi}) \Re \left(\frac{e^{i\varphi} + re^{i\theta}}{e^{i\varphi} - re^{i\theta}} \right) d\varphi \\
&= \frac{1}{2\pi} \int_0^{2\pi} u(Re^{i\varphi}) \Re \left(\frac{e^{i(\varphi-\theta)} + r}{e^{i(\varphi-\theta)} - r} \right) d\varphi \\
&= \frac{1}{2\pi} \int_0^{2\pi} u(Re^{i\varphi}) P_r(\theta - \varphi) d\varphi
\end{aligned}$$

$\square$

13. Lemma 1. Let $S \subset \mathbb{C}$ be a subset of the plane with no accumulation points. Then S is at most countable.

Proof. For each $x \in S$, since x is not an accumulation point of S , $\exists r_x > 0$ such that $B_{r_x} \cap S = \{x\}$. Then $\{B_{r_x/2}(x) : x \in S\}$ is a disjoint family of open sets; since each contains a distinct rational point, it is at most countable. But this set is bijective with S , so S is at most countable.

Now suppose that f is not a polynomial. Then none of its derivatives can be identically zero, because if $f^{(n)}$ were identically zero, then $f^{(k)}$ would be zero for $k \geq n$ and f would be a polynomial of degree $\geq n - 1$. Since the derivatives of f are entire functions that are not everywhere zero, the set of zeros of $f^{(n)}$ has no accumulation points, so it is at most countable by the lemma. The set of zeros of any derivative of f must then be countable since it is a

countable union of countable sets. But by hypothesis, every point $z \in \mathbb{C}$ is a zero of some derivative of f , since if $f(z) = \sum c_n (z - z_0)^n$ and $c_k = 0$, then $\left. \frac{d^k}{dz^k} f(z) \right|_{z_0} = 0$. Since $\mathbb{C}$ is uncountable, this is a contradiction, so f must be a polynomial. $\square$

14. Since z_0 is the only pole on the unit circle, we can decompose f into $f = \frac{c_{-1}}{z - z_0} + g$ which g is holomorphic in the closed unit disc. Hence in $|z| < R, R > 1$, we have $f - \frac{c_{-1}}{z - z_0}$ is analytic. Let $f - \frac{c_{-1}}{z - z_0} = \sum_{i=0}^{\infty} b_i z^i, |z| < R$. Since it converges at z_0 , we have $b_n z_0^n \rightarrow 0$. Notice that

$$\begin{aligned} \frac{c_{-1}}{z - z_0} &= \sum_{n=0}^{\infty} \frac{-c_{-1}}{z_0^{n+1}} z^n, |z| < 1 \Rightarrow \\ f(z) - \frac{c_{-1}}{z - z_0} &= \sum_{n=0}^{\infty} \left(a_n + \frac{-c_{-1}}{z_0^{n+1}} \right) z^n, |z| < 1 \Rightarrow \\ b_n &= a_n + \frac{c_{-1}}{z_0^{n+1}} \end{aligned}$$

Hence we have

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1}} = \lim_{n \rightarrow \infty} \frac{b_n z_0^{n+2} - c_{-1} z_0}{b_{n+1} z_0^{n+2} - c_{-1}} = z_0$$

$\square$

15. solution 1. Define

$$F(z) = \begin{cases} f(z) & |z| \leq 1 \\ \frac{1}{\overline{f(1/\bar{z})}} & |z| > 1 \end{cases}$$

Then F is obviously continuous for both $|z| < 1$ and $|z| > 1$; for $|z| = 1$, we clearly have continuity from the inside, and if $\omega \rightarrow z$ with $|\omega| > 1$, then $\frac{1}{\bar{\omega}} \rightarrow \frac{1}{\bar{z}} = z$ and $F(\omega) = \frac{1}{\overline{f(1/\bar{\omega})}} \rightarrow \frac{1}{\overline{f(z)}} = f(z) = F(z)$. Hence F is continuous everywhere. It is known to be holomorphic for $|z| < 1$. For $|z| > 1$ we can compute $\frac{\partial f}{\partial \bar{z}} = 0$; alternatively, if Γ is any contour lying in the region $|z| > 1$, let Γ' be the image of Γ under the map $\omega = \frac{1}{\bar{z}}$. Then Γ' is a contour lying in the region $|\omega| < 1$ and excluding the region from its interior (since the point at infinity does not lie within Γ), so

$$\oint_{\Gamma} F(z) dz = \oint_{\Gamma'} \frac{1}{\overline{f(\bar{\omega})}} \frac{-d\omega}{\omega^2} = 0$$

since $\frac{1}{\overline{f(\bar{\omega})}\omega^2}$ is analytic on and inside Γ' . To show F is analytic at points on the unit circle we follow the same procedure as with the Schwarz reflection principle, by subdividing a triangle which crosses the circle into triangles which either have a vertex on the circle or an edge lying

along the circle (i.e. a chord of the circle). In the former case we may move the vertex by ϵ to conclude that the integral around the triangle is zero. In the case where a side of the triangle is a chord of the circle, we subdivide into smaller triangles (take the midpoint of the circular arc spanned by the chord) until the chord lies within ϵ of the circle and apply the same argument. The result is that F is entire. But F is bounded since $f(\overline{\mathbb{D}})$ is a compact set which excludes 0 and hence exclude a neighborhood of zero, so $\frac{1}{f}$ is bounded on $\mathbb{D}$. Since F is a bounded entire function, it is constant, so f is constant. (Liouville's theorem)

solution 2. Since f is non-vanishing on $\overline{\mathbb{D}}$, $\frac{1}{f}$ is analytic in $\mathbb{D}$. If f is not constant, by maximal modulus principle we have $|f| < 1$ in $\mathbb{D} \Rightarrow |\frac{1}{f}| > 1$ in $\mathbb{D}$. However, in the unit circle we have $|\frac{1}{f}| = 1$, contradiction! $\square$

4.2 Problem

1. (a) It suffices to prove that the singular point is dense in the unit circle. Let $z = x, 0 < x < 1$. Observe that

$$\lim_{x \rightarrow 1} f(x) = x^2 + \dots + x^{2^n} + \dots > \lim_{x \rightarrow 1} x^2 + \dots + x^{2^n} = n \Rightarrow \lim_{x \rightarrow 1} f(x) = \infty$$

So 1 is the singular point. Notice that

$$f(z) = z^2 + \dots + z^{2^n} + (z^{2^{n+1}} + \dots) = z^2 + \dots + z^{2^n} + f(z^{2^n})$$

So every point ω satisfies $\omega^{2^n} = 1$ is the singular point. Obviously the singular point is dense. $\square$

(b) Since $|f(z)| \leq \sum_{n=0}^{\infty} 2^{-n\alpha} = \frac{1}{1-2^{-\alpha}}, |z| = 1 \Rightarrow f(z)$ converge for $|z| = 1$. Let $|z_0| = 1, \forall \epsilon > 0$, we can choose $N, \forall n > N, \sum_{i=n}^{\infty} 2^{-i\alpha} < \epsilon$. Hence

$$|f(z) - f(z_0)| \leq \sum_{n=0}^N 2^{-n\alpha} |z^{2^n} - z_0^{2^n}| + \sum_{n=N}^{\infty} 2^{-n\alpha} |z^{2^n} - z_0^{2^n}| < \sum_{n=0}^N 2^{-n\alpha} |z^{2^n} - z_0^{2^n}| + 2\epsilon \rightarrow 0, \\ z \rightarrow z_0, \epsilon \rightarrow 0$$

.

So the extension is continuous.

Since in Book 1 we prove that

$$g(x) = \sum_{n=0}^{\infty} 2^{-n\alpha} e^{ix2^n}$$

is not differentiable (page 114, theorem 3.1), so this extension cannot be analytic. $\square$

2. Let $n = \prod_{i=1}^k p_i^{t_i}$, then $d(n) = \prod_{i=1}^k (1 + t_i)$. We can give a simple estimate of t_i . Since $p_i^{t_i} \leq n \rightarrow t_i \leq \log_{p_i} n = \frac{\ln n}{\ln p_i} \leq \frac{\ln n}{\ln 2}$. Obviously $k \leq \sum_{i=1}^k t_i \leq \log_2 n + 1 \leq 2 \log_2 n$ for $n \geq 2$. Hence by *L'Hospital's* rule,

$$\begin{aligned} \overline{\lim}_{n \rightarrow \infty} \sqrt[n]{d(n)} &= \overline{\lim}_{n \rightarrow \infty} \sqrt[n]{\prod_{i=1}^k (1 + t_i)} \leq \overline{\lim}_{n \rightarrow \infty} \sqrt[n]{\prod_{i=1}^k (1 + \frac{\ln n}{\ln p_i})} \leq \\ &= \overline{\lim}_{n \rightarrow \infty} e^{\frac{2 \log_2 n}{n} \ln(1 + \frac{\ln n}{\ln 2})} = \overline{\lim}_{n \rightarrow \infty} e^{\frac{2}{\ln 2} \cdot \frac{n \ln 2 + 2 \ln n}{n^2 \ln 2 + n \ln^2 n}} = 1 \end{aligned}$$

The sup can be reach by the sequence $\{d(2^n)\}_{n \in \mathbb{N}_+}$. Hence the ridus of convergence is 1.

By Abel theorem we know we can change the order of sum when $|z| < 1$. Let $g(z) = \sum_{k=1}^{\infty} \sum_{n=1}^{\infty} z^{kn}$. We observe that if $m = kn \Leftrightarrow k|m$ and $n|m$, so the term z^m occurs $d(m)$ times $\Rightarrow$

$$g(z) = \sum_{k=1}^{\infty} \sum_{n=1}^{\infty} z^{kn} = \sum_{n=1}^{\infty} \frac{z^n}{1 - z^n} = \sum_{n=1}^{\infty} d(n) z^n = F(z)$$

. Notice that

$$\log\left(\frac{1}{1-r}\right) = \sum_{n=1}^{\infty} \frac{r^n}{n}, r < 1$$

$\Rightarrow$

$$(1-r)|F(r)| = (1-r) \sum_{n=1}^{\infty} \frac{r^n}{1-r^n} = (1-r) \sum_{i=1}^{\infty} \frac{r^i}{1+r+\dots+r^{i-1}} \geq \sum_{n=1}^{\infty} \frac{r^n}{n} = \log\left(\frac{1}{1-r}\right), r \rightarrow 1$$

We assume $(p, q) = 1$ and note that for n not a multiple of q , one has $2\pi - 2\pi/q \geq \arg n\theta \geq 2\pi/q$ so $|1 - z^n| \geq a_{p,q} > 0$ (Since z^n will occurs in some discrete lines and for the minmal modulus we can consider the vertical lines from 1 to them), hence

$$\left| \sum_{n=1, n \neq rq}^{\infty} \frac{z^n}{1 - z^n} \right| \leq \frac{1}{a_{p,q}} \sum_{n=1}^{\infty} r^n \leq \frac{r}{a_{p,q}(1-r)}$$

On the other hand for $n = mq$ one has $\frac{z^n}{1-z^n} = \frac{r^{mq}}{1-r^{mq}}$ and

$$1 - r^{mq} = (1 - r)(1 + r + \dots + r^{mq-1}) \leq mq(1 - r) \Rightarrow$$

$$\sum_{m=1}^{\infty} \frac{z^{mq}}{1 - z^{mq}} = \sum_{m=1}^{\infty} \frac{r^{mq}}{1 - r^{mq}} \geq \frac{1}{1 - r} \sum_{m=1}^{\infty} \frac{r^{mq}}{mq} = \frac{1}{1 - r} \log \frac{1}{1 - r^q}$$

Now

$$\log \frac{1}{1 - r^q} = \log \frac{1}{1 - r} - \log (1 + r + \dots + r^{q-1}) \geq \log \frac{1}{1 - r} - \log q$$

So putting all together we get

$$\left| \sum_{n=1}^{\infty} \frac{z^n}{1 - z^n} \right| \geq \left| \sum_{n=rq}^{\infty} \right| - \left| \sum_{n \neq rq}^{\infty} \right| \geq \frac{1}{1 - r} \log \frac{1}{1 - r} - \frac{1}{1 - r} \log q - \frac{r}{a_{p,q}(1 - r)} \geq c_{p,q} \frac{1}{1 - r} \log \frac{1}{1 - r}$$

The last step is due to

$$\lim_{r \rightarrow 1} \frac{\frac{1}{1-r} \log \frac{1}{1-r}}{\frac{1}{1-r} \log \frac{1}{1-r} - \frac{1}{1-r} \log q - \frac{r}{a_{p,q}(1-r)}} = 1$$

So we conclude that F cannot be continued analytically past the unit circle since it will be unbounded as $r \rightarrow 1$. $\square$

3. I only prove (b) since I cannot understand the hint of (a)

(b) We assume φ is smooth and satisfies $\int \int_{|\omega| \leq 1} \varphi(\omega) dV(\omega) = 1$ and $\varphi(\omega) = 0, |\omega| > 1$.

Since $\int \int_{|\omega| \leq 1} \varphi(\omega) dV(\omega) = 1$, we obtain that $\forall \delta > 0, \exists \epsilon, \forall \epsilon' > 0, \int \int_{|\omega| \geq \delta} \frac{1}{\epsilon^2} \varphi(\omega/\epsilon) dV(\omega) < \epsilon'$. Then we have

$$|f_{\epsilon}(z) - f(z)| = \left| \int \int_{|\omega| \leq \epsilon} f(z - \omega) \varphi_{\epsilon}(\omega) dV(\omega) - \int \int_{|\omega| \leq \epsilon} f(z) \varphi_{\epsilon}(\omega) dV(\omega) \right| \leq$$

$$\int \int_{|\omega| \leq \epsilon} |f(z) - f(z - \omega)| |\varphi(\omega)| dV(\omega) \leq$$

$$\int \int_{|\omega| < \delta} |f(z) - f(z - \omega)| |\varphi(\omega)| dV(\omega) + \int \int_{\delta \leq |\omega| \leq \epsilon} |f(z) - f(z - \omega)| |\varphi(\omega)| dV(\omega)$$

Take any compact subset K of $\mathbb{C}$, since f is uniformly continuous in K we can choose proper δ s.t. $|f(z) - f(z - \omega)| < \epsilon', \forall |\omega| < \delta$. Then we can choose a ϵ sufficiently small and we have

$$\int \int_{|\omega| \geq \delta} \varphi(\omega) dV(\omega) \leq \frac{\epsilon'}{2 \sup_{u \in \{z - \omega | z \in K, |\omega| \leq \epsilon\}} |f(u)|} \Rightarrow$$

$$|f_{\epsilon}(z) - f(z)| \leq 2\epsilon', \forall z \in K$$

Hence $f_\epsilon \rightarrow f$ uniformly. Since

$$\iint_{\mathbb{R}^2} f(z - \omega) \varphi_\epsilon(\omega) dV(\omega) = \iint_{\mathbb{R}^2} f(\omega) \varphi_\epsilon(z - \omega) dV(\omega)$$

we can easily verify that f_ϵ is smooth. Let $f_\epsilon = u + iv$, by Green's formula we have

$$\begin{aligned} \int_\gamma f_\epsilon(z) dz &= \int_\gamma u dx - v dy + i \int_\gamma u dy + v dx = \\ \iint_{D, \partial D = \gamma} \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) dx dy + i \iint_{D, \partial D = \gamma} \left(\frac{\partial v}{\partial y} - \frac{\partial u}{\partial x}\right) dx dy &= 0 \end{aligned}$$

$\forall z \in \mathbb{C}$ we choose a proper curve γ and we can let D shrink to z . By equation above we know f_ϵ satisfies C.-R. equation. Hence f_ϵ is analytic. Since $f_\epsilon \rightarrow f$ uniformly, we know f is holomorphic. $\square$

4. Let z_0 in a bounded component U of K^c and let $f(z) = \frac{1}{z - z_0}$. If $\frac{1}{z - z_0}$ can be approximated uniformly by polynomials on K , then there exists $p(z)$ s.t. $|(z - z_0)p(z) - 1| < 1$. Since connected components are closed and disjoint, we have

$$\partial U = \overline{U} \cap \overline{\mathbb{C} - U} = U \cap \overline{(\cup_i U_i) \cup K} = U \cap K \subset K.$$

(U_i denotes the connected component of K^c)

So we have $|(z - z_0)p(z) - 1| < 1, z \in \partial U$. By maximal modulus principle, we have $|(z - z_0)p(z) - 1| < 1, z \in U$. But $1 = |(z_0 - z_0)p(z) - 1| < 1$, contradiction! $\square$

5. (a) Since h is entire, there exist a sequence $\{n_k\}$ s.t. $\lim_{k \rightarrow +\infty} p_{n_k}(z) = h(z)$. If $|F(z) - p_n(z - M_n)| < \frac{1}{n}, \forall z \in D_n$, we have $|F(z + M_{n_k}) - h(z)| < |F(z + M_{n_k}) - p_{n_k}(z)| + |p_{n_k}(z) - h(z)| \rightarrow 0, \forall z \in D_n$. Hence $\lim_{n \rightarrow \infty} F(z + N_k) = h(z)$.

(b) First we assume the polynomials are arranged by this diagram

$$\begin{array}{ll} \text{deg} & \text{polynomials} \\ 0 & p_{11}, p_{12}, p_{13}, \dots \\ 1 & p_{21}, p_{22}, p_{23}, \dots \\ 2 & p_{31}, p_{32}, p_{33}, \dots \end{array} \tag{11}$$

we number it by diagonal lines ($p_{11}, p_{12}, p_{21}, p_{31}, p_{22}, \dots$). Hence easily we have $\deg p_{\frac{n(n+1)}{2}} \leq n \Rightarrow \deg p_n \leq 2\sqrt{n}$. We have $|p_n(z)| = o(|z|^{2\sqrt{n}+1})$.

We can choose $M_n = 4^{n+1}, c_n = 2^{-n}$. Since

$$\begin{aligned} |F(z) - p_n(z - M_n)| &\leq \sum_{k < n} |u_k(z)| + |p_n(z - M_n)| |1 - e^{-c_n(z - M_n)^2}| + \sum_{k > n} |u_k(z)| = \\ &I_1 + I_2 + I_3 \end{aligned}$$

For I_2 , let $z = M_n + Re^{i\theta}$, $0 \leq R \leq n$, we have

$$I_2 = |p_n(Re^{i\theta})| |c_n R^2 e^{2i\theta} - \frac{1}{2} c_n^2 R^4 e^{4i\theta} + \dots| \leq |p_n(Re^{i\theta})| \sum_{i=1}^{+\infty} \frac{n^{2i}}{2^{ni}} \leq C_{n1} \frac{n^{2\sqrt{n}+3}}{2^n} \left(\sum_{i=1}^{+\infty} \frac{n^{2i}}{2^{ni}} \right) \leq K_n \frac{n^2}{2^n}$$

For I_1 , we know

$$\begin{aligned} |\arg(z - 4^{i+1})^2| &\leq |\arg(z - 4^n)^2| \leq 2 \arcsin \frac{n}{3 \cdot 4^n} \leq \frac{\pi}{4} \\ |(z - 4^{i+1})^2| &\geq (4^{n+1} - 4^{i+1} - n)^2 \end{aligned}$$

So

$$\begin{aligned} I_1 &\leq \sup_{i \in [1, n-1], z \in D_i} |p_i(z - M_i)| \sum_{i=1}^{n-1} |e^{-2^{-i}(4^{n+1} - 4^{i+1} - n)^2 \cos \arg(z - 4^{i+1})^2}| \leq \\ &\sup_{i \in [1, n-1], z \in D_i} |p_i(z - M_i)| \sum_{i=1}^{n-1} |e^{-2^{-i}(4^{n+1} - 4^{i+1} - n)^2 \frac{\sqrt{2}}{2}}| \leq \frac{C_{n2} 4^{(2\sqrt{n}+1)(n+1)} (n-1)}{e^{2^n}} \end{aligned}$$

For I_3

$$\begin{aligned} I_3 &\leq \sum_{i=n+1}^{+\infty} |p_i(z - M_i)| |e^{-2^{-i}(4^{i+1} - 4^{n+1} - n)^2 \cos \arg(z - 4^{i+1})^2}| \leq \\ &C_{n3} \sum_{i=n+1}^{+\infty} 4^{(2\sqrt{i}+1)(i+1)} |e^{-2^{-i}(4^{i+1} - 4^{n+1} - n)^2 \frac{\sqrt{2}}{2}}| \leq \\ &C_{n3} \sum_{i=n+1}^{+\infty} \frac{4^{(2\sqrt{i}+1)(i+1)}}{e^{2^i}} \end{aligned}$$

By $|p_n(z)| = o(|z|^{2\sqrt{n}+1})$ we know $C_{n1}, C_{n2}, C_{n3} \rightarrow 0, n \rightarrow 0$. So for large n , we have

$$|F(z) - p_n(z - M_n)| < \frac{1}{n}, \forall z \in D_n$$

□

5 Chapter 3

5.1 Exercise

1. As we know, $\sin \pi z = 0 \Leftrightarrow e^{i\pi z} = e^{-i\pi z}$. Suppose $z = x + iy$, $\sin \pi z = 0$, then we have

$$\begin{aligned}\cos \pi x \cdot [e^{2\pi y} - 1] - i \sin \pi x \cdot [e^{2\pi y} + 1] &= 0, x, y \in \mathbb{R} \\ \cos \pi x \cdot [e^{2\pi y} - 1] &= 0; \sin \pi x \cdot [e^{2\pi y} + 1] = 0\end{aligned}$$

Hence $\sin \pi x = 0 \Rightarrow x \in \mathbb{Z}; e^{2\pi y} - 1 = 0, \Rightarrow y = 0$, so $z \in \mathbb{Z}$. To find the order of the zero, we note that

$$2i \frac{d}{dz} \Big|_{z=n} \sin \pi z = i\pi (e^{in\pi} + e^{-in\pi}) = 2i\pi e^{in\pi} \neq 0$$

so the zero is of order 1. Finally, the residue at n of $\frac{1}{\sin \pi z}$ is

$$\lim_{z \rightarrow n} (z - n) \frac{1}{\sin \pi z} = \frac{1}{\pi \cos \pi n} = \frac{(-1)^n}{\pi}$$

2. Suppose $f(z) = \frac{1}{1+z^4}$, to find out the poles of it, we suppose $z^4 + 1 = 0$ and obviously by De Moivre's formula, we know it has two poles of order 1.

$$a_k = e^{\frac{\pi+2k\pi}{4}i}, k = 0, 1, 2, 3$$

Hence we have $1 + z^4 = (z - a_1)(z - a_2)(z - a_3)(z - a_4)$.

$$\text{Res}(f; a_k) = \lim_{z \rightarrow a_k} (z - a_k) \cdot \frac{1}{1 + z^4} = \frac{1}{4z^3} \Big|_{z=a_k} = \frac{1}{4a_k^3} = -\frac{a_k}{4}, k = 0, 1, 2, 3$$

thus the Residue Theorem guarantees that

$$\int_{-\infty}^{+\infty} \frac{dx}{x^4 + 1} = -2\pi i \cdot \frac{1}{4} \cdot (e^{\frac{\pi}{4}i} + e^{\frac{3}{4}\pi i}) = \pi \sin \frac{\pi}{4} = \frac{\pi}{\sqrt{2}}$$

3. Let $f(z) = \frac{e^{iz}}{z^2 + a^2}$. We integrate f around a semicircular contour of radius R in UHP , so $|f(z)| \leq \frac{1}{|z^2 + a^2|} \leq \frac{1}{R^2 - a^2}$. Hence the integral over the curved part is at most $\frac{\pi R}{R^2 - a^2} \rightarrow 0$ as $R \rightarrow \infty$. Hence

$$\int_{-\infty}^{+\infty} f(z) dz = 2\pi i \sum_{z \in UHP} \text{Res}(f; z)$$

The residues of f are $z = \pm ai$, so the only pole in the UHP is at ai with residue

$$\lim_{z \rightarrow ai} (z - ai) \frac{e^{iz}}{(z - ai)(z + ai)} = \frac{e^{-a}}{2ai}$$

Hence $\int_{-\infty}^{+\infty} f(z)dz = \pi \frac{e^{-a}}{a}$. Taking the real parts of both sides,

$$\int_{-\infty}^{+\infty} \frac{\cos x}{x^2 + a^2} dx = \pi \frac{e^{-a}}{a}$$

4. Here we begin with Jordan's Lemma

Lemma. If Γ_R is the semicircular contour $z(\theta) = Re^{i\theta}$, $0 \leq \theta \leq \pi$, in the upper half-plane, and $P(z)$ and $Q(z)$ are polynomials with $\deg Q(z) \geq \deg P(z) + 1$, then

$$\lim_{R \rightarrow \infty} \int_{\Gamma_R} \frac{P(z)e^{iz}}{Q(z)} dz = 0$$

Proof. As $R \rightarrow +\infty$, we have

$$\begin{aligned} \left| \int_{\Gamma_R} \frac{P(z)e^{iz}}{Q(z)} dz \right| &\leq O\left(\frac{1}{R}\right) \left| \int_{\Gamma_R} e^{iz} dz \right| \leq RO\left(\frac{1}{R}\right) \int_0^\pi |e^{iRe^{i\theta}}| d\theta = \\ &RO\left(\frac{1}{R}\right) \int_0^\pi e^{-R \sin \theta} d\theta = 2RO\left(\frac{1}{R}\right) \int_0^{\frac{\pi}{2}} e^{-R \sin \theta} d\theta \leq \\ &2RO\left(\frac{1}{R}\right) \int_0^{\frac{\pi}{2}} e^{-2R\theta/\pi} d\theta = O\left(\frac{\pi}{R}\right) \rightarrow 0 \end{aligned}$$

Now back to our problem, let $f(z) = \frac{ze^{iz}}{z^2 + a^2}$ and integrate f around a semicircular contour of radius R . By Jordan's Lemma, the integral over the curved part tend to 0 as $R \rightarrow \infty$. Hence

$$\int_{-\infty}^{\infty} f(z)dz = 2\pi i \sum_{z \in UHP} \text{Res}(f; z)$$

The poles of f are at $\pm ai$; the residue at ai is

$$\lim_{z \rightarrow ai} (z - ai) \frac{ze^{iz}}{z^2 + a^2} = \frac{e^{-a}}{2}$$

Hence $\int_{-\infty}^{\infty} f(z)dz = \pi i e^{-a}$, taking imaginary parts, we have

$$\int_{-\infty}^{+\infty} \frac{x \sin x}{x^2 + a^2} dx = \pi e^{-a}$$

5. First we note that if $\xi > 0$, substituting $u = -x$ reveals that

$$\int_{-\infty}^{+\infty} \frac{e^{-2\pi i x \xi}}{(1 + x^2)^2} dx = \int_{+\infty}^{-\infty} \frac{e^{-2\pi i u(-\xi)}}{(1 + u^2)^2} d(-u) = \int_{-\infty}^{+\infty} \frac{e^{-2\pi i u(-\xi)}}{(1 + u^2)^2} du$$

so that we may assume WLOG that $\xi \leq 0$, we integrate $f(z) = \frac{e^{-2\pi i z \xi}}{(1 + z^2)^2}$ along a semicircular

contour of radius R , since $|e^{-2\pi iz\xi}| \leq 1$ for $z \in UHP$ and $\xi \leq 0$, the integral over the curved part is

$$\left| \int_{\Gamma_R} f(z) dz \right| \leq \int_0^\pi \left| \frac{R}{(1 + R^2 e^{2i\theta})^2} \right| d\theta \leq \frac{R}{(R^2 - 1)^2} \cdot \pi$$

while $R \rightarrow \infty$, it tends to 0. Hence $\int_{-\infty}^{+\infty} f(z) dz = 2\pi i \sum_{z \in UHP} \text{Res}(f; z)$ The only pole in the UHP is at $z = i$, because this is a double pole, the residue is

$$\lim_{z \rightarrow i} \frac{d}{dz} (z - i)^2 f(z) = \lim_{z \rightarrow i} \frac{d}{dz} (z - i)^2 \cdot \frac{e^{-2\pi iz\xi}}{(1 + z^2)^2} = \frac{(1 - 2\pi\xi)e^{2\pi\xi}}{4i}$$

Hence

$$\int_{-\infty}^{+\infty} \frac{e^{-2\pi ix\xi}}{(1 + x^2)^2} dx = 2\pi i \cdot \frac{(1 - 2\pi\xi)e^{2\pi\xi}}{4i} = \frac{\pi}{2} (1 + 2\pi|\xi|) e^{-2\pi|\xi|}$$

6. Let $f(z) = \frac{dz}{(z^2+1)^{n+1}}$, then the pole of $f(z)$ in the UHP is at $z = i$, because this is a pole of order $n + 1$, its residue is

$$\begin{aligned} \frac{1}{n!} \lim_{z \rightarrow i} \frac{d^n}{dz^n} \frac{1}{(z + i)^{n+1}} &= \frac{1}{n!} \lim_{z \rightarrow i} (-n-1)(-n-2) \cdots (-2n) \frac{1}{(z + i)^{2n+1}} \\ &= \frac{(2n!)(-1)^n}{(n!)^2} \cdot \frac{1}{2i^{2n+1}} = \frac{(2n)!}{(2^n n!)^2} \cdot \frac{1}{2i} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot \frac{1}{2i} \end{aligned}$$

Integrating f around a semicircular contour of radius R , the integral over the curved part is bounded by $\frac{\pi R}{(R^2-1)^{n+1}} \rightarrow 0$. So

$$\int_{-\infty}^{+\infty} f(z) dz = 2\pi i \text{Res}(f; i) = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \pi$$

7. Letting $z = e^{i\theta}$ turns this into a contour integral around the unit circle C . which means

that.

$$\begin{aligned}
\int_0^{2\pi} \frac{d\theta}{(a + \cos \theta)^2} &= \oint_C \frac{dz/iz}{\left(a + \left(\frac{z+1/z}{2}\right)\right)^2} \\
&= \oint_C \frac{4zdz}{i(z^2 + 2za + 1)^2} \\
&= 2\pi i \operatorname{Res} \left(\frac{4z}{i(z^2 + 2za + 1)^2}; -a + \sqrt{a^2 - 1} \right) \\
&= 2\pi i \lim_{z \rightarrow -a + \sqrt{a^2 - 1}} \frac{d}{dz} \frac{4z}{i(z + a + \sqrt{a^2 - 1})^2} \\
&= 2\pi i \lim_{z \rightarrow -a + \sqrt{a^2 - 1}} \frac{4(a + \sqrt{a^2 - 1} - z)}{i(z + a + \sqrt{a^2 - 1})^3} \\
&= 2\pi i \frac{8a}{i(2\sqrt{a^2 - 1})^3} \\
&= 2\pi a (a^2 - 1)^{-3/2}
\end{aligned}$$

8. Again, we convert into a contour integral around the unit circle by substituting $z = e^{i\theta}$

$$\begin{aligned}
\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} &= \oint_C \frac{dz/iz}{a + b\left(\frac{z+1/z}{2}\right)} = \oint_C \frac{2dz}{i(bz^2 + 2az + b)} = \\
4\pi \operatorname{Res} \left(\frac{1}{bz^2 + 2az + b}; \frac{-a + \sqrt{a^2 - b^2}}{b} \right) &= 2\pi \frac{1}{2bz + 2a} \Big|_{z = \frac{-a + \sqrt{a^2 - b^2}}{b}} = \\
&= 2\pi \frac{1}{2\sqrt{a^2 - b^2}} = \frac{\pi}{\sqrt{a^2 - b^2}}
\end{aligned}$$

9. We need a lemma.

Lemma. Let $f(z)$ be a continuous function around the arc $S : z - z_0 = re^{i\theta}, \theta_1 \leq \theta \leq \theta_2$ and satisfy $\lim_{z \rightarrow z_0} (z - z_0)f(z) = \lambda$. uniformly in the arc. Then we have

$$\lim_{r \rightarrow 0} \int_{S_r} f(z) dz = i(\theta_2 - \theta_1)\lambda$$

proof. Since

$$\lim_{r \rightarrow 0} \left| \int_{S_r} f(z) dz - i(\theta_2 - \theta_1)\lambda \right| = \lim_{r \rightarrow 0} \left| \int_{S_r} f(z) - \frac{\lambda}{z - z_0} dz \right| \leq \int_{S_r} \left| \frac{(z - z_0)f(z) - \lambda}{z - z_0} \right| dz \rightarrow 0, r \rightarrow 0$$

We draw the conclusion. If condition turns to $z_0 = 0$ and $z \rightarrow +\infty$, we have similar result.

Consider $f(z) = 1 - e^{2\pi iz} = -2ie^{\pi iz} \sin(\pi z)$. We will integrate $\log f(z)$ along the rectangular contour bounded by $0, 1, iY, 1 + iY$, with quarter-circle indentations of radius ϵ at 0 and 1 . On this region, $e^{\pi iz}$ is in the upper half plane, and $\sin(\pi z)$ is in the right half plane. Hence, if we take the principal branch Arg of the argument function, $0 \leq \operatorname{Arg}(e^{\pi iz}) < \pi$ and $-\frac{\pi}{2} \leq$

$\text{Arg}(\sin(\pi z)) < \frac{\pi}{2}$. Thus

$$\begin{aligned}
-\pi &\leq \text{Arg}(-2i) + \text{Arg}(e^{\pi iz}) + \text{Arg}(\sin(\pi z)) \\
\Rightarrow \text{Arg}(f(z)) &= \text{Arg}(-2i) + \text{Arg}(e^{\pi iz}) + \log(\sin(\pi z)) \\
\Rightarrow \log(f(z)) &= \log(-2i) + \log(e^{\pi iz}) + \log(\sin(\pi z)) \\
\Rightarrow \log(f(z)) &= \log 2 - \frac{\pi}{2}i + \pi iz + \log(\sin(\pi z))
\end{aligned}$$

where $\log$ is the principal branch of the logarithm function, corresponding to the principal branch Arg of the argument function.

Returning to our contour integral, we note that the two vertical pieces out by the periodicity of the sine function. Moreover, the integral over the upper half of the rectangle approaches zero as $Y \rightarrow \infty$ since for $z = t + iY$ with $0 \leq t < 1$,

$$e^{2\pi iz} \rightarrow 0 \Rightarrow f(z) \rightarrow 1 \Rightarrow \log f(z) \rightarrow 0$$

when $Y \rightarrow \infty$. What's more, the integrals over the quarter-circle indentations also approach zero: Since $f(z) = z$ for z near 0, $\log f(z) \approx \log z$ so the integral over the quarter-circle near zero is $O(\epsilon |\log \epsilon|) \rightarrow 0$. Similar analysis holds near π . Since $\log f(z)$ is analytic on and inside our contour, we use the lemma above and we are left with

$$\int_0^1 \log f(z) dz = 0$$

By our above comments concerning the branches of the logarithm, this implies

$$0 = \int_0^1 \left(\log 2 - \frac{\pi}{2}i + \pi iz + \log(\sin(\pi z)) \right) dz = \log 2 + \int_0^1 \log \sin(\pi x) dx$$

10. Use the figure 10. We use residue formula and we have

$$\int_{-R}^{-\epsilon} + \int_{\epsilon}^R + \int_{C_{\epsilon}} + \int_{C_R} = 2\pi i \frac{\log ai}{2ai}$$

By lemma above and let $\epsilon \rightarrow 0$ and $R \rightarrow +\infty$, we know

$$\int_{-\infty}^0 + \int_0^{+\infty} = \pi \frac{\log ai}{a}$$

Hence compare the real part we know

$$\int_0^{\infty} \frac{\log x}{x^2 + a^2} dx = \frac{\pi}{2a} \log a$$

11. See page 136 step 4.

12. Since by ex 1 we know that $z = n$ is the pole of order 1 of f , let $N > |u|$, we have

$$\frac{1}{2\pi i} \int_{|z|=N+\frac{1}{2}} f(z) dz = \sum_{z=-N}^N \text{Res}(f; z) + \text{Res}(f; -u) = \sum_{n=-N}^N \frac{1}{(u+n)^2} - \frac{\pi^2}{(\sin \pi u)^2}$$

Let $N \rightarrow +\infty$, we have to show that the integral is equal to 0. For convenience we consider the square integral path $|x| = N + \frac{1}{2}$ and $|y| = N + \frac{1}{2}$.

Hence we have

$$|\cot \pi(x + iy)| = \left| \frac{e^{\pi i x - \pi y} + e^{-\pi i x + \pi y}}{e^{\pi i x - \pi y} - e^{-\pi i x + \pi y}} \right| = \left| \frac{e^{-2\pi i x} + e^{-2\pi y}}{e^{-2\pi i x} - e^{-2\pi y}} \right|$$

When $|y| \geq 1$ and $|x| = N + \frac{1}{2}$ or $|y| = N + \frac{1}{2}$, we easily know the $|\cot \pi(x + iy)|$ is bounded. When $|y| \leq 1$ we know

$$|\cot \pi(x + iy)| = |\cot \pi(N + \frac{1}{2} + iy)| = |\tanh(y)| \leq \tanh \frac{\pi}{2} < +\infty$$

Let $N \rightarrow +\infty$, by above argument we know the integral tends to 0. Hence we draw the conclusion.

13. Let $g(z) = (z - z_0) f(z)$. Then g is analytic in the punctured disc and $g(z) \rightarrow 0$ as $z \rightarrow z_0$. Since g is bounded, the singularity at z_0 is removable, so we can take g to be analytic at z_0 . Since g is a holomorphic function with $g(z_0) = 0$, $g(z) = (z - z_0) h(z)$ in some neighborhood of z_0 where h is holomorphic at z_0 . Then $f(z) = h(z)$ in a deleted neighborhood of z_0 so by defining $f(z_0) = h(z_0)$ we can extend f to a holomorphic function at z_0 . Thus the singularity is removable.

14. Define $g : \mathbb{C} \setminus \{0\} \rightarrow \mathbb{C}$ by $g(z) = f\left(\frac{1}{z}\right)$. Then g is holomorphic on the punctured plane. It cannot have a removable singularity at 0, because this would imply that f is bounded as $|z| \rightarrow \infty$ and therefore constant (so certainly not injective!) by Liouville's theorem. Moreover, g cannot have an essential singularity at 0. Suppose it did, let $z \in \mathbb{C} \setminus \{0\}$, and let $r < |z|$ and $r' < |z| - r$. By the Open Mapping Theorem, $g(B_r(z))$ contains a neighborhood of $g(z)$. But by the Casorati-Weierstrass theorem, $g(B_{r'}(0) \setminus \{0\})$ is dense and hence contains a point in said neighborhood of $g(z)$. This implies that g , and therefore f , is not injective. Thus, g must have a pole at 0, so

$$g(z) = \frac{a_m}{z^m} + \cdots + \frac{a_1}{z} + h(z)$$

where h is analytic at zero. But then

$$f(z) = a_m z^m + \cdots + a_1 z + h\left(\frac{1}{z}\right)$$

cannot be analytic unless h is constant. So f is a polynomial. The only polynomials without

distinct roots are powers, so $f(z) = a(z - z_0)^m$ for some $a, z_0 \in \mathbb{C}$ and $m \in \mathbb{N}$. If $m > 1$ then $f(z_0 + 1) = f(z_0 + e^{2\pi/m})$, so f is not injective. Hence $n = 1$ and $f = a(z - z_0)$. Note that we cannot have $a = 0$ since then f is constant and obviously not injective.

15. (a) By the Cauchy inequalities,

$$|f^{(n)}(0)| \leq \frac{n! (AR^k + B)}{R^n}$$

For $n > k$, taking the limit as $R \rightarrow \infty$ implies $f^{(n)}(0) = 0$. Since f is entire and all derivatives higher than k vanish, it is a polynomial of degree at most k .

(b) Without loss of generality, we assume $\theta = 0$

solution 1. We can use the same method as we do in the ex 15. of chapter 2. Let

$$F(z) = \begin{cases} f(z), & z \in \{z = re^{it} | r \in (0, 1), t \in (0, \varphi)\} \\ 0, & z \in \{z = re^{it} | r = 1, t \in (0, \varphi)\} \\ 1/\overline{f(1/\bar{z})}, & z \in \{z = re^{it} | r \in (1, +\infty), t \in (0, \varphi)\} \end{cases} \quad (12)$$

Then we can check it is holomorphic in the unbounded sector $\{z = re^{it} | r \in (0, +\infty), t \in (0, \varphi)\}$. However, we observe that F has non-isolated zero point. Hence $F = 0 \Rightarrow f = 0$. (Actually in this way the uniform convergence can be omitted. I see it in Math.stackexchange and I don't think it is correct but I think the idea is great.) $\square$

solution 2. Choose a proper n s.t. $n\varphi > 2\pi$. Consider $g = \prod_{k=0}^n f(ze^{ik\varphi})$. Let $|f| < M, |z| < 1$. Since f converges uniformly in the given sector, $\forall \epsilon, \exists r', |f| < \epsilon, \forall z \in \{z = re^{it} | r \in (r', 1), t \in (0, \varphi)\} \Rightarrow |g| < M^n \epsilon, \forall |z| > r'$. Let $\epsilon \rightarrow 0$, by maximal modulus principle we have $g = 0 \Rightarrow f = 0$. $\square$

(c) We prove the stronger proposition: there are n points $\{z_k\}_{k=1}^n$ in the unit closed disc, prove that there exists a point z on the unit circle, s.t. the product of their distance $\prod_{k=1}^n |z_k - z|$ is greater than 1.

solution 1. Define $f(z) = \prod_{k=1}^n (z - z_k)$. By Cauchy inequality, take $R = 1, |f^{(n)}(0)| \leq n! \max_{|z|=1} |f(z)|$.

Then $\max_{|z|=1} \frac{|f^{(n)}|}{n!} = 1$ $\square$

solution 2. Define $g(t) = \prod_{k=1}^n (1 - tz_k)$, then $|g(0)| = 1$. Consider the whole unit closed disc, by maximal modulus principle, there exists t_0 on the unit circle s.t. $|g(t_0)| \geq 1$. Then take $z = \frac{1}{t_0}$, $|g(t_0)| = \prod_{k=1}^n |1 - t_0 z_k| = \prod_{k=1}^n |z - z_k| \geq 1$. $\square$

solution 3. $g(z) = \prod_{k=1}^n (1 - zz_k)$ is a polynomial of degree n , take $z = \omega^k$ with $k = 0, 1, 2, \dots, n$ and ω be the $n+1$ -th primitive unit root. $\sum_{k=0}^n g(\omega^k) = n+1$, then there exists a k , $|g(\omega^k)| \geq 1$. $\square$

(d) In fact, it is sufficient for the real part of f to be bounded from one side. Suppose $\operatorname{Re}(f) \leq M$ everywhere. Let $g(z) = e^{f(z)}$. Then g is entire, and $|g| = e^{\operatorname{Re}(f)} < e^M$. By Liouville's theorem, $g(z) = c$ for some constant c . Then $f(z) = \log c$ everywhere. Although there are multiple branches of the log function, they differ by $2\pi i$ and f is continuous, so f must be constant.

16. (a) Since the closed unit disc is compact and $|g|$ is continuous, it attains a maximum M on it. Similarly, $|f|$ is continuous and nonzero on the unit circle, so it has a minimum value m . Then for $\epsilon < \frac{m}{M}$, $|f(z)| > |\epsilon g(z)|$ on the unit circle, so f and f_c have the same number of zeros inside the circle by Rouché's theorem. $\square$

(ii) Fix ϵ and z_ϵ . For $r > 0$, let N_r be the open neighborhood of z_ϵ of radius r . Then $|f_\epsilon| \geq \delta_r$ on $\bar{\mathbb{D}} \setminus N_r$ for some $\delta_r > 0$, because $\bar{\mathbb{D}} \setminus N_r$ is compact and $|f_\epsilon| > 0$ on $\bar{\mathbb{D}} \setminus \{z_\epsilon\}$. Then for $|\epsilon' - \epsilon| < \frac{\delta_r}{M}$, $|f_{\epsilon'}(z) - f_\epsilon(z)| = |\epsilon' - \epsilon||g(z)| < \delta_r$ so $f_{\epsilon'}(z) \neq 0$ for $\bar{\mathbb{D}} \setminus N_r$. Hence $z_{\epsilon'} \in N_r$ if $|\epsilon' - \epsilon| < \frac{\delta_r}{M}$. $\square$

17.(a) By Rouché's theorem, $f(z)$ and $f(z) - \omega_0$ have the same number of zeros inside the unit circle provided $|\omega_0| < 1$. Hence, if f has in zero, its image includes the unit disc. If f is nonzero, then $\frac{1}{f}$ is holomorphic, so $\left|\frac{1}{f(z)}\right| \leq 1$ for $z \in \bar{D}$ by the maximum modulus principle. But then $|f(z)| \geq 1$ for $z \in D$, which contradicts the maximal modulus principle. $\square$

(ii) Let $\omega_0 = f(z_0)$ where $|z_0| < 1$ and $|\omega_0| < 1$. By Rouché's theorem, $f(z)$ and $f(z) - \omega$ have the same number of zeros for all ω with $|\omega| < 1$. Since there exists a ω (namely ω_0) for which $f(z) - \omega$ has a zero, it has a zero for all $\omega \in \mathbb{D}$. So the image of f contains $\mathbb{D}$. $\square$

18. Trivial, just follow the hint and ex 6 of chapter 2.

19. (a) *solution 1.* Suppose u achieve its maximum at z_0 , $z_0 \in \Omega$, let f be a holomorphic function in the neighborhood of z_0 which satisfies $u = \text{Re}(f)$.

$$f(z_0) = u(z_0) + iv(z_0)$$

Because u is not constant all the time, f cannot be constant all the time. By Open Mapping Theorem, f will map from $U_\delta(z_0) \subset \Omega$ to an open set. Thus we can choose a δ' , s.t. $U_\delta(f(z_0)) \subset f(U_{\delta'}(z_1))$. Moreover, $\forall z \in U_\delta(z_0)$, $u(z) \leq u(z_0)$, thus $\forall \delta' > 0$, $u(z_0) + \frac{\delta'}{2} + iv(z_0) \in U_{g_r}(f(z_0))$ and $u(z_0) + \frac{\delta'}{2} + iv(z_0) \notin f(U_{g'}(z_0))$, which means that u cannot get its maximum in Ω , similarly, u cannot get its minimum in Ω

solution 2. We just use

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} u(z + re^{i\theta}) d\theta$$

and follow the proof of maximal modulus principle in 《复变函数论》钟玉泉.

(b) u is continuous in Ω . $|u|$ can get its maximum in $\bar{\Omega}$. From (a), we know that $|u|$ cannot get its maximum in Ω , or u will be a constant, which means $\sup_{z \in \Omega} |u(z)| \leq \sup_{z \in \bar{\Omega} - \Omega} |u(z)|$

20. (a) Define $d > 0$, and let $g(z)$ be a holomorphic function on some neighborhood of

$\overline{D_d(z_0)}$. The mean-value theorem shows

$$g(z) = \frac{1}{2\pi} \int_0^{2\pi} g(z + re^{i\theta}) d\theta$$

for any $0 < r < d$, so we take the modulus of both sides we have

$$\begin{aligned} |g(z)| &\leq \frac{1}{2\pi} \int_0^{2\pi} |g(z + re^{i\theta})| d\theta \\ &\Leftrightarrow r|g(z)| \leq \frac{r}{2\pi} \int_0^{2\pi} |g(z + re^{i\theta})| d\theta \\ &\Rightarrow \int_0^d r|g(z)| dr \leq \frac{1}{2\pi} \int_0^d \int_0^{2\pi} r |g(z + re^{i\theta})| d\theta dr \\ &\Leftrightarrow \frac{d^2}{2} |g(z)| \leq \frac{1}{2\pi} \iint_{|z-\omega| \leq d} |g(\omega)| dx dy \end{aligned}$$

Letting $g = f^2$, we have

$$|f(z)|^2 \leq \frac{1}{\pi d^2} \iint_{|z-\omega| \leq d} |f^2(\omega)| dx dy \Rightarrow |f(z)| \leq \frac{1}{\sqrt{\pi}d} \|f\|_{L^2(D_d(z))}$$

Let $d = r - s$, we have

$$|f(z)| \leq \frac{1}{\sqrt{\pi}(r-s)} \|f\|_{L^2(D_{(r-s)}(z))} \leq \frac{1}{\sqrt{\pi}(r-s)} \|f\|_{L^2(D_r(z_0))}$$

for all $z \in D_s(z_0)$. Hence

$$\|f\|_{L^\infty(D_s(z_0))} \leq \frac{1}{\sqrt{\pi}(r-s)} \|f\|_{L^2(D_r(z_0))}$$

□

(ii) Let $K \subset U$ be a compact set. Choose a positive d smaller than the distance from K to $\mathbb{C} - U$, so that $\overline{D_d(z)} \subset U$ for all $z \in K$. From the same proof from (1) we know that

$$\|f\|_{L^\infty(K)} \leq \frac{1}{\sqrt{\pi}d} \|f\|_{L^2(U)}$$

for every square-integrable holomorphic f . As a result, if $\{f_n\}_{n \in \mathbb{N}}$ is an $L^2(U)$ -Cauchy sequence of holomorphic functions, the restricted sequence $\{f_n\}_{n \in \mathbb{N}}$ is an $L^\infty(K)$ -Cauchy sequence. Hence $\{f_n\}_{n \in \mathbb{N}}$ converges uniformly. Note that the compact subsets of U cover U , $\{f_n\}_{n \in \mathbb{N}}$ converges pointwise on all of U , and by the above, the convergence is uniform on all compact subsets of U . By Weierstrass convergence theorem, the limit function is holomorphic on U . □

21. (a) Suppose otherwise that Ω is not simply connected, as a result of which there exists two curves $\gamma_1(s)$ and $\gamma_2(s)$ (with common end-points z_1 and z_2) not homotopic. So there is an exact $t_0 \in (1, 2)$ that $\gamma_{t_0}(s)$ is not contained in Ω . So there must be a $z_0 \in \Omega$ such that the

line segment from z_1 to z_0 is not in Ω , So we have achieved a contraction. So a convex set in $\mathbb{C}$ is simply connected.

(b) Let A be star-shaped with respect to z_0 and let $\gamma : [a, b] \rightarrow A$ be a closed curve in A , therefore $\gamma(a) = \gamma(b)$. WLOG, we assume that $a = 0$ and $b = 1$ (a reparametrization will do). Define a homotopy $H : [0, 1] \times [a, b] \rightarrow A$ by

$$H(s, t) = s\gamma(t) + (1 - s)z_0$$

Obviously it is a continuous function mapping A into A by the definition of starshaped given in the problem. This gives the homotopy between γ and z_0 . Hence A is simply connected. Take any $z_0 > 0$, we know $\mathbb{C} - (-\infty, 0]$ is simply connected.

22. Assume f exists. Since $g = zf(z) = 1$ and it is holomorphic in $\mathbb{D}$, by mean value theorem

$$0 = g(0) = \frac{1}{2\pi} \int_0^{2\pi} g(e^{i\theta}) d\theta = 1$$

Contradiction!

□

5.2 Problem

1. (a)(b) $f_\varepsilon = \varepsilon(e^{\frac{z}{\varepsilon}} - 1)$. □

$$\begin{aligned}
& (c) \overline{Aera(h(D_\rho(0) - \{0\}))} \\
&= \int_{C_\rho(0)} y dx = \int_{|z|=\rho} \Im(h(z)) d\Re(h(z)) \\
&= \int_0^{2\pi} \Im(h(\rho e^{\theta i})) d\Re(h(\rho e^{\theta i})) \\
&= \int_0^{2\pi} \frac{h(\rho e^{\theta i}) - h(\rho e^{-\theta i})}{2i} d \frac{h(\rho e^{\theta i}) + h(\rho e^{-\theta i})}{2} \\
&= \int_0^{2\pi} \left(\frac{\rho^{-1} e^{-\theta i} - \rho^{-1} e^{\theta i}}{2i} + \frac{c_1 \rho e^{\theta i} - \overline{c_1} \rho e^{-\theta i}}{2i} + \sum_{k=2}^{\infty} \frac{c_k \rho^k e^{k\theta i} - \overline{c_k} \rho^k e^{-k\theta i}}{2i} \right) \\
&\quad \left(\frac{\rho^{-1} e^{-\theta i} + \rho^{-1} e^{\theta i}}{2} + \frac{c_1 \rho e^{\theta i} + \overline{c_1} \rho e^{-\theta i}}{2} + \sum_{k=2}^{\infty} \frac{c_k \rho^k e^{k\theta i} + \overline{c_k} \rho^k e^{-k\theta i}}{2} \right)' d\theta \\
&= \int_0^{2\pi} \left(\frac{\rho^{-1} e^{-\theta i} - \rho^{-1} e^{\theta i}}{2i} + \frac{c_1 \rho e^{\theta i} - \overline{c_1} \rho e^{-\theta i}}{2i} + \sum_{k=2}^{\infty} \frac{c_k \rho^k e^{k\theta i} - \overline{c_k} \rho^k e^{-k\theta i}}{2i} \right) \\
&\quad \left(i \frac{-\rho^{-1} e^{-\theta i} + \rho^{-1} e^{\theta i}}{2} + i \frac{c_1 \rho e^{\theta i} - \overline{c_1} \rho e^{-\theta i}}{2} + \sum_{k=2}^{\infty} k i \frac{c_k \rho^k e^{k\theta i} - \overline{c_k} \rho^k e^{-k\theta i}}{2} \right) d\theta \\
&= 2\pi \left(\frac{\rho^{-2}}{4} + \frac{c_1}{4} + \frac{\rho^{-2}}{4} + \frac{\overline{c_1}}{4} - \frac{c_1}{4} - \frac{\rho^2 |c_1|^2}{2} - \frac{\overline{c_1}}{4} - \frac{\rho^2 |c_1|^2}{4} - \sum_{k=2}^{\infty} \frac{k |c_k|^2 \rho^{2k}}{4} + \frac{k |c_k|^2 \rho^{2k}}{4} \right) \\
&= \pi \left(\rho^{-2} - \sum_{k=1}^{\infty} k |c_k| \rho^{2k} \right) > 0
\end{aligned}$$

i.e. $\rho^{-2} - \sum_{k=1}^{\infty} k |c_k| \rho^{2k} > 0$, then let $\rho \rightarrow 1$ we get

$$\sum_{k=1}^{\infty} k |c_k|^2 \leq 1$$

□

(d) $f(z) = z + a_2 z^2 + \dots = 0$ iff $z = 0$, then $\frac{f(z)}{z}$ vanishes nowhere and has a "square root" $\psi^2(z) = \frac{f(z)}{z}$ and $\psi(0) = 1$. Let $g(z) = z\psi(z^2)$ and it is trivially verified that $g(0) = 0$, $g'(0) = 1$ and $g^2(z) = z\psi(z^2)$. If $g(z_1) = g(z_2)$ then $f(z_1^2) = g^2(z_1) = g^2(z_2) = f(z_2^2)$. We have $z_1 = \pm z_2$. Suppose $z_1 = -z_2$, $z_1\psi(z_1^2) = g(z_1) = g(z_2) = z_2\psi(z_2^2)$, contradiction, so g is injective. □

(e)

$$f(z) = z + a_2 z^2 + a_3 z^3 + \dots$$

$$g(z) = z + b_2 z^2 + b_3 z^3 + \dots$$

$$(z + b_2 z^2 + b_3 z^3 + \dots)^2 = z^2 + a_2 z^4 + a_3 z^6 + \dots$$

since $g^2(z) = f(z^2)$. Then we have $2b_2 = 0$ and $2b_3 + b_2^2 = a_2$ i.e. $g(z) = z(1 + b_3z^2 + \dots)$. Remember $1 + b_3z^2 + \dots$ vanishes nowhere then $\frac{1}{1+b_3z^2+\dots}$ is holomorphic.

$$\frac{1}{g(z)} = \frac{1}{z} \frac{1}{1 + b_3z^2 + \dots} = \frac{1}{z}(1 - b_3z^2 + \dots)$$

Then $|b_3| \leq 1$ i.e. $|a_2| \leq 2$. The equality holds when $|b_3| = 1$ and all following coefficients vanish, then $\frac{1}{g(z)} = \frac{1}{z}(1 - b_3z^2)$. i.e. $f(z) = \frac{z}{(1 - e^{i\theta}z)^2}$ $\square$

(We omit (f) and prove (g) directly)

(g) Take any $w \notin f(D)$, we shall prove $|w| \geq \frac{1}{4}$. Let $F(z) = \frac{1}{\frac{1}{f(z)} - w} = \frac{wf(z)}{w - f(z)}$

$$F'(z) = \frac{w^2 f'(z)}{(w - f(z))^2}$$

$$F''(z) = \frac{w^2 f''(z)(w - f(z))^2 - w^2 f'(z) \cdot 2(w - f(z))(-f'(z))}{(w - f(z))^4}$$

Then we have $F(0) = 0, F'(0) = 1, F$ is injective (F satisfies the hypotheses of the theorem). And $\frac{F''(0)}{2} = \frac{f''(0)}{2} + \frac{1}{w}$, then $|\frac{1}{w}| \leq |\frac{F''(0)}{2}| + |\frac{f''(0)}{2}| \leq 4$ i.e. $|w| \geq \frac{1}{4}$. $\square$

2. By ex 12 we know that there exists a holomorphic function f in the unit disc such that $f = u + iv$. Let $T(z) = \frac{z_0 - z}{1 - \overline{z_0}z}$, by ex 7 of chapter 1 we know $f(T(z))$ is holomorphic $\Rightarrow u(T(z))$ is harmonic. Hence by mean value theorem

$$u(z_0) = u(T(0)) = \frac{1}{2\pi} \int_0^{2\pi} u(T(e^{i\theta})) d\theta$$

by ex 7 of chapter 1 we know $T \circ T = 1$. Let $e^{it} = T(e^{i\theta}) \Rightarrow \theta = -i \log T(e^{it}) \Rightarrow |\frac{d\theta}{dt}| = \frac{1 - |z|^2}{|z_0 - e^{it}|^2}$ (just calculate it directly). Hence

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(T(e^{i\theta})) d\theta = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - |z|^2}{|z_0 - e^{i\theta}|^2} u(T(e^{i\theta})) d\theta$$

$\square$

3. Given $z \in D$, choose r_1, r_2 s.t.

$$r < r_1 < |z - z_0| < r_2 < R.$$

Let $\gamma_j = \{\zeta : |\zeta - z_0| = r_j\}, j = 1, 2$. We have

$$f(z) = \frac{1}{2\pi i} \int_{\gamma_2} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \int_{\gamma_1} \frac{f(\zeta)}{\zeta - z} d\zeta$$

We denote $M_j = \sup \{|f(\zeta)| : \zeta \in \gamma_j\}$, $j = 1, 2$. $\left| \frac{\zeta - z_0}{z - z_0} \right| = \frac{r_1}{|z - z_0|} < 1$ when $\zeta \in \gamma_1$, so we have

$$\begin{aligned} \frac{1}{\zeta - z} &= -\frac{1}{z - z_0} \left(1 - \frac{\zeta - z_0}{z - z_0} \right)^{-1} \\ &= -\sum_{n=0}^{\infty} \frac{(\zeta - z_0)^n}{(z - z_0)^{n+1}} = -\sum_{n=1}^{\infty} \frac{(\zeta - z_0)^{n-1}}{(z - z_0)^n} \end{aligned}$$

Hence

$$\frac{f(\zeta)}{\zeta - z} = -\sum_{n=1}^{\infty} f(\zeta) \frac{(\zeta - z_0)^{n-1}}{(z - z_0)^n}, \zeta \in \gamma_1$$

Since

$$\left| \frac{f(\zeta) (\zeta - z_0)^{n-1}}{(z - z_0)^n} \right| \leq \frac{M_1}{|z - z_0|} \left(\frac{r_1}{|z - z_0|} \right)^{n-1},$$

and the right series converges uniformly, so the former series converges uniformly in γ_1 . Hence we can do integral term by term:

$$\frac{1}{2\pi i} \int_{\gamma_1} \frac{f(\zeta)}{\zeta - z} d\zeta = -\sum_{n=1}^{\infty} \left(\frac{1}{2\pi i} \int_{\gamma_1} \frac{f(\zeta)}{(\zeta - z_0)^{-n+1}} d\zeta \right) (z - z_0)^{-n}$$

If $\zeta \in \gamma_2$, $\left| \frac{z - z_0}{\zeta - z_0} \right| = \frac{|z - z_0|}{r_2} < 1$, we use the same way and get

$$\frac{1}{2\pi i} \int_{\gamma_2} \frac{f(\zeta)}{\zeta - z} d\zeta = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_{\gamma_2} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta \right) (z - z_0)^n$$

By Cauchy theorem, we have

$$\frac{1}{2\pi i} \int_{\gamma_1} \frac{f(\zeta)}{(\zeta - z_0)^{-n+1}} d\zeta = \frac{1}{2\pi i} \int_{\gamma_0} \frac{f(\zeta)}{(\zeta - z_0)^{-n+1}} d\zeta = a_{-n}$$

$$\frac{1}{2\pi i} \int_{\gamma_2} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta = \frac{1}{2\pi i} \int_{\gamma_0} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta = a_n$$

So we have

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n,$$

Then we prove it is there is no other expansion. Assume it exists,

$$f(z) = \sum_{n=-\infty}^{\infty} a'_n (z - z_0)^n,$$

Since

$$\frac{1}{2\pi i} \int_{\gamma_\rho} \frac{f(\zeta)}{(\zeta - z_0)^{m+1}} d\zeta = \sum_{n=-\infty}^{\infty} a'_n \frac{1}{2\pi i} \int_{\gamma_\rho} (\zeta - z_0)^{n-m-1} d\zeta = a_{m'} = a_m,$$

we draw the conclusion. $\square$

4. It is a useful conclusion in topology, here we prove the stronger theorem:

Theorem. Let X be a space which can be written as the union of two simply connected open sets U, V in such a way that U, V is path-connected. Then X is simply connected.

proof. We show that any loop in X is homotopic to a product of loops each of which is contained in either U or V . This is enough to prove the theorem since U and V are both simply connected. Choose a base point $p \in U \cap V$, and let $\alpha : I \rightarrow X$ be a loop based at p . Using Lebesgue's lemma, we can find points $0 = t_0 < t_1 < t_2 < \dots < t_n = 1$ in I such that $\alpha([t_{k-1}, t_k])$ is always contained in U or V . Write α_k for the path $s \mapsto \alpha((t_k - t_{k-1})s + t_{k-1})$, $0 \leq s \leq 1$. Join p to each point $\alpha(t_k)$, $1 \leq k \leq n-1$, by a path γ_k which lies in U if $\alpha(t_k) \in U$, and which lies in V if $\alpha(t_k) \in V$. If $\alpha(t_k) \in U \cap V$ we need to find γ_k in $U \cap V$. This is not a problem since we have assumed $U \cap V$ to be path-connected. Our loop α is homotopic to the product

$$(\alpha_1 \cdot \gamma_1^{-1}) \cdot (\gamma_1 \cdot \alpha_2 \cdot \gamma_2^{-1}) \cdot (\gamma_2 \cdot \alpha_3 \cdot \gamma_3^{-1}) \dots (\gamma_{n-1} \cdot \alpha_n)$$

Where

$$a \cdot b = \begin{cases} a(2t), 0 \leq t \leq \frac{1}{2} \\ b(2t-1), \frac{1}{2} \leq t \leq 1 \end{cases}$$

$\square$

5. (a) Let $z = a + bi \in \mathbb{C} - [-M, M]$. Then we have

$$g(z) = \frac{1}{2\pi i} \int_{-M}^M \frac{h(x)}{x - a - bi} dx = \frac{1}{2\pi} \int_{-M}^M \frac{h(x)b}{(x-a)^2 + b^2} dx - \frac{i}{2\pi} \int_{-M}^M \frac{h(x)(x-a)}{(x-a)^2 + b^2} dx = u(a, b) + iv(a, b)$$

Then we can check that u, v satisfy the C.-R. equation when $z \in \mathbb{C} - [-M, M]$. So we know g is holomorphic. Since h is continuous in the closed interval, we assume $|h| < L$. Then we have

$$|g(z)| \leq \frac{ML}{\pi} \frac{1}{|z| - M} \rightarrow 0, |z| \rightarrow \infty$$

Notice that

$$g(t + i\epsilon) - g(t - i\epsilon) = \frac{\epsilon}{\pi} \int_{-M}^M \frac{h(x)}{(x-t)^2 + \epsilon^2} dx, t \in [-M, M]$$

and

$$\frac{\epsilon}{\pi} \int_{-M}^M \frac{1}{(x-t)^2 + \epsilon^2} dx = \frac{1}{\pi} \int_{-M}^M \frac{1}{\left(\frac{x-t}{\epsilon}\right)^2 + 1} d\left(\frac{x-t}{\epsilon}\right) \rightarrow 1, \epsilon \rightarrow 0$$

So

$$\left| \lim_{\epsilon \rightarrow 0+} g(t+i\epsilon) - g(t-i\epsilon) - h(t) \right| = \lim_{\epsilon \rightarrow 0+} \left| \frac{\epsilon}{\pi} \int_{-M}^M \frac{h(x) - h(t)}{(x-t)^2 + \epsilon^2} dx \right|$$

Since h is uniformly continuous in $[-M, M]$, $\forall \epsilon' > 0$, we can choose δ' s.t. $\sup_{|x-t| < \delta'} |h(x) - h(t)| < \epsilon'$. Let $\delta = \min\{\delta', \epsilon'\}$. Then we have

$$\lim_{\epsilon \rightarrow 0+} \left| \frac{\epsilon}{\pi} \int_{-M}^M \frac{h(x) - h(t)}{(x-t)^2 + \epsilon^2} dx \right| < \left| \int_{-M}^{t-\delta} \right| + \left| \int_{t-\delta}^{t+\delta} \right| + \left| \int_{t+\delta}^M \right| < \frac{2\delta}{\pi} + \frac{8ML}{\pi} \frac{\epsilon}{\delta^2 + \epsilon^2}$$

Since $\frac{8ML}{\pi} \frac{\epsilon}{\delta^2 + \epsilon^2} \rightarrow 0, \epsilon \rightarrow 0$, we can choose $\epsilon'', \forall \epsilon < \epsilon'', \frac{8ML}{\pi} \frac{\epsilon}{\delta^2 + \epsilon^2} < \epsilon'$. Hence $\forall \epsilon < \epsilon''$,

$$|g(x+i\epsilon) - g(x-i\epsilon) - h(t)| < \frac{2\delta}{\pi} + \epsilon' < \left(\frac{2}{\pi} + 1\right)\epsilon'$$

□

(b) Assume G is another function, let $h = g - G$. Since

$$|h(x+i\epsilon) - h(x-i\epsilon)| < |[g(x+i\epsilon) - g(x-i\epsilon)] - [G(x+i\epsilon) - G(x-i\epsilon)]| \rightarrow 0, \epsilon \rightarrow 0$$

We just extend h s.t. $h = 0, z \in [-M, M]$. Then we use the same method as we do in the symmetry principle. Let T be triangle across the closed interval. Since the convergence is uniform, we can and then by Morera' theorem, we know h is entire. By (i) we know $h = \text{constant}$. □

6 Chapter 4

6.1 Exercise

1. (a)

$$\begin{aligned}
 & A(\xi) - B(\xi) \\
 &= \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi(x-t)} dx \\
 &= \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi x} dx \\
 &= \hat{f}(\xi) = 0
 \end{aligned}$$

(b) By symmetry principle

$$F(z) = \begin{cases} A(z) & \text{when } \Im(z) \geq 0 \\ B(z) & \text{when } \Im(z) < 0 \end{cases}$$

is holomorphic.

$$A(z) \leq \int_{-\infty}^t |f(x)| |e^{-2\pi i z(x-t)}| dt \leq \int_{-\infty}^t \frac{A}{1+x^2} e^{2\pi \Im(z)(x-t)} dt$$

is bounded when $\Im(z) \geq 0$, so is $B(z)$.

Then $F(z)$ is bounded, constant by Liouville's theorem, Let $z = is$ and $s \rightarrow \infty$ we shall see $F(z) \rightarrow 0$ then $F(z)$ is identically 0.

(c) Take $z = 0$.

2. $|f^{(n)}(z)|$ is clearly bounded near the origin, for the large $|z|$.

$$\begin{aligned}
 |f^{(n)}(z)| &= \frac{n!}{2\pi} \left| \int_C \frac{f(\zeta)}{(z-\zeta)^{n+1}} d\zeta \right| \\
 &= \frac{n!}{2\pi} \int_C \frac{A}{r^{n+1}(1+\Re(\zeta)^2)} d\zeta \\
 &= \frac{n!}{r^{n+1}} \frac{A}{(1+(x-r)^2)} \\
 &= \frac{n!}{r^{n+1}} \frac{A'}{(1+x^2)}
 \end{aligned}$$

where C is a circle centered at z and of radius $r < a - b$.

3. The case $\xi = 0$ is trivial. Let $f(z) = \frac{a}{a^2+z^2} e^{-2\pi i z \xi}$ for $\xi < 0$, use upper semicircle contour.

$$\text{Res}(f, ia) = \frac{1}{2i} e^{2\pi a \xi} = \frac{1}{2\pi i} \int_{-\infty}^{\infty} f(z) dz$$

Then $\frac{1}{\pi} \int_{-\infty}^{\infty} f(z) dz = e^{-2\pi a|\xi|}$. For $\xi < 0$, use lower semicircle contour instead.

By Fourier inversion formula, $f \in \mathcal{F}$, so $\int_{-\infty}^{\infty} e^{-2\pi a|\xi|} e^{2\pi i \xi x} d\xi = \frac{1}{\pi} \frac{a}{a^2 + x^2}$.

4. If $\xi = 0$

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{Q(x)} dx = \int_{-\infty}^{\infty} \frac{1}{Q(x)} dx$$

If $\xi > 0$, use the upper semicircle, let z are roots of $Q(z)$:

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{Q(x)} dx = 2\pi i \sum_{\Im(z) > 0} \text{Res}(z)$$

If $\xi < 0$, use the lower semicircle, let z are roots of $Q(z)$:

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{Q(x)} dx = 2\pi i \sum_{\Im(z) < 0} \text{Res}(z)$$

5. (a)

$$\int_{-\infty}^{\infty} R(x) e^{-2\pi i x \xi} dx = 2\pi i \sum_{i=1}^k \text{Res}(\alpha_i)$$

$$\begin{aligned} \text{Res}(\alpha_i) &= \lim_{z \rightarrow \alpha_i} \frac{1}{(n-1)!} \left(\frac{d}{dz} \right)^{n-1} (z - \alpha_i)^n R(z) e^{-2\pi i z \xi} \\ &= \frac{1}{(n-1)!} \lim_{z \rightarrow \alpha_i} \sum_{k=0}^{n-1} \binom{n-1}{k} \left(\frac{d}{dz} \right)^{n-1-k} (z - \alpha_i)^n R(z) \cdot \left(\frac{d}{dz} \right)^k e^{-2\pi i z \xi} \\ &= \frac{1}{(n-1)!} \left[\sum_{k=0}^{n-1} \binom{n-1}{k} \lim_{z \rightarrow \alpha_i} \left(\frac{d}{dz} \right)^{n-1-k} (z - \alpha_i)^n R(z) (-2\pi i \xi)^k \right] e^{-2\pi i \alpha_i \xi} \end{aligned}$$

Then $P_i(\xi) = \frac{2\pi i}{(n-1)!} \sum_{k=0}^{n-1} \binom{n-1}{k} \lim_{z \rightarrow \alpha_i} \left(\frac{d}{dz} \right)^{n-1-k} (z - \alpha_i)^n R(z) (-2\pi i \xi)^k$.

(b) trivial.

(c) Use roots in the lower half plane.

(d)

$$|e^{-2\pi i \alpha_j \xi}| = e^{-2\pi |\Im(\alpha_j)| |\xi|}$$

so $a < \min 2\pi |\Im(\alpha_j)|$.

6. Use Poisson's summation formula on $f(z) = \frac{1}{\pi} \frac{a}{a^2 + z^2}$, we have $\hat{f}(\xi) = e^{-2\pi a|\xi|}$.

$$\sum_n f(n) = \sum_n \hat{f}(n)$$

$$\frac{1}{\pi} \sum_n \frac{a}{a^2 + n^2} = \sum_n e^{-2\pi a|n|}$$

To prove the sum equals $\coth a$, use the method of section 3.2 of chapter 5 i.e. their difference is holomorphic and bounded.

7. (a)

$$\hat{f}(\xi) = \begin{cases} \frac{(-2\pi i)^k}{(k-1)!} \xi^{k-1} e^{2\pi i \xi \tau} & \text{when } \xi > 0 \\ 0 & \text{when } \xi < 0 \end{cases}$$

then by the Poisson's formula we get

$$\sum_n \frac{1}{(\tau + n)^k} = \frac{(2\pi i)^k}{(k-1)!} \sum_{m=1}^{\infty} m^{k-1} e^{2\pi i m \tau}$$

(b)

$$\sum_n \frac{1}{(\tau + n)^2} = -4\pi^2 \sum_{m=1}^{\infty} m e^{2\pi i m \tau} = \frac{\pi^2}{\sin^2(\pi \tau)}$$

(c) For $\Im(\tau) > 0$, see (b); For $\Im(\tau) < 0$, similar argument holds; For $\Im(\tau) = 0$, use limit argument.

Remark:

$$\pi \cot \pi z = \sum_{n=-\infty}^{\infty} \frac{1}{z + n}$$

8.

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi$$

By Paley-Wiener theorem:

$$f(z) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i z \xi} d\xi$$

$$\begin{aligned} a_n &= \frac{f^{(n)}(0)}{n!} \\ &= \frac{1}{n!} \int_{-\infty}^{\infty} \hat{f}(\xi) \left(\frac{d}{dz} \right)^n e^{2\pi i z \xi} \Big|_{z=0} d\xi \\ &= \frac{(2\pi i)^n}{n!} \int_{-\infty}^{\infty} \hat{f}(\xi) \xi^n d\xi \\ &= \frac{(2\pi i)^n}{n!} \int_{-M}^M \hat{f}(\xi) \xi^n d\xi \end{aligned}$$

for the second statement

$$\sqrt[n]{n! |a_n|} = 2\pi \sqrt[n]{\left| \int_{-M}^M \hat{f}(\xi) \xi^n d\xi \right|} \leq 2\pi \sqrt[n]{\int_{-M}^M A \xi^n d\xi} \rightarrow 2\pi M$$

9. (b) Define

$$F_{\epsilon}(z) = F(z) e^{-\epsilon z^r}$$

where $\alpha < r < \beta$ and r is rational.

$$|e^{-\epsilon z^r}| = e^{-\epsilon \Re(z^r)} = e^{-\epsilon |z|^r \cos r\theta}$$

$$|F_\epsilon(z)| \leq C e^{c|z|^\alpha - \epsilon|z|^\gamma \cos r\theta}$$

Choose the sector

$$S = \{z : -\frac{\pi}{2\beta} < \arg z < \frac{\pi}{2\beta}\}$$

then $r\theta \in (-\pi, \pi)$, then $\cos r\theta > 0$, so $|F_\epsilon(z)| \rightarrow 0$ as $|z| \rightarrow \infty$. Let

$$M = \sup_{z \in \bar{S}} |F_\epsilon(z)|$$

and

$$\omega_j : |F(\omega_j)| \rightarrow M$$

$\{\omega_j\}$ cannot escape to the infinity, so $\omega_j \rightarrow \omega$. By the maximum module principle, ω is not a interior point. So ω is at the boundary. Then $|F_\epsilon(z)| \leq 1$. Let $\epsilon \rightarrow 0$, $|F(z)| \leq 1$.

10.

$$\begin{aligned} |\hat{f}(\xi + i\eta)| &= \left| \int_{-\infty}^{\infty} f(x) e^{-2\pi i x(\xi + i\eta)} dx \right| \\ &= \left| \int_{-\infty}^{\infty} f(x - iy) e^{-2\pi i (x - iy)(\xi + i\eta)} dx \right| \\ &\leq \int_{-\infty}^{\infty} c e^{-ax^2 + by^2} e^{2\pi(x\eta - y\eta)} dx \\ &= c e^{by^2 - 2\pi y\xi} \int_{-\infty}^{\infty} e^{-ax^2 + 2\pi xy} dx \\ &= c e^{by^2 - 2\pi y\xi + \frac{\pi^2 \eta^2}{a^2}} \int_{-\infty}^{\infty} e^{-a(x - \frac{\pi\eta}{a})^2} dx \end{aligned}$$

Let $y = \epsilon\xi$, $by^2 - 2\pi y\xi = (b\epsilon^2 - 2\pi\epsilon)\xi^2 = -a'\xi^2$, $\frac{\pi\eta^2}{a} = b'\eta^2$ and

$$c \int_{-\infty}^{\infty} e^{-a(x - \frac{\pi\eta}{a})^2} dx = c'$$

Then

$$|\hat{f}(\xi + i\eta)| \leq c' e^{-a'\xi^2 + b'\eta^2}$$

6.2 Problem

1. Lemma without proof: $u^q + \xi^p \geq u\xi$ when $\xi, u \geq 0$ and $\frac{1}{p} + \frac{1}{q} = 1$

Let $f_n = \int_{-n}^n \hat{f}(\xi) e^{2\pi i \xi z} d\xi$, $f = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i |\xi|z} d\xi$. Then $f_n \rightarrow f$ uniformly, since:

$$\begin{aligned} |f - f_n| &= \left| \int_{|\xi| \leq n} \hat{f}(\xi) e^{2\pi i \xi z} d\xi \right| \\ &\leq A \int_{|\xi| \leq n} e^{-a|\xi|^p} e^{-2\pi \xi y} dy \rightarrow 0 \end{aligned}$$

Then f is holomorphic.

$$\begin{aligned}
|f(z)| &= \left| \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i \xi z} d\xi \right| \\
&\leq A \left(\int_{|\xi| \leq n} + \int_{|\xi| \geq n} \right) e^{-a|\xi|^p} e^{-2\pi \xi y} d\xi \\
&\leq A M e^{a'|y|^q} + C \leq A' e^{a|z|^q}
\end{aligned}$$

Remark: Ref: Thm 3.1

2. (a) Suppose f is supported on $[-M, M]$, and $|f''(t)| \leq A$.

$$\begin{aligned}
|\hat{f}(z)| &= \left| \frac{\int_{-M}^M f(t) d e^{-2\pi i z t}}{-2\pi i z} \right| \\
&= \left| \frac{\int_{-M}^M e^{-2\pi i z t} f'(t) dt}{2\pi z} \right| \\
&= \left| \frac{\int_{-M}^M e^{-2\pi i z t} f''(t) dt}{4\pi^2 z^2} \right| \\
&\leq \frac{\int_{-M}^M |e^{-2\pi i z t}| |f''(t)| dt}{4\pi^2 |z|^2} \\
&\leq A \frac{\int_{-M}^M |e^{2\pi y t}| dt}{|z|^2} \\
&\leq \frac{A'}{|z|^2} \leq \frac{A''}{1+x^2}
\end{aligned}$$

since $\int_{-M}^M |e^{2\pi y t}| dt$ is bounded when $|y| \leq a$.

(b) $P(z)$ as a polynomial, only has finitely number of zeros. Obviously there is a line missing these zeros.

(c)

$$\left(\frac{d}{dt} \right)^k u(t) = \int_L \frac{(2\pi i z)^k}{P(z)} e^{2\pi i z t} \hat{f}(z) dz$$

Sum up by k and we get the first desired conclusion.

As for the second, integral $e^{2\pi i z t} \hat{f}(z)$ through the toy contour, the rectangle with vertices $N, -N, -N + ic, N + ic$ and let $N \rightarrow \infty$, where the integral on the vertical sides vanishes.

3. (a) Let

$$\begin{aligned}
F_\epsilon(z) &= F(z) e^{-\epsilon z^2} \\
|F_\epsilon(z)| &\leq M e^{-\epsilon \Re(x^2 - y^2)} \rightarrow 0
\end{aligned}$$

as $x \rightarrow \infty$. Then use the statement of the proof Phragmen Lindelof theorem (or Exercise 9).

(b) Consider

$$G(z) = M_0^{z-1} M_1^{-z} F(z)$$

Then $G(z)$ satisfies the condition in (a)

(c) Let $M_y = \sup_x |F(x + iy)|$, then: ($y_2 > y_1$)

$$\log \sup_x |F(x + iy)| \leq \log M_{y_1}^{\frac{y_2 - y}{y_2 - y_1}} M_{y_2}^{\frac{y - y_1}{y_2 - y_1}} = \frac{y_2 - y}{y_2 - y_1} \log M_{y_1} + \frac{y - y_1}{y_2 - y_1} \log M_{y_2}$$

4. (a) Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$, $g(w) = \sum_{n=0}^{\infty} A_n w^{-n-1}$.

$$\begin{aligned} f(z) &= \int_C e^{2\pi i z w} g(w) dw \\ &= \sum_{n=0}^{\infty} A_n \int_C \frac{e^{2\pi i z w}}{w^{n+1}} dw \\ &= \sum_{n=0}^{\infty} A_n \frac{(2\pi i z)^n}{n!} \\ &= \sum_{n=0}^{\infty} a_n z^n \end{aligned}$$

So take $A_n = \frac{n!}{(2\pi i)^{n+1}} a_n$.

If $f(z) = \int_C e^{2\pi i z w} g(w) dw$ with given w , then

$$|f(z)| \leq 2\pi(M + \epsilon) \left(\sup_{xi \in C_{M+\epsilon}} |g(\xi)| \right) e^{2\pi(M+\epsilon)|z|}$$

(b) We know $\hat{f}(\xi)$ is supported on $[-M, M]$ by Paley-Wiener theorem.

$$\begin{aligned} g(z) &= \frac{1}{2\pi i} \int_{-M}^M \frac{\hat{f}(\xi)}{\xi - z} d\xi \\ &= \frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{\hat{f}(\xi)}{\xi - z} d\xi \\ &= \int_{-\infty}^{\infty} \hat{f}(\xi) K(\xi - z) d\xi \\ &= \int_{-\infty}^{\infty} f(x) \hat{K}(x) e^{-2\pi i \xi z} dx \\ &= \int_{-\infty}^{\infty} f(x) \chi_{[0, \infty)}(x) e^{-2\pi i \xi z} dx \\ &= \int_0^{\infty} f(x) e^{-2\pi i \xi z} dx \end{aligned}$$

and obviously

$$\int_0^{\infty} x^n e^{-2\pi i x z} dx = \frac{n!}{(2\pi i z)^{n+1}}$$

Remark:

$$K(x) = -\frac{1}{2\pi i x}$$

$$\hat{K}(\xi) = \chi_{[0,\infty)}$$

7 Chapter 5

7.1 Exercise

1. We follow the proof of Jensen's formula that is given in the book. We keep step 1 exactly the same. Following step 2, we have set $g = f/\psi_1 \cdot \psi_N$, then g is holomorphic and bounded near each z_j . So it suffices to prove the theorem for Blaschke factors and for bounded functions that vanish nowhere. Functions that vanish nowhere are treated in Step 3. It remains to show the result for Blaschke factors. We have

$$\log |\psi_\alpha(0)| = \log |\alpha| = \log |\alpha| + \frac{1}{2\pi} \int_0^{2\pi} \log |\psi_\alpha(e^{i\theta})| d\theta$$

since $|\psi_\alpha(z)| = 1$ for $z \in \partial\mathbb{D}$.

2. (a) Noting that $|z^n| = O(e^{z^\epsilon})$ for all $\epsilon > 0$ we see that the order of growth of p is 0. (b) Clearly the order of growth is n . (c) The order of growth is e^{e^z} is infinite. Since $|z|^k = O(e^z)$ we know that there is no k and constants A and B , such that $e^{e^{|z|}} \leq Ae^{Bz^k}$.

3. For $t = \Im(\tau) > 0$, $-n^2 + 2n|z| \leq -\frac{n^2 t}{2}$ when $n \geq \frac{4|z|}{t}$.

$$\begin{aligned} |\Theta(z|\tau)| &\leq \sum_{n=-\infty}^{\infty} \left| e^{\pi i n^2 \tau} e^{2\pi i n z} \right| \\ &\leq \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t + 2\pi n |z|} \\ &= \sum_{n < \frac{4|z|}{t}} e^{-\pi n^2 t + 2\pi n |z|} + \sum_{n \geq \frac{4|z|}{t}} e^{-\pi n^2 t + 2\pi n |z|} \\ &= C e^{\frac{8\pi |z|^2}{t}} + \sum_{n \geq \frac{4|z|}{t}} e^{-\frac{n^2 t}{2}} \\ &= C e^{\frac{8\pi |z|^2}{t}} + C' \end{aligned}$$

if of order 2.

4. For (a) see the hint. For (b): let $r = \min 1, t$ and $R = \max 1, t$. So there are eight (m, n) such that $r \leq |nit + m| \leq R$, namely $(0, \pm 1), (\pm 1, 0), (\pm 1, \pm 1)$. There are 16 (m, n) such that $2r \leq |nit + m| \leq 2R$. In general there are $8k$ tuples with $kr \leq |nit + m| \leq kR$. Thus with

$$S(k) := \sum_{(m,n) \text{ with } \max(m,n) \leq k} \frac{1}{|nit + m|^\alpha},$$

we have

$$\frac{8}{R^\alpha} \sum_{j=1}^k \frac{1}{j^{\alpha-1}} \leq S(k) \leq \frac{8}{r^{\alpha-1}} \sum_{j=1}^k \frac{1}{j^{\alpha-1}}.$$

5. To show entire let $C_R(z_0)$ be a circle of radius R centered at z_0 . Then we have

$$\begin{aligned}\int_{C_1(z_0)} F_\alpha(z) dz &= \int_0^{2\pi} \int_{-\infty}^{\infty} e^{-|t|^\alpha + 2\pi i z_0 t + e^{i\theta} 2\pi i t} dt i e^{i\theta} d\theta \\ &= \int_{-\infty}^{\infty} e^{-|t|^\alpha + 2\pi i z_0 t} \int_0^{2\pi} e^{2\pi i e^{i\theta} t} i e^{i\theta} d\theta dt \\ &= 0.\end{aligned}$$

We can switch the order of integration by Fubini's theorem since the integrand is L^1 . The final equality is established since $\int_{C(z_0)} e^{2\pi i z t} dz = 0$. This shows that the function is holomorphic in any disc by Morera's theorem. Thus it is entire.

6. By the product formula for sine we have

$$\sin(\pi/2) = \frac{\pi}{2} \prod_{n=1}^{\infty} \left(1 - \frac{1}{4n^2}\right) = \frac{\pi}{2} \prod_{n=1}^{\infty} \left(\frac{(2n+1)(2n-1)}{(2n)(2n)}\right).$$

7. 8. 9. Trivial, which can be easily found in textbooks on mathematical analysis.

10. (a) $e^z - 1$ is order 1 and has zeros precisely when $z = 2\pi i n$ for $n \in \mathbb{Z}$. Thus we have

$$e^z - 1 = e^{Az+B} z \prod_{n \neq 0} \left(1 - \frac{z}{2\pi i n}\right) e^{z/2\pi i n} = e^{Az+B} z \prod_{n \geq 1} \left(1 + \frac{z^2}{4\pi^2 n^2}\right).$$

Multiplying this equation by $e^{-z/2}$ we are left with two expressions both of which must be odd. Thus we see that we must have $A = 1/2$. Also considering $\lim_{z \rightarrow 0} (e^z - 1)/z = 1$ we see that we must have $B = 0$. (b) $\cos(\pi z)$ is also order 1 and has zeros at $\frac{2n+1}{2}$ for all $n \in \mathbb{Z}$. Thus we are left with

$$\cos(\pi z) = e^{Az+B} \prod_{n \in \mathbb{Z}} \left(1 - \frac{2z}{(2n+1)}\right) e^{2z/(2n+1)} = e^{Az+B} \prod_{n \geq 1} \left(1 - \frac{4z^2}{(2n+1)^2}\right)$$

Since cosine is even we see that $A = 0$. Also letting $z = 0$ we see that we must have $B = 0$.

11. By Hadamard's theorem if it omits a and b , then we have $f(z) - a = e^{p(z)}$ and $f(z) - b = e^{q(z)}$ for polynomials p and q . Then $e^{p(z)} - e^{q(z)} = C$ for some constant C . Letting z tend to infinity we see that the leading terms of the polynomials p and q must be the same. Say it is $a_n z^n$. Then, considering the limit as z tends to infinity of

$$e^{p(z) - a_n z^n} - e^{q(z) - a_n z^n} = C e^{-a_n z^n},$$

we see that the next leading terms are also equal. Proceeding by induction shows that $p = q$, but then this is a contradiction since it would imply that $b = a$. But we assume they are distinct.

12. If f has finite order and never vanishes then we have $f(z) = C e^{p(z)}$ for some polynomial, this follows from Hadamard's theorem. So then $f'(z) = C p'(z) e^{p(z)}$. Since we

assume that the derivatives are never 0, we see that p' is a constant function.

13. $e^z - z$ is entire of order 1. So $e^z - z = e^{Az+B} \prod_n \left(1 - \frac{z}{a_n}\right) e^{z/a_n}$. If there are only finitely many zeros then we have $e^z - z = e^{\tilde{A}z+\tilde{B}}Q(z)$ for some polynomial Q . But then

$$Q(z) = \frac{e^z - z}{e^{\tilde{A}z+\tilde{B}}} = O\left(e^{(1-\tilde{A})z}\right).$$

This is only possible if Q is a constant and $1 - \tilde{A} = 0$. In that case we have $z = Ce^z$ for some constant $C = 1 - e^{\tilde{B}}$, but this is clearly not possible.

14. Let $k < \rho < k + 1$, and say we have only finitely many zeros, then as in the previous problem, $F(z) = e^{p(z)}Q(z)$ for some polynomial p of degree at most k and Q a polynomial. Now F has order of growth less than or equal to k , since Q has order of growth 0. This contradicts the assumption that F has order of growth ρ .

15. Every meromorphic function, f , is holomorphic with poles at some sequence $p_1, p_2, \dots$. Then there is an entire function with zeros precisely at the p_j with desired multiplicity. Call this function g . Then fg has no poles and is holomorphic except possibly at the p_j , but since there are no poles at the p_j , we see that we have an entire function, say h . Then $f = g/h$. For the second part of the problem construct two entire functions and take their quotient.

7.2 Problem

1. By Jensen's formula, take $R < 1$.

$$\sum_{z_n < R} (1 - |z_n|) < - \sum_{z_n < R} \log |z_n| = \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta - \log |f(0)|$$

Then let $R \rightarrow 1^-$

2. (a)

$$|\alpha + |\alpha|z| \leq |\alpha| + |\alpha||z| = |\alpha|(1 + |z|) \leq |\alpha|(1 + r)$$

$$|1 - \bar{\alpha}z| \geq 1 - |\bar{\alpha}||z| \geq 1 - r$$

Then

$$\left| \frac{\alpha + |\alpha|z}{(1 - \bar{\alpha}z)\alpha} \right| \leq \frac{1 + r}{1 - r}$$

(b)

$$\begin{aligned} & \left| 1 - \frac{\alpha_n - z}{1 - \bar{\alpha}_n z} \frac{|\alpha_n|}{\alpha_n} \right| \\ &= \left| \frac{\alpha_n - |\alpha|^2 z - |\alpha_n| \alpha_n + |\alpha_n| z}{(1 - \bar{\alpha}_n z) \alpha_n} \right| \\ &= \left| \frac{\alpha_n + |\alpha_n| z}{(1 - \bar{\alpha}_n z) \alpha_n} \right| (1 - |\alpha_n|) \\ &\leq \frac{1 + r}{1 - r} (1 - |\alpha_n|) \end{aligned}$$

By Proposition 3.2, we can conclude the uniform convergence.

$$\sum_{n=1}^{\infty} \left| 1 - \frac{\alpha_n - z}{1 - \overline{\alpha_n} z} \frac{|\alpha_n|}{\alpha_n} \right| \leq \frac{1+r}{1-r} \sum_{n=1}^{\infty} (1 - |\alpha_n|) < \infty$$

Then

$$|f(z)| = \prod_{n=1}^{\infty} \left| \frac{\alpha_n - z}{1 - \overline{\alpha_n} z} \right| < 1$$

3. Assume that $\alpha = \frac{p}{q} > 1$ is rational, suppose we prove the result for all rational α , the result holds for all real α by continuity. WLOG, let z in real positive axis.

First, we prove for large $|z|$:

$$\sum \frac{|z|^n}{(n!)^\alpha} = \sum \left(\frac{|z|^{\frac{n}{\alpha}}}{n!} \right)^{\frac{p}{q}} \geq A \sum \frac{|z|^{\frac{n}{\alpha}}}{(n!)} = A e^{|z|^{\frac{1}{\alpha}}}$$

Let $x = z^{\frac{1}{\alpha}}$

$$\sum_{n=(k-1)q+1}^{kq} \left(\frac{x^n}{n!} \right)^{\frac{p}{q}} \geq q \left(\frac{x^{kq}}{(kq)!} \right)^{\frac{p}{q}} \geq \frac{q}{p} \cdot p \cdot \frac{x^{kp}}{(kp)!} \geq \frac{q}{p} \sum_{n=kp}^{(k+1)p-1} \frac{x^n}{n!}$$

Then sum by k , we get the desired estimate.

Then, for large $|z|$:

$$\sum \frac{|z|^n}{(n!)^{\frac{p}{q}}} = \sum \left(\frac{|z|^{\frac{nq}{p}}}{n!} \right)^{\frac{p}{q}} \leq B \sum \frac{|z|^{\frac{nq}{p-1}}}{(n!)} = B e^{|z|^{\frac{q}{p-1}}}$$

$$\sum_{n=kq}^{(k+1)q-1} \left(\frac{|z|^{\frac{nq}{p}}}{n!} \right)^{\frac{p}{q}} \leq q \left(\frac{|z|^{\frac{kq^2}{p}}}{(kq)!} \right)^{\frac{p}{q}} \leq \frac{q}{p-1} \cdot (p-1) \cdot \frac{|z|^{kq}}{(k(p-1))!} \leq \frac{q}{p-1} \sum_{n=(k-1)(p-1)+1}^{k(p-1)} \frac{|x|^{\frac{nq}{p-1}}}{n!}$$

4. (a) By Cauchy's inequality, $|a_n| \leq \frac{A}{r^n} e^{ar^\rho}$ for all r . Take $r = \left(\frac{n}{\rho}\right)^{\frac{1}{\rho}}$:

$$\begin{aligned} |a_n| &\leq \frac{A}{\left(\frac{n}{\rho}\right)^{\frac{n}{\rho}}} e^{\frac{an}{\rho}} \\ |a_n|^{\frac{1}{n}} n^{\frac{1}{\rho}} &\leq \frac{\sqrt[n]{A}}{\left(\frac{n}{\rho}\right)^{\frac{1}{\rho}}} \cdot e^{\frac{a}{\rho}} \cdot n^{\frac{1}{\rho}} \\ &\leq \sqrt[n]{A} \rho^{\frac{1}{\rho}} e^{\frac{a}{\rho}} \end{aligned}$$

(b) Since $\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} n^{\frac{1}{\rho}} < \infty$, then $|a_n|^{\frac{1}{n}} n^{\frac{1}{\rho}} < M$ for large n .

$$\begin{aligned}
 |F(z)| &\leq \sum |a_n| |z|^n \\
 &\leq \sum M^n n^{-\frac{n}{\rho}} |z|^n \\
 &= \sum \frac{|Mz|^n}{(n^n)^{\frac{1}{\rho}}} \\
 &\leq \sum \frac{|Mz|^n}{(n!)^{\frac{1}{\rho}}}
 \end{aligned}$$

By Problem 3, the last term is of order ρ .

8 Chapter 6

8.1 Exercise

1.

$$\begin{aligned}
 \frac{1}{\Gamma(s)} &= e^{\gamma s} s \prod_{n \geq 1} \left(1 + \frac{s}{n}\right) e^{-s/n} \\
 &= \lim_{N \rightarrow \infty} e^{s(\sum_{n=1}^N \frac{1}{n} - \log N)} s \prod_{n=1}^N \frac{n+s}{n} e^{-s/n} \\
 &= \lim_{N \rightarrow \infty} \frac{e^{\log N^{-s}}}{N!} s(s+1) \cdots (s+N)
 \end{aligned}$$

2.

$$\begin{aligned}
 \frac{\Gamma(a+1)\Gamma(b+1)}{\Gamma(a+b+1)} &= \lim_{n \rightarrow \infty} \frac{\frac{n^{a+1}n!}{(a+1)(a+2)\cdots(a+n+1)} \cdot \frac{n^{b+1}n!}{(b+1)(b+2)\cdots(b+n+1)}}{\frac{n^{a+b+1}n!}{(a+b+1)(a+b+2)\cdots(a+b+n+1)}} \\
 &= \lim_{n \rightarrow \infty} \frac{n \cdot n! \cdot (a+b+1) \cdots (a+b+n+1)}{(a+1)(a+2) \cdots (a+n+1)(b+1)(b+2) \cdots (b+n+1)} \\
 &= \lim_{n \rightarrow \infty} \prod_{k=1}^n \frac{k(k+a+b)}{(k+a)(k+b)} \cdot \left(\frac{n(a+b+n+1)}{(a+n+1)(b+n+1)} \right) \\
 &\rightarrow \prod_{k=1}^{\infty} \frac{k(k+a+b)}{(k+a)(k+b)}
 \end{aligned}$$

3.

$$\begin{aligned}
 \frac{\pi}{2} &= \lim_{N \rightarrow \infty} \prod_{n=1}^N \frac{(2n)^2}{(2n)^2 - 1} \\
 &= \lim_{N \rightarrow \infty} \frac{2^{4N} (N!)^4 (2N+1)}{((2N+1)!)^2}
 \end{aligned}$$

Then $\sqrt{\frac{\pi}{2}} = \lim_{N \rightarrow \infty} \frac{2^{2N} (N!)^2 \sqrt{2N+1}}{(2N+1)!}$.

$$\begin{aligned}
 \Gamma(s)\Gamma(s + \frac{1}{2}) &= \lim_{n \rightarrow \infty} \frac{n^s \cdot n!}{s(s+1) \cdots (s+n)} \frac{n^{s+\frac{1}{2}} \cdot n!}{(s\frac{1}{2})(s+\frac{3}{2}) \cdots (s+n+\frac{1}{2})} \\
 &= \lim_{n \rightarrow \infty} \frac{n^{2s+\frac{1}{2}} (n!)^2 2^{2n+2}}{(2s)(2s+2) \cdots (2s+2n)(2s+1)(2s+3) \cdots (2s+2n+1)} \\
 &= \lim_{n \rightarrow \infty} \left(\frac{2^{2n} (n!)^2 \sqrt{2n+1}}{(2n+1)!} \right) \left(\frac{(2n)^{2s} (2n)!}{(2s) \cdots (2s+2n)} \right) \frac{\sqrt{n} (2n+1) 2^{2-2s}}{\sqrt{2n+1} (2s+2n+1)} \\
 &= \sqrt{\frac{\pi}{2}} \cdot \Gamma(2s) \cdot \frac{1}{\sqrt{2}} \cdot 2^{2-2s} \\
 &= \sqrt{\pi} 2^{1-2s} \Gamma(2s)
 \end{aligned}$$

4. We have $a_n(\alpha) = \alpha(\alpha+1)\cdots(\alpha+n-1)\frac{1}{n}$.

5. This follows immediately from using the fact that $\overline{\Gamma(\sigma+it)} = \Gamma(\sigma-it)$ (by Schwartz reflection principle) and $\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin \pi s}$.

6. Notice that $\gamma + \log(n) = \sum_{m=1}^n \frac{1}{m} + o(1)$. Thus

$$1 + \frac{1}{3} + \cdots + \frac{1}{2n-1} - \frac{1}{2}(\log(n) + \gamma) = \sum_{m=1}^{2n} \frac{(-1)^m}{m} \rightarrow \log(2).$$

7. For (a) follow the hint to obtain

$$\begin{aligned} \Gamma(\alpha)\Gamma(\beta) &= \int_0^\infty \int_0^\infty t^{\alpha-1} s^{\beta-1} e^{-t-s} dt ds \\ &= \int_0^\infty \int_0^1 (ur)^{\beta-1} (u(1-r))^{\alpha-1} e^{-u} u dr du \\ &= \Gamma(\alpha+\beta)B(\alpha, \beta). \end{aligned}$$

For (b) use

$$B(\alpha, \beta) = \int_0^\infty \left(\frac{u}{1+u} \right)^{\alpha-1} \left(\frac{1}{1+u} \right)^{\beta-1} \frac{du}{(1+u)^2}$$

8. Follow the hint to get

$$J_\nu(x) = \frac{(x/2)^\nu}{\Gamma(\nu+1/2)\sqrt{\pi}} \int_{-1}^1 \sum_{n=0}^\infty \frac{(ixt)^n}{n!} (1-t^2)^{\nu-1/2} dt$$

Switching the integral and sum we see that we need to compute

$$\int_{-1}^1 t^n (1-t^2)^{\nu-1/2} dt$$

This is 0 when n is odd because it is an odd function. In the case $n = 2m, m \in \mathbb{Z}$, we have

$$\int_{-1}^1 t^{2m} (1-t^2)^{\nu-1/2} dt = 2 \int_0^1 t^{2m} (1-t^2)^{\nu-1/2} dt = \frac{\Gamma(\nu+1/2)\Gamma(m+1/2)}{\Gamma(\nu+m+1)}$$

by the previous problem with the change of variables $u = t^2$. From this we see that we need to prove something like

$$\frac{1}{2^{2m}m!} = \frac{\Gamma(m+1/2)}{(2m)!\sqrt{\pi}}$$

which is because

$$\Gamma(m + \frac{1}{2}) = \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{2m-1}{2} \cdot \Gamma(\frac{1}{2}) = \frac{(2m)!}{2^{2m}m!} \cdot \sqrt{\pi}$$

Remark: The definition of Bessel function in this question

$$J_\nu(x) = \frac{(x/2)^\nu}{\Gamma(\nu + 1/2)\sqrt{\pi}} \int_{-1}^1 e^{ixt} (1-t^2)^{\nu-1/2} dt$$

is an extension of the definition

$$J_n(\rho) = \frac{1}{2\pi} \int_0^{2\pi} e^{i\rho \sin \theta} e^{-in\theta} d\theta$$

in Chapter 6, Book I (page 197).

9. The hypergeometric series is initially defined as

$$F(\alpha, \beta, \gamma; z) = \sum_{n=0}^{\infty} \frac{\alpha \cdots (\alpha + n - 1) \beta \cdots (\beta + n - 1)}{n! \gamma \cdots (\gamma + n - 1)} z^n$$

Use the facts that

$$\frac{1}{(1-zt)^\alpha} = \sum_{n=0}^{\infty} \frac{\alpha \cdots (\alpha + n - 1)}{n!} t^n z^n$$

for $|tz| < 1$ and

$$\begin{aligned} & \frac{\Gamma(\gamma)}{\Gamma(\beta)\Gamma(\gamma-\beta)} \int_0^1 t^{\beta-1} (1-t)^{\gamma-\beta-1} (1-zt)^{-\alpha} dt \\ &= \frac{\Gamma(\gamma)}{\Gamma(\beta)\Gamma(\gamma-\beta)} \int_0^1 t^{\beta-1} (1-t)^{\gamma-\beta-1} \left[\sum_{n=0}^{\infty} \frac{\alpha \cdots (\alpha + n - 1)}{n!} \right] t^n z^n dt \\ &= \sum_{n=0}^{\infty} \frac{\Gamma(\gamma)}{\Gamma(\beta)\Gamma(\gamma-\beta)} \frac{\alpha \cdots (\alpha + n - 1)}{n!} \left(\int_0^1 t^{\beta+n-1} (1-t)^{\gamma-\beta-1} dt \right) z^n \\ &= \sum_{n=0}^{\infty} \frac{\Gamma(\gamma) B(\beta+n, \gamma-\beta)}{\Gamma(\beta)\Gamma(\gamma-\beta)} \frac{\alpha \cdots (\alpha + n - 1)}{n!} z^n \\ &= \sum_{n=0}^{\infty} \frac{\Gamma(\gamma)}{\Gamma(\gamma+n)} \frac{\Gamma(\beta+n)}{\Gamma(\beta)} \frac{\alpha \cdots (\alpha + n - 1)}{n!} z^n \\ &= \sum_{n=0}^{\infty} \frac{\alpha \cdots (\alpha + n - 1) \beta \cdots (\beta + n - 1)}{n! \gamma \cdots (\gamma + n - 1)} z^n \end{aligned}$$

The continuation is obvious by the formula.

10. (a) Integral as in the hint:

$$\begin{aligned}\Gamma(z) &= \int_0^\infty e^{-t} t^{z-1} dt \\ &= \int_0^\infty e^{-it} (it)^{z-1} d(it) \\ &= e^{\frac{\pi}{2}iz} \int_0^\infty e^{-it} t^{z-1} dt\end{aligned}$$

Then

$$\int_0^\infty e^{-it} t^{z-1} dt = \Gamma(z) e^{-\frac{\pi}{2}iz}$$

Similarly, use the contour reflecting over the real axis.

$$\int_0^\infty e^{it} t^{z-1} dt = \Gamma(z) e^{\frac{\pi}{2}iz}$$

$$\int_0^\infty \cos t \cdot t^{z-1} dt = \int_0^\infty \frac{e^{it} + e^{-it}}{2} \cdot t^{z-1} dt = \Gamma(z) \frac{e^{\frac{\pi}{2}iz} + e^{-\frac{\pi}{2}iz}}{2} = \Gamma(z) \cos\left(\frac{\pi}{2}z\right)$$

$$\int_0^\infty \sin t \cdot t^{z-1} dt = \int_0^\infty \frac{e^{it} - e^{-it}}{2i} \cdot t^{z-1} dt = \Gamma(z) \frac{e^{\frac{\pi}{2}iz} - e^{-\frac{\pi}{2}iz}}{2i} = \Gamma(z) \sin\left(\frac{\pi}{2}z\right)$$

For (b) the defect integral near the origin converges. notice that

$$\mathcal{M}(\sin)(0) = \int_0^\infty \sin(t) t^{-1} dt = \lim_{z \rightarrow 0} \Gamma(z) \sin(\pi z/2) = \frac{\pi}{2}$$

and

$$\mathcal{M}(\sin)(-1/2) = \int_0^\infty \sin(t) t^{-3/2} dt = \Gamma(-1/2) \sin(-\pi/4) = (-2\sqrt{\pi}) \left(-\frac{1}{\sqrt{2}}\right) = \sqrt{2\pi}$$

11.

$$|f(z)| = e^{ax} e^{\Re(-e^z)} = e^{ax} e^{-e^x \frac{x}{\sqrt{x^2+y^2}}}$$

When $x \rightarrow \infty$

$$|f(z)| \leq e^{ax - \frac{1}{2}e^x} \leq e^{-cx}$$

When $x \rightarrow -\infty$

$$|f(z)| \leq e^{ax} \leq e^{-cx}$$

$$\begin{aligned}
\hat{f}(\xi) &= \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx \\
&= \int_{-\infty}^{\infty} e^{ax - e^x - 2\pi i x \xi} dx \\
&= \int_0^{\infty} t^{a - 2\pi i \xi - 1} e^{-t} dt \\
&= \Gamma(a - 2\pi i \xi)
\end{aligned}$$

with changing the variable $x = \ln t$

12. For (a) Follow the hint and notice that We essentially have

$$\frac{1}{|\Gamma(s)|} \geq k!/\pi \geq C(k/e)^k \geq C e^{k \log k}$$

when $s = -k - 1/2$, k a positive integer. So we have essentially, $\frac{1}{|\Gamma(s)|} \geq C e^{c|s| \log |s|}$ for some constants c and C .

For (b) notice that if we had such an F , then $F(s)\Gamma(s) = e^{P(z)}$ for some polynomial P of degree at most 1. But this would imply that $\frac{1}{|\Gamma(s)|} = O(e^{c|s|})$. Which we know from part (a) is a contradiction.

13. We have

$$\log(1/\Gamma(s)) = -\gamma s + \log s + \sum_{n=1}^{\infty} \left(-\frac{s}{n} + \log \left(1 + \frac{s}{n} \right) \right).$$

Hence we have

$$\frac{d}{ds} \log \frac{1}{\Gamma(s)} = -\frac{d}{ds} \log(\Gamma(s)) = -\gamma + \frac{1}{s} + \sum_{n=1}^{\infty} \left(\frac{1}{s+n} - \frac{1}{n} \right).$$

Where we are allowed to pass the derivative inside the sum because of absolute convergence. Differentiating again we have

$$\left(\frac{d}{ds} \right)^2 \log \Gamma(s) = \frac{1}{s^2} + \sum_{n=1}^{\infty} \frac{1}{(n+s)^2},$$

which is justified again by absolute convergence. Notice that Γ'/Γ is holomorphic away from $s = 0, -1, -2, \dots$ so its derivative is also holomorphic in this region.

14. (a) follows from

$$\frac{d}{dx} \int_x^{x+1} \log \Gamma(t) dt = \log \Gamma(x+1) - \log \Gamma(x) = \log(x\Gamma x) - \log \Gamma x = \log x.$$

Integrating $\log x$ gives the result. Since Γt is increasing for $t \geq 1$, we see that $\log \Gamma(t)$ is

increasing in that range. Thus

$$\log \Gamma x \leq \int_x^{x+1} \log \Gamma(t) dt \leq \log \Gamma(x+1) = \log x + \log \Gamma(x).$$

Therefore we have

$$1 - \frac{1}{\log x} + \frac{c}{x \log x} \leq \frac{\log \Gamma(x)}{x \log x} \leq 1 - \frac{\log x + x + c}{x \log x}.$$

Letting $x \rightarrow \infty$ gives the result.

Since Γt is increasing for $t \geq 1$, we see that $\log \Gamma(t)$ is increasing in that range. Thus

$$\log \Gamma x \leq \int_x^{x+1} \log \Gamma(t) dt \leq \log \Gamma(x+1) = \log x + \log \Gamma(x)$$

Therefore we have

$$1 - \frac{1}{\log x} + \frac{c}{x \log x} \leq \frac{\log \Gamma(x)}{x \log x} \leq 1 - \frac{\log x + x + c}{x \log x}$$

Letting $x \rightarrow \infty$ gives the result.

15. Use the suggested substitution and then change variables $t = nx$ to get $\int_0^\infty x^{s-1} e^{-nx} dx = \frac{1}{n^s} \int_0^\infty t^{s-1} e^{-t} dt = \frac{\Gamma(s)}{n^s}$. After substituting the Taylor series the change of integral and summation is justified by using absolute convergence and the fact that for $\operatorname{Re}(s) > 1$ the integral $\int_0^1 x^{s-1} dx$ converges.

16. Proceed as in the proof they give and follow the hint.

17. First observe that $\int_0^\infty f(x) x^{s-1} dx$ is holomorphic for $\operatorname{Re}(s) > 1$ because $\int_0^1 x^{s-1} dx$ converges for these s and for $x \rightarrow \infty$ f decays faster than any polynomial. Next we may apply the same argument to

$$I(s) = \frac{(-1)^k}{\Gamma(s+k)} \int_0^\infty f^{(k)}(x) x^{s+k-1} dx$$

to see that $I(s)$ is holomorphic in the region $\operatorname{Re}(s) > k-1$. This formula is justified by integration by parts. Finally using this formula for $k = n+1$ gives

$$I(-n) = (-1)^{n+1} f^{(n)}(0)$$

There seems to be a typo in the book.

8.2 Problem

1. (a)

$$\begin{aligned}
 \zeta(s) &= \sum_{1 \leq n < N} n^{-s} + \sum_{n \geq N} n^{-s} \\
 &= \sum_{1 \leq n < N} n^{-s} + \sum_N^{\infty} x^{-s} dx + \sum_{n \geq N} \delta_n(s) \\
 &= \sum_{1 \leq n < N} n^{-s} + \frac{N^{1-s}}{1-s} + \sum_{n \geq N} \delta_n(s)
 \end{aligned}$$

(b) For every given t take $N = \lfloor |t| \rfloor$:

$$\begin{aligned}
 |\zeta(1+it)| &= \left| \sum_{1 \leq n < N} n^{-1-it} + \frac{N^{-it}}{-it} + \sum_{n \geq N} \delta_n(1+it) \right| \\
 &\leq \left| \sum_{1 \leq n < N} n^{-1-it} \right| + \left| \frac{N^{-it}}{-it} \right| + \left| \sum_{n \geq N} \delta_n(1+it) \right| \\
 &\leq \sum_{1 \leq n < N} n^{-1} + \frac{1}{|t|} + \sum_{n \geq N} \frac{|1+it|}{n^2} \\
 &\sim \log N + \frac{1}{|t|} + \frac{|1+it|}{N} \\
 &\sim \log |t| + \frac{1}{|t|} + \frac{|1+it|}{|t|} \\
 &\sim \log |t|
 \end{aligned}$$

Remark: Then notation $\delta_n(s)$ is defined in Prop 2.5, and " $\sim$ " means "of the same order".

2.

$$\begin{aligned}
 &\frac{s}{s-1} - s \int_1^{\infty} \frac{\{x\}}{x^{s+1}} dx \\
 &= \frac{s}{s-1} - s \sum_{n=1}^{\infty} \int_n^{n+1} \frac{x-n}{x^{s+1}} dx \\
 &= \frac{s}{s-1} - s \left[\int_1^{\infty} \frac{dx}{x^s} - \sum_{n=1}^{\infty} \int_n^{n+1} \frac{n}{x^{s+1}} dx \right] \\
 &= \frac{s}{s-1} - s \left[\frac{1}{1-s} - \sum_{n=1}^{\infty} \frac{n}{s} ((n+1)^{-s} - n^{-s}) \right] \\
 &= \sum_{n=1}^{\infty} n((n+1)^{-s} - n^{-s}) \\
 &= \sum_{n=1}^{\infty} n^{-s} = \zeta(s)
 \end{aligned}$$

3. $\zeta(s) = \frac{s}{s-1} - s \int_1^{\infty} \frac{\{x\}}{x^{s+1}} dx = \frac{s}{s-1} - \frac{1}{2} + s \int_1^{\infty} \frac{\{x\}}{x^{s+1}} dx$ is obvious, and $Q_k(x)$ can be well

defined to be continuous with exception on integers. We only need to prove:

$$\zeta(s) = \frac{s}{s-1} - \frac{1}{2} + s \int_1^\infty \left(\frac{d^k}{dx^k} Q_k(x) \right) x^{-s-1} dx$$

is a well defined analytic continuation when $\Re(s) > -k$. By induction:

Obviously, above formula is well-defined when $\Re(s) > 0$, By integration by parts:

$$\begin{aligned} & \int_1^\infty Q_k^{(k)}(x) x^{-s-1} dx \\ &= \sum_{n=1}^\infty \int_n^{n+1} Q_k^{(k)}(x) x^{-s-1} dx \\ &= \sum_{n=1}^\infty \left(\frac{Q^{(k-1)}(x)}{x^{s+1}} \Big|_n^{n+1} + (s+1) \int_n^{n+1} Q_k^{(k-1)}(x) x^{-s-2} dx \right) \\ &= \sum_{n=1}^\infty \frac{Q^{(k-1)}(x)}{x^{s+1}} \Big|_n^{n+1} + (s+1) \int_1^\infty Q_k^{(k-1)}(x) x^{-s-2} dx \end{aligned}$$

Then we can find when $\Re(s) > -1$, the first term vanishes and the second term is well-defined. Then $\zeta(s)$ is extended to $\Re(s) > -1$. Every times we do integration by parts, we can extend $\zeta(s)$ by one unit left.

9 Chapter 7

9.1 Exercise

1. To obtain convergence use summation by parts to obtain

$$\sum_{n=1}^N \frac{a_n}{n^s} = \frac{A_N}{N^s} + \sum_{n=1}^N A_n (n^{-s} - (n+1)^{-s}).$$

Then $|n^{-s} - (n+1)^{-s}| \leq Cn^{Re(s)+1}$. Thus letting $N \rightarrow \infty$ we see that

$$\left| \sum_{n=1}^N A_n (n^{-s} - (n+1)^{-s}) \right| \leq B \sum_{n=1}^N n^{Re(s)+1}$$

will converge for $Re(s) > 0$. As the hint indicates we can show absolute convergence on any closed half-plane $\{z : Re(z) \geq \delta > 0\}$ thus showing that the series defines a holomorphic function on the open half-plane $\{z : Re(z) > 0\}$.

2. The sequences are bounded thus $B_N = \sum_{n=1}^N b_n$ and $A_N = \sum_{n=1}^N a_n$ satisfy $|A_N| \leq AN$ and $|B_N| \leq BN$ where A and B are the bounds for the respective sequences. Using this and an argument as in the first exercise we see that the series define holomorphic functions for $Re(s) > 1$. The produce formula is easily verified by first considering the partial sums and then letting them tend to infinity. To prove the convergence for the sequence $\sum_{n \geq 1} \frac{c_n}{n^s}$ in the range $Re(s) > 1$, we use the estimate $|c_n| \leq ABd(n)$. Now $d(n) \leq c \log(n)$ for some constant c . With this in hand we apply partial summation again just as in the previous exercise. (b) The first desired identity is a direct consequence since $a_n = b_n = 1$ for that identity. For the second identity notice that $\zeta(s-a) = \sum_{n \geq 1} \frac{n^a}{n^s}$. In this case we don't have boundedness of the sequences under consideration, but we can apply the same argument so long as we assume that $Re(s) > Re(a)+1$, since we will need this to get the convergence in our partial summation trick.

3. We actually prove (b) first. For $n > 1$, write $n = p_1^{e_1} \cdots p_m^{e_m}$. Notice that

$$\begin{aligned} \sum_{k|n} \mu(k) &= \mu(1) + \mu(p_1) + \cdots + \mu(p_m) + \mu(p_1 p_2) + \cdots + \mu(p_1 \cdots p_m) \\ &= 1 - \binom{m}{1} + \binom{m}{2} - \cdots + (-1)^m \binom{m}{m} \\ &= (1-1)^m = 0. \end{aligned}$$

Now to prove (a) we prove that $\zeta(s) \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s} = 1$. To do this we apply exercise 2, part (a) directly and use the identity we just proved.

4.

$$\begin{aligned}
& \frac{1}{\Gamma(s)} \int_0^\infty \frac{Q(x)x^{s-1}}{e^{qx}-1} dx \\
&= \frac{1}{\Gamma(s)} \int_0^\infty \left(\sum_{m=0}^{q-1} a_{-m} e^{mx} \right) x^{s-1} \left(\sum_{n=1}^\infty e^{-nqx} \right) dx \\
&= \frac{1}{\Gamma(s)} \sum_{m=0}^{q-1} \sum_{n=1}^\infty a_{-m} \int_0^\infty e^{-(nq-m)x} x^{s-1} dx \\
&= \frac{1}{\Gamma(s)} \sum_{k=1}^\infty a_k \int_0^\infty e^{-kx} x^{s-1} dx \\
&= \frac{1}{\Gamma(s)} \sum_{k=1}^\infty \frac{a_k}{k^s} \int_0^\infty e^{-kx} (kx)^{s-1} d(kx) \\
&= \sum_{k=1}^\infty \frac{a_k}{k^s} = L(s)
\end{aligned}$$

Since $M(x) = \sum_{m=0}^{q-1} a_{q-m} e^{mx}$ is a finite sum, we only need to prove that for each m in $\{0, 1, \dots, q-1\}$, $\frac{1}{\Gamma(s)} \int_0^\infty \frac{e^{mx} x^{s-1}}{e^{qx}-1} dx$ is continuable into the complex plane. We split the integral into two parts

$$\frac{1}{\Gamma(s)} \int_0^{1/q} \frac{e^{mx} x^{s-1}}{e^{qx}-1} dx + \frac{1}{\Gamma(s)} \int_{1/q}^\infty \frac{e^{mx} x^{s-1}}{e^{qx}-1} dx$$

The second integral defines an entire function, while for any fixed $s > 1$

$$\begin{aligned}
& \int_0^{1/q} \frac{e^{mx} x^{s-1}}{e^{qx}-1} dx \\
&= \int_0^1 \frac{e^{mx/q} x^{s-1}}{q^s (e^x - 1)} dx \\
&= \frac{1}{q^s} \int_0^1 \sum_{n=0}^\infty \frac{B_n e^{mx/q} x^{n+s-2}}{n!} dx \\
&= \frac{1}{q^s} \sum_{n=0}^\infty \frac{B_n}{n!} \int_0^1 e^{mx/q} x^{n+s-2} dx \\
&= \frac{1}{q^s} \sum_{n=0}^\infty \frac{B_n}{n!} \int_0^1 \sum_{k=0}^\infty \frac{m^k x^{n+k+s-2}}{k! q^k} dx \\
&= \frac{1}{q^s} \sum_{n=0}^\infty \sum_{k=0}^\infty \frac{B_n}{n!} \frac{m^k}{k! q^k} \int_0^1 x^{n+k+s-2} dx \\
&= \frac{1}{q^s} \sum_{n=0}^\infty \sum_{k=0}^\infty \frac{B_n}{n!} \frac{m^k}{k! q^k} \frac{1}{n+k+s-1}
\end{aligned}$$

We were allowed to move the sums outside the integral because each time, the series inside the integral converges uniformly.

What remains is to prove this double sum, multiplied by $\frac{1}{\Gamma(s)}$, defines a meromorphic

function on the complex plane with the only possible singularity a pole at $s = 1$. For any $|s| \leq R$, this double sum can be split into two parts.

$$\begin{aligned} & \frac{1}{q^s} \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} \frac{B_n}{n!} \frac{m^k}{k!q^k} \frac{1}{n+k+s-1} \\ &= \frac{1}{q^s} \sum_{n+k < R+2} \frac{B_n}{n!} \frac{m^k}{k!q^k} \frac{1}{n+k+s-1} + \frac{1}{q^s} \sum_{n+k \geq R+2} \frac{B_n}{n!} \frac{m^k}{k!q^k} \frac{1}{n+k+s-1}. \end{aligned}$$

The first part is the sum of a finite number of holomorphic functions of s , with simple holes at $\{1, 0, -1, \dots\}$, all which except the pole at 1 are cancelled by the zeros of $\frac{1}{\Gamma(s)}$. The second part absolutely and uniformly converges and is therefore entire on $|s| \leq R$. Finally we let $R \rightarrow \infty$ and this concludes the proof.

Remark: Solution to the second statement is from mathematics stack exchange.

5. (a) By the alternating series test, the sum converges for real $s > 0$. More generally, the partial sums of $a_n = (-1)^{n+1}$ are bounded, thus we may apply exercise 1 of this chapter to get $\tilde{\zeta}$ is holomorphic in $\operatorname{Re}(s) > 0$. (b) Use the identity

$$\zeta(s) + \tilde{\zeta}(s) = 2 \sum_{n=1}^{\infty} \frac{1}{2^n n^s} = 2^{1-s} \zeta(s),$$

which is justified in $\operatorname{Re}(s) > 1$ by absolute convergence of the series involved.

(c) Since $\tilde{\zeta}$ is holomorphic for $\operatorname{Re}(s) > 0$ and ζ extends to a holomorphic function in all of $\mathbb{C}$ we know they must agree on $\operatorname{Re}(s) > 0$. Additionally $\tilde{\zeta}$ does not vanish for $s \in (0, 1)$ because it is an alternating sum. If $\zeta(0) = 0$ then by the functional equation

$$\zeta(s) \Gamma(s/2) \pi^{-s/2} = \zeta(1-s) \Gamma((1-s)/2) \pi^{-(1-s)/2},$$

we see that if $\zeta(0) = 0$, then $\zeta(1)$ is finite because Γ has a simple pole at 0, but this contradicts the fact that ζ has a simple pole at $s = 1$.

6. In the case $0 \leq a < 1$ shift the contour to $c \rightarrow +\infty$. In the case $a > 1$ shift the contour to $c \rightarrow -\infty$ and use the residue theorem to pick up the desired value because of the pole. Use the principal value in the case $a = 1$.

7. For real s use the definition of ξ to see that it is real. Also for $s > 0$ we can see it easily from $\tilde{\zeta}$ defined in exercise 5. For s with real part $1/2$, we use the functional equation and the facts that $\zeta(1/2 - it) = \overline{\zeta(1/2 + it)}$, which can be observed from $\tilde{\zeta}$ in exercise 5, and the fact that $\Gamma(\bar{z}) = \overline{\Gamma(z)}$, which follows from the definition of Γ .

8. (a) $\xi(s) = \xi(1-s)$ so $F(s) = \xi(\frac{1}{2} + s)$.

(b)

$$\xi(s) = \frac{1}{s-1} - \frac{1}{s} + \int_1^{\infty} (u^{(-s/2)-1/2} + u^{(s/2)-1}) \psi(u) du$$

where

$$\psi(u) = \sum_{n=1}^{\infty} e^{-\pi n^2 u}$$

Let $s = \sigma + it$ and $|\sigma| \leq 2N - 2 < |\sigma| + 2$. For the integral, we have

$$\begin{aligned} & \left| \int_1^{\infty} (u^{(-s/2)-1/2} + u^{(s/2)-1}) \psi(u) du \right| \\ & \leq \int_1^{\infty} (u^{(|\sigma|/2)-1/2} + u^{(|\sigma|/2)-1}) \sum_{n=1}^{\infty} e^{-\pi n^2 u} du \\ & \leq 2 \sum_{n=1}^{\infty} \int_1^{\infty} u^{(|\sigma|+1)/2-1} e^{-\pi n^2 u} du \\ & \leq 2 \sum_{n=1}^{\infty} \int_0^{\infty} u^{(|\sigma|+2)/2-1} e^{-\pi n^2 u} du \\ & \leq 2\Gamma\left(\frac{|\sigma|}{2} + 1\right) \sum_{n=1}^{\infty} \frac{1}{(\pi n^2)^{|\sigma|/2+1}} \\ & \leq 2\Gamma\left(\frac{|\sigma|}{2} + 1\right) \sum_{n=1}^{\infty} \frac{1}{\pi n^2} \\ & = \frac{\pi}{3}\Gamma\left(\frac{|\sigma|}{2} + 1\right) \\ & < \frac{\pi}{3}N! \leq \frac{\pi}{3}N^N = \frac{\pi}{3}e^{N \log N} \leq \frac{\pi}{3}e^{(|\sigma|+2) \log(|\sigma|+2)} \end{aligned}$$

Since $e^{|\sigma| \log |\sigma|}$ is of growth order 1, this proves that except for the two simple poles at o and 1, $\xi(s)$ is of growth order 1. Furthermore, we have

$$(s-1)\zeta(s) = (s-1)\pi^{s/2} \frac{\xi(s)}{\Gamma(s/2)}$$

Now the two poles at o and 1 of $\xi(s)$ are cancelled by $\frac{1}{\Gamma(s/2)}$ and $(s-1)$ respectively. Both $\pi^{s/2}$ and $\frac{1}{\Gamma(s/2)}$ are of growth order 1. Therefore $(s-1)\zeta(s)$ is also of growth order 1.

(c) $\xi(s)$ has exact zeros as $\zeta(s)$ in the critical strip. $G(s)$ is of fractional order, then it has infinitely many zeros, so is $\xi(s)$.

Remark: Solution to (b) is from mathematics stack exchange.

9. (a) Recall that $|\delta_n(\sigma + it)| \leq \frac{|s|}{n^{\sigma+1}}$ and $|\delta_n(\sigma + it)| \leq \frac{2}{n^{\sigma}}$. For $\sigma = 1$, $|\delta_n(1 + it)| \leq \frac{|s|}{n^2}$

and $|\delta_n(1+it)| \leq \frac{2}{n}$.

$$\begin{aligned}
|\zeta(1+it)| &= \left| \frac{1}{it} + H(1+it) \right| \\
&\leq O(1) + \sum_{n=1}^{\infty} |\delta_n(1+it)| \\
&\leq O(1) + \sum_{n < \frac{|t|}{2}} |\delta_n(1+it)| + \sum_{n \geq \frac{|t|}{2}} |\delta_n(1+it)| \\
&\leq O(1) + \sum_{n < \frac{|t|}{2}} \frac{2}{n} + \sum_{n \geq \frac{|t|}{2}} \frac{|1+it|}{n^2} \\
&\leq O(1) + \sum_{n < \frac{|t|}{2}} \frac{2}{n} + \sum_{n \geq \frac{|t|}{2}} \frac{|it|}{n^2} \\
&\sim O(1) + \int_1^{\frac{|t|}{2}} \frac{2}{x} dx + \int_{\frac{|t|}{2}}^{\infty} \frac{|t|}{x^2} dx \\
&\sim O(1) + 2 \log x \Big|_1^{\frac{|t|}{2}} + \frac{|t|}{x^2} \Big|_{\frac{|t|}{2}}^{\infty} \\
&\sim O(1) + O(\log |t|) + O(1) = O(\log |t|)
\end{aligned}$$

(b)

$$\begin{aligned}
\delta_n(s) &= \int_n^{n+1} \frac{1}{n^s} - \frac{1}{x^s} dx \\
\delta'_n(s) &= \int_n^{n+1} \frac{\log n}{n^s} - \frac{\log x}{x^s} dx
\end{aligned}$$

Then $|\delta'_n(s)| \leq 2 \frac{\log n}{n^\sigma}$ and $|\delta'_n(s)| \leq \frac{|s| \log n}{n^\sigma}$

$$\begin{aligned}
|\zeta'(1+it)| &= \left| \frac{1}{(it)^2} + H'(1+it) \right| \\
&\leq O(1) + \sum_{n=1}^{\infty} |\delta'_n(1+it)| \\
&\leq O(1) + \sum_{n < \frac{|t|}{2}} |\delta'_n(1+it)| + \sum_{n \geq \frac{|t|}{2}} |\delta'_n(1+it)| \\
&\leq O(1) + \sum_{n < \frac{|t|}{2}} \frac{2 \log n}{n} + \sum_{n \geq \frac{|t|}{2}} \frac{|1+it| \log n}{n^2} \\
&\leq O(1) + \sum_{n < \frac{|t|}{2}} \frac{2 \log n}{n} + \sum_{n \geq \frac{|t|}{2}} \frac{|it| \log n}{n^2} \\
&\sim O(1) + \int_1^{\frac{|t|}{2}} \frac{2 \log x}{x} dx + \int_{\frac{|t|}{2}}^{\infty} \frac{|t| \log x}{x^2} dx \\
&\sim O(1) + \log^2 x \Big|_1^{\frac{|t|}{2}} + O(1) \\
&\sim O(\log^2 |t|)
\end{aligned}$$

(c) It can be found that estimates in (a)(b) is still valid for $\sigma > 1$. By Corollary 1.5

$$|\zeta(\sigma+it)| \geq |\zeta(\sigma)|^{-\frac{3}{4}} |\zeta(\sigma+2it)|^{-\frac{1}{4}} \sim c'(\sigma-1)^{\frac{3}{4}} \log^{-\frac{1}{4}} |t|$$

When $\sigma-1 \geq \log^a |t|$,

$$|\zeta(\sigma+it)| \geq c' \log^{\frac{3}{4}a - \frac{1}{4}} |t|$$

When $\sigma-1 \leq \log^a |t|$, let $\sigma' = 1 + \log^a |t|$, by mean value theorem:

$$|\zeta(\sigma'+it) - \zeta(\sigma+it)| \leq c''(\sigma' - \sigma) \cdot A \log^2 |t| \leq c''(\sigma' - 1) \cdot A \log^2 |t|$$

$$\begin{aligned}
|\zeta(\sigma+it)| &\geq |\zeta(\sigma'+it)| - |\zeta(\sigma'+it) - \zeta(\sigma+it)| \\
&\geq c'(\sigma'-1)^{\frac{3}{4}} \log^{-\frac{1}{4}} |t| - c''(\sigma'-1) \cdot A \log^2 |t| \\
&= c' \log^{\frac{3}{4}a - \frac{1}{4}} |t| + c'' \log^{a+2} |t|
\end{aligned}$$

Let $\frac{3}{4}a - \frac{1}{4} = a + 2$, $a = -9$, then $|\zeta(\sigma+it)| \geq \log^{-7} |t|$ i.e.

$$\frac{1}{|\zeta(\sigma+it)|} \leq \log^7 |t|$$

Then let $\sigma \rightarrow 1$

$$\frac{1}{|\zeta(1+it)|} \leq \log^7 |t|$$

10. Integration by parts gives

$$\int_2^x \frac{dt}{\log(t)^m} = \frac{x}{\log(x)^m} - \frac{2}{(\log 2)^m} + m \int_2^x \frac{dt}{(\log t)^{m+1}}.$$

Additionally, we have

$$\int_2^x \frac{dt}{(\log t)^m} = \left(\int_2^{\sqrt{x}} + \int_{\sqrt{x}}^x \right) \frac{dt}{(\log t)^m} \leq C\sqrt{x} + C' \frac{1}{(\log \sqrt{x})^m} (x - \sqrt{x}) \leq \tilde{C} \frac{x}{(\log x)^m}.$$

Using these calculations with induction gives the results we wish to obtain.

11. (iv) implies (iii) implies (ii) all appear in the text. Assume (ii). Partial summation gives

$$\phi(x) = \log(x)\pi(x) + \sum_{y \leq x} \pi(y)(\log(y+1) - \log(y)).$$

Notice that

$$\left| \sum_{y \leq x} \pi(y)(\log(y+1) - \log(y)) \right| \leq C \sum_{y \leq x} \frac{\pi(y)}{y} \leq C \sum_{y \leq x^\epsilon} 1 + C' \sum_{x^\epsilon < y \leq x} \frac{1}{\log(y)} \leq Cx^\epsilon + C' \frac{x}{\log x}.$$

Thus we see that $\phi(x) \sim x$. We will show that (i) implies (iii) and that (iii) implies (iv). This finishes the result. Assume (iii). Then

$$\psi_1(x) = \int_1^x \psi(u) du = \int_1^{x^\epsilon} \psi(u) du + \int_{x^\epsilon}^x \psi(u) du = O(x^{2\epsilon} \log x) + \frac{1}{2}x^2 + O(x^{2\epsilon}) + o(x^2).$$

Assume (i) then we have $\sum_{p^n \leq x} \log(p) = \sum_{p \leq x^{1/n}} \log(p) \sim x^{1/n}$. So we have

$$\psi(x) \sim x + x^{1/2} + \dots + x^{1/n} + \dots$$

12. We have $\pi(p_n) = n \sim \frac{p_n}{\log p_n}$. Taking the logarithm of both sides we see that $\log n \sim \log p_n - \log \log p_n$. Hence $\log n \sim \log p_n$, because as $n \rightarrow \infty$, $\frac{\log \log p_n}{\log p_n} \rightarrow 0$. Returning to our first expression for n we have

$$n \sim \frac{p_n}{\log p_n} \sim \frac{p_n}{\log n}.$$

This gives $n \log n \sim p_n$, which is the desired result.

9.2 Problem

1. (a)

$$\frac{1}{2T} \int_{-T}^T (nm)^{-\sigma} n^{-it} m^{it} dt \rightarrow \begin{cases} n^{-2\sigma} & \text{for } n = m \\ 0 & \text{for } n \neq m \end{cases}$$

$$\begin{aligned}
& \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |F((\sigma + it))|^2 dt \\
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \left(\sum_{n=1}^{\infty} \frac{a_n}{n^{\sigma+it}} \right) \left(\sum_{m=1}^{\infty} \frac{\overline{a_m}}{m^{\sigma-it}} \right) dt \\
&= \lim_{T \rightarrow \infty} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{a_n \overline{a_m}}{2T} \int_{-T}^T (nm)^{-\sigma} n^{-it} m^{it} dt \\
&= \sum_{n=1}^{\infty} \frac{|a_n|^2}{n^{-2\sigma}}
\end{aligned}$$

(b) Since $F(s) \equiv 0$, $\sum_{n=1}^{\infty} \frac{|a_n|^2}{n^{-2\sigma}} = 0$, then $a_n = 0$. The condition $|a_n| \leq cn^k$ is to guarantee the convergence of $F(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$ for large s .

2. Recall:

$$\psi_1 = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{x^{s+1}}{s(s+1)} \left(-\frac{\zeta'(s)}{\zeta(s)} \right)$$

Choose a semi-circle contour like this and let the radius tends to infinity:

Calculate the residues:

Residue at $s = 1: \frac{x^2}{2}$, residue at $s = 0: -x \cdot \frac{\zeta'(0)}{\zeta(0)}$, residue at $s = -1: -\frac{\zeta'(-1)}{\zeta(-1)}$,

Residue at $s = -2n: \frac{x^{1-2n}}{2n(2n+1)}$, residue in critical strip at $\rho: -\frac{x^{\rho+1}}{\rho(\rho+1)}$.

Sum these residues we get the desired:

$$\psi_1(x) = \frac{x^2}{2} - \sum_{\rho} \frac{x^{\rho+1}}{\rho(\rho+1)} - \left(\frac{\zeta'(0)}{\zeta(0)} x + \frac{\zeta'(-1)}{\zeta(-1)} + \sum_{k=1}^{\infty} \frac{x^{1-2k}}{2k(2k-1)} \right)$$

Then we give some comments on why the integral vanishes on the half circle. $(z-1)\zeta(z)$ is of order 1, by Hademard factorization theorem, $(z-1)\zeta(z) = e^{az+b} \prod_{\rho} E_1\left(\frac{z}{\rho}\right)$. Then

$$\frac{1}{z-1} + \frac{\zeta'}{\zeta}(z) = a + \sum_{\rho} \frac{1}{\frac{z}{\rho} - 1} + \sum_{\rho} \frac{1}{\rho}$$

So $\frac{x^{s+1}}{s(s+1)} \left(-\frac{\zeta'(s)}{\zeta(s)} \right)$ integral on the half circle vanishes.

3.

$$\psi_1(x) = \frac{x^2}{2} - \sum_{\rho} \frac{x^{\rho+1}}{\rho(\rho+1)} - E(x)$$

Then

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - O(1)$$

Following statements are equivalent:

(1). $\pi(x) = Li(x) + O(x^{\alpha+\epsilon})$ for any $\epsilon > 0$;

(2). $\psi(x) = x + O(x^{\alpha+\epsilon})$ for any $\epsilon > 0$;

(3). $\zeta(s)$ has no zeros in $\alpha < \Re(s) < 1$.

Recall: $\pi(x) = \sum_{1 \leq p \leq x} 1$, $\psi(x) = \sum_{1 \leq p^m \leq x} \log p$, $\theta(x) = \sum_{1 \leq p \leq x} \log p$. Then $\psi(x) =$

$\sum_{n=1}^{\infty} \theta(x^{\frac{1}{n}})$, which indicates that

$$\psi(x) = x + O(x^{\alpha+\epsilon}) \Leftrightarrow \theta(x) = x + O(x^{\alpha+\epsilon})$$

(1) $\Rightarrow$ (2), Use Abel summation formula on $a(n) = \mathbb{K}_p(n)$ and $f(t) = \log t$

$$\begin{aligned} \theta(x) &= \sum_{1 \leq p \leq x} \log p \\ &= \sum_{1 \leq n \leq x} \mathbb{K}_p(n) \log n \\ &= \pi(x)f(x) - \int_2^x \pi(t)f'(t)dt \\ &= \pi(x)f(x) - \left[\pi(t)f(t) \Big|_2^x - \int_2^x \log t d\pi(x) \right] \\ &= O(1) + \int_2^x \log t d \left[\int_2^t \frac{ds}{\log s} + O(t^{\alpha+\epsilon}) \right] \\ &= O(1) + \int_2^x 1 + \log t O(t^{\alpha+\epsilon-1}) dt \\ &= x + O(1) + \int_2^x \log t O(t^{\alpha+\epsilon-1}) dt \\ &= x + O(x^{\alpha+\epsilon}) \end{aligned}$$

Then so is $\psi(x) = x + O(x^{\alpha+\epsilon})$.

(2) $\Rightarrow$ (1), Use Abel summation formula on $a(n) = \mathbb{K}_p(n) \log n$ and $f(t) = \frac{1}{\log t}$

$$\begin{aligned} \pi(x) &= \sum_{1 \leq p \leq x} \mathbb{K}_p(n) \\ &= \sum_{1 \leq n \leq x} \mathbb{K}_p(n) \log n \cdot \frac{1}{\log n} \\ &= \theta(x)f(x) - \int_2^x \theta(t)f'(t)dt \\ &= \theta(x)f(x) - \left[\theta(t)f(t) \Big|_2^x - \int_2^x \frac{1}{\log t} d\theta(t) \right] \\ &= O(1) + \int_2^x \frac{1}{\log t} d[t + O(t^{\alpha+\epsilon})] \\ &= O(1) + \int_2^x \frac{1}{\log t} (1 + O(t^{\alpha+\epsilon-1})) dt \\ &= Li(x) + O(1) + \int_2^x \frac{1}{\log t} O(t^{\alpha+\epsilon-1}) dt \\ &= Li(x) + O(x^{\alpha+\epsilon}) \end{aligned}$$

□□□

(2) $\Rightarrow$ (3), Let $R(x) = \psi(x) - x$

$$\begin{aligned} -\frac{\zeta'(s)}{\zeta(s)} &= \int_1^\infty t^{-s} d\psi(t) \\ &= \frac{1}{s-1} + \int_1^\infty t^{-s} dR(t) \\ &= \frac{1}{s-1} + 1 + s \int_1^\infty R(t) t^{-s-1} dt \end{aligned}$$

Since $R(t) = O(t^\alpha)$, then the formula above gives a meromorphic extension of $\zeta(s)$ in $\Re(s) > \alpha$ with only simple pole at $s = 1$.

(3) $\Rightarrow$ (2), $\psi(x) = x - \sum_\rho \frac{x^\rho}{\rho} - O(1)$ and $\Re(\rho) \leq \alpha$.

$$\begin{aligned} |\psi(x) - x| &= \left| \sum_\rho \frac{x^\rho}{\rho} \right| \\ &\leq |x^\rho| \sum_\rho \frac{1}{|\rho|} \\ &\leq Mx^\alpha \end{aligned}$$

$\sum_\rho \frac{1}{|\rho|} \leq M$ since ρ are zeros of $\zeta(s)$ and $(1-s)\zeta(s)$ is of order 1.

Remark 1: Able summation formula in this problem refers to:

$$\sum_{2 \leq n \leq x} a_n f(n) = \sum_{2 \leq n \leq x} a_n f(x) - \int_2^x \left(\sum_{2 \leq n \leq t} a_n \right) f'(t) dt$$

Remark 2: $-\frac{\zeta'(s)}{\zeta(s)} = \int_1^\infty t^{-s} d\psi(t)$ is because $-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^\infty \frac{\Lambda(n)}{n^s}$ and $\psi(x) = \sum_{1 \leq n \leq x} \Lambda(n)$.

10 Chapter 8

10.1 Exercise

1. $\Rightarrow$:by prop 1.1.

$\Leftarrow$:We denote $B_r(z_0) = \{z \mid |z - z_0| < r\}$. Let $z_0 \in U$. If $f'(z_0) \neq 0$, $f(z) - f(z_0)$ have only one root in $B_\delta(z_0)$ for small δ . Then we choose ω satisfies $|\omega - f(z_0)| < \min_{|z - z_0| = \delta} |f(z) - f(z_0)| = t$. Then by Rouché's Theorem, we know $f(z) - \omega = f(z) - f(z_0) - (\omega - f(z_0))$ have only one root in $B_\delta(z_0)$. Since f is continuous, we know there exists δ' satisfies $\forall |z - z_0| < \delta', |f(z) - f(z_0)| < t$. Let $\delta'' = \min\{\delta, \delta'\}$, then we know $B_{\delta''}(z_0)$ satisfies the requirements. $\square$

2. Since $f'' \neq 0$ near z_0 , we have $f = (z - z_0)^2 h(z)$ where $h \neq 0$ near z_0 . Let $g = (z - z_0)e^{\frac{1}{2} \text{Log} h} \Rightarrow g^2 = f$. Observe that $f''(z_0) = 2g'(z_0)^2 + 2g(z_0)g''(z_0) \neq 0 \Rightarrow g'(z_0) \neq 0$. Hence by ex 1 we know g is local bijection near z_0 , then define

$$\begin{cases} \Gamma_1 = g^{-1}(t), t \geq 0 \\ \Gamma_2 = g^{-1}(it), t \geq 0 \end{cases} \Rightarrow \begin{cases} F(\Gamma_1(t)) = t^2, t \geq 0 \\ F(\Gamma_2(t)) = -t^2, t \geq 0 \end{cases} \quad (13)$$

Then we draw the conclusion. $\square$

3. $f : U \rightarrow V$ is a conformal equivalence and let γ_1 and γ_2 be two paths from $z_0 \rightarrow z_1$ in V . Then use the conformal equivalence to pull back the paths to paths in U , then find a homotopy in U and then push them forward with f to a homotopy in V . The proof of the condition that f is a continuous bijection is similar. $\square$

4. Follow the hint. First use F given in the chapter to find a map from $\mathbb{D} \rightarrow \mathbb{H}$, then shift H down with the map $z \mapsto z - i$ and then compose with squaring. Recall that squaring will double the argument and thus fill out the entire plane. So the map is $z \mapsto (F(z) - i)^2$. $\square$

5. From the hint we know that the solutions to $f(z) = w$ are $z = -w \pm \sqrt{w^2 - 1}$. Let these solutions be z_+ and z_- . Since their product is 1, we know that one of them lies inside of $\bar{\mathbb{D}}$. Suppose z^* is that solution, then since $w \in \mathbb{H}$ we wish to prove that $z^* \in \bar{\mathbb{D}} \cap \mathbb{H}$. To see this notice that

$$\text{Im}(f(z)) = -1/2 (\text{Im}(z) + \text{Im}(1/z)) = -1/2 \left(r - \frac{1}{r} \right) \sin(\theta)$$

where $z = re^{i\theta}$. Therefore, if $f(z) = w \in \mathbb{H}$, then $\text{Im}(f(z)) > 0$, so we see that for $z^* \in \bar{\mathbb{D}}$, we have $r \leq 1$, thus $\sin(\theta) > 0$, so $\theta \in (0, \pi)$. We also observe from this argument that we must have $z^* \in \mathbb{D}$, thus $z^* \in \mathbb{D} \cap \mathbb{H}$. $\square$

6. Just calculate it directly. $\square$

7. (a) For first part we just verify it directly. Since $r = \pm \frac{\cos \pi y}{1 + \sin \pi y}$, we have $r^2 = \frac{1 - \sin^2 \pi y}{(1 + \sin \pi y)^2} = \frac{1 - \sin \pi y}{1 + \sin \pi y}$. And for $\theta = \frac{\pi}{2}$ we have

$$P_r(\theta - \varphi) = \frac{\frac{2 \sin \pi y}{1 + \sin \pi y}}{1 - 2 \sin \varphi \frac{\cos \pi y}{1 + \sin \pi y} + \frac{1 - \sin \pi y}{1 + \sin \pi y}} = \frac{\sin \pi y}{1 - \cos \pi y \sin \varphi}$$

Similarly for $\theta = -\frac{\pi}{2}$. □

(b) Since $t = F(e^{i\varphi}) \Rightarrow e^{i\varphi} = G(t) = \frac{i - e^{\pi t}}{i + e^{\pi t}}$. Take the imaginary part we get

$$\begin{aligned} \sin \varphi &= \frac{1}{\cosh \pi t} \Rightarrow \varphi = \pi - \arcsin \frac{1}{\cosh \pi t}, \\ \frac{d\varphi}{dt} &= \frac{1}{\sqrt{1 - \frac{1}{\cosh^2 \pi t}}} \frac{1}{\cosh^2 \pi t} \frac{\pi(e^{\pi t} - e^{-\pi t})}{2} = \frac{\pi}{\cosh \pi t} \end{aligned}$$

(Here $\varphi = \pi - \arcsin \frac{1}{\cosh \pi t}$ because we assume $\frac{d\varphi}{dt} > 0$.)

Hence

$$\begin{aligned} \frac{1}{2\pi} \int_0^\pi P_r(\theta - \varphi) \tilde{f}_0(\varphi) d\varphi &= \frac{1}{2\pi} \int_0^\pi \frac{\sin \pi y}{1 - \cos \pi y \sin \varphi} \tilde{f}_0(\varphi) d\varphi = \\ &= \frac{\sin \pi y}{2} \int_{-\infty}^{+\infty} \frac{f_0(t)}{\cosh \pi t - \cos \pi y} dt \end{aligned}$$

□

(c) The same as (b) □

8. Using the hint we have $F_1(z) = \frac{z-1}{z+1}$, $F_2(z) = \log(z)$, $F_3(z) = e^{-i\pi/2}z - \frac{\pi}{2}$, $F_4(z) = \sin(z)$, and $F_5(z) = z - 1$ □

9. u is the real part of a holomorphic function on $\mathbb{D}$, thus u is harmonic. A calculation shows that for $|z| = 1$, we have $\bar{z} = \frac{1}{z}$, a straight forward calculation gives

$$2 \operatorname{Re} \left(\frac{i+z}{i-z} \right) = \frac{i+z}{i-z} + \frac{\overline{i+z}}{\overline{i-z}} = \frac{i+z}{i-z} + \frac{i+z}{-i+z} = 0.$$

□

10. Let $T : \mathbb{D} \rightarrow \mathbb{H}$ be given by $T(z) = i \frac{1+z}{1-z}$. Then we see that $F \circ T : \mathbb{D} \rightarrow \mathbb{D}$ and it is holomorphic in $\mathbb{D}$, with $F(T(0)) = F(i) = 0$. Thus by the Schwarz Lemma we have $|F(T(z))| \leq |z|$, for $z \in \mathbb{D}$. Then we have

$$|F(z)| = |F \circ T \circ T^{-1}(z)| \leq |T^{-1}(z)| = \left| \frac{z-i}{z+i} \right|$$

□

11. Consider $g(z) = \frac{1}{M}f(Rz) : \mathbb{D} \rightarrow \mathbb{D}$, then it suffices to show that:

$$\left| \frac{g(z) - g(0)}{1 - \overline{g(0)}g(z)} \right| \leq |z|$$

and this inequality is a direct conclusion of Schwarz lemma. $\square$

12. (a) Suppose we have two fixed points z_1 and z_2 in the unit disc. Consider the automorphism of disc $h_1(z) = \frac{z_1 - z}{1 - \overline{z_1}z}$, then $g(z) = h_1 \circ f \circ h_1^{-1}$ has two fixed points 0 and $\omega = h_1(z_2)$. By Schwarz lemma, $g(0) = 0$ and $|g(\omega)| = |\omega|$, then g is a rotation, then since $g(\omega) = \omega$, g is an identity map, so is f . $\square$

(b) No, by the conformal mapping between $\mathbb{D}$ and $\mathbb{H}$, we consider the case in the upper half-plane. However, an automorphism of the upper half-plane without a fixed point is trivial: $h(z) = z + 1$. Recall two conformal mappings: $F(\omega) = i\frac{1-\omega}{1+\omega}$ and $G(z) = \frac{i-z}{i+z}$. The $\tilde{g} = F \circ g \circ G$ is the desired automorphism of $\mathbb{D}$ without the fixed point. $\square$

13. (a) Consider the automorphism of the unit disc $\psi_\alpha(z) = \frac{z-\alpha}{1-\overline{\alpha}z}$. Then the inequality we want to prove is $|\psi_{f(w)} \circ f(z)| \leq |\psi_w(z)|$, which is equivalent to $|\psi_{f(w)} \circ f \circ \psi_w^{-1}(z)| \leq |z|$. Above inequality is obvious by Schwarz lemma. As for the automorphism case, we both have $\rho(f(z), f(w)) \leq \rho(z, w)$ and $\rho(f^{-1}(z), f^{-1}(w)) \leq \rho(z, w)$. so the equality holds. $\square$

(b) In (1), it shows that $\left| \frac{f(z)-f(w)}{1-\overline{f(w)}f(z)} \right| \leq \left| \frac{z-w}{1-\overline{w}z} \right|$ i.e. $\left| \frac{f(z)-f(w)}{z-w} \right| \leq \left| \frac{1-\overline{f(w)}f(z)}{1-\overline{w}z} \right|$ and let $w \rightarrow z$ we get the desired conclusion. $\square$

14. Since we have known that every conformal mapping from $\mathbb{D} \rightarrow \mathbb{D}$ can be written as the form $e^{i\theta} \frac{\alpha-z}{1-\overline{\alpha}z}$. Take any conformal mapping $f : \mathbb{H} \rightarrow \mathbb{D}$, we composite it with $g = i\frac{1-\omega}{1+\omega} : \mathbb{D} \rightarrow \mathbb{H}$. Then $f \circ g$ can be written as $e^{i\theta} \frac{\alpha-z}{1-\overline{\alpha}z}$. By simple computation we draw the conclusion. $\square$

15. (a) Φ has the form $\Phi(z) = \frac{az+b}{cz+d}$. If x_1, x_2, x_3 is three fixed point. Then we have

$$ax_1 + b = x_1(cx_1 + d)$$

$$ax_2 + b = x_2(cx_2 + d)$$

$$ax_3 + b = x_3(cx_3 + d)$$

Then x_1, x_2, x_3 is three distinct roots for $cx^2 + (d-a)x - b = 0$. So $c = b = 0$ and $d = a$, then Φ is the identity. $\square$

(b) Given x_i and y_i , solve for a :

$$\frac{y_3 - y_1}{y_2 - y_1} = \frac{-\frac{1}{x_3+a} + \frac{1}{x_1+a}}{-\frac{1}{x_2+a} + \frac{1}{x_1+a}}$$

This is a linear function for a so the solution is trivial. Consider:

$$\Phi(z) = \frac{y_2 - y_1}{-\frac{1}{x_2 + a} + \frac{1}{x_1 + a}} \left(-\frac{1}{z + a} + \frac{1}{x_1 + a} \right) + y_1$$

By the choice of a , Φ maps x_i to y_i , and the construction of Φ shows that Φ is the composition of a sequence of automorphism on the upper half plane. $\square$

16. It is trivial by Theorem 2.4. $\square$

17. The first part follows since

$$\pi = \text{Area}(\mathbb{D}) = \iint_{\psi(\mathbb{D})} dx dy = \iint_{\mathbb{D}} |\psi'_\alpha(z)|^2 dx dy.$$

For the second we need a lemma.

Lemma. Let f be an analytic function in $|z| < R$, $f(z) = \sum_{n=0}^{\infty} a_n z^n$, then $\forall 0 < r < R$,

$$\frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^2 d\theta = \sum_{n=0}^{\infty} |a_n|^2 r^{2n} \quad (14)$$

proof. Notice that

$$\frac{1}{2\pi} \int_{|z|=r} z^m \bar{z}^n dz = \frac{r^{m+n}}{2\pi} \int_0^{2\pi} e^{i(m-n)\theta} d\theta = \begin{cases} 0, m \neq n \\ r^{2n}, m = n \end{cases} \quad (15)$$

By Abel theorem we have $f(z) = \sum_{n=0}^{\infty} a_n z^n$ converging absolutely so we can multiply $f \cdot \bar{f}$ in Cauchy and do a termwise integral, hence we have

$$\begin{aligned} \frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^2 d\theta &= \frac{1}{2\pi} \int_{|z|=r} f(z) \overline{f(z)} dz = \\ &= \frac{1}{2\pi} \int_{|z|=r} \sum_{n=0}^{\infty} a_n z^n \sum_{m=0}^{\infty} \bar{a}_m \bar{z}^m dz = \\ &= \frac{1}{2\pi} \sum_{m,n=0}^{\infty} a_m \bar{a}_n \int_{|z|=r} z^m \bar{z}^n dz = \sum_{n=0}^{\infty} |a_n|^2 r^{2n} \end{aligned} \quad (16)$$

So use this lemma, we have

$$\begin{aligned} \frac{1}{\pi} \iint_{\mathbb{D}} |\psi'| dx dy &= \frac{1 - |\alpha|^2}{\pi} \int_0^1 \int_0^{2\pi} \frac{r}{|1 - \bar{\alpha} r e^{i\theta}|^2} d\theta dr = \\ 2(1 - |\alpha|^2) \int_0^1 \sum_{n=0}^{\infty} |\alpha|^{2n} r^{2n+1} dr &= (1 - |\alpha|^2) \int_0^1 \frac{dr^2}{1 - |\alpha|^2 r^2} = \\ &= \frac{1 - |\alpha|^2}{|\alpha|^2} \log \frac{1}{1 - |\alpha|^2} \end{aligned} \quad (17)$$

18. Follow the proof and lemmas of Theorem 4.2 □

19. Choose a closed curve γ , we can divide it into pieces by $A_k + iy, y \in \mathbb{R}$. If γ is in UHP , there is nothing to prove. Otherwise, we denote $\gamma_k = \gamma \cap \{x + iy | A_k \leq x \leq A_{k+1}, y \leq 0\}$. It is obviously the strip given by A_k and A_{k+1} is simply connected. Hence we can raise γ_k to UHP . Then we know the closed curve can be retracted into a point. Hence the former space is simply connected. □

20. (a) Following the notation of Proposition 4.1 we have $A_1 = 0, A_2 = 1$, and $A_3 = \lambda$, when $\lambda > 1$, the other cases $0 < \lambda < 1$ and $\lambda < 0$ are handled similarly. In any case we have $\beta_1 = \beta_2 = \beta_3 = \beta_\infty = \frac{1}{2}$. To see that we have a rectangle it is enough to notice that $\alpha_1 = \alpha_2 = \alpha_3 = \alpha_\infty = \frac{1}{2}$ □

(b)

$$\begin{aligned} \int_0^1 \frac{d\zeta}{\sqrt{\zeta(\zeta^2 - 1)}} &= \frac{1}{2} \int_0^1 (1-u)^{-1/2} u^{-3/4} du \\ &= \frac{1}{2} \frac{\Gamma(1/2)\Gamma(1/4)}{\Gamma(3/4)} \\ &= \frac{1}{2\sqrt{2}\pi} \Gamma(1/4)^2, \end{aligned}$$

where we use the calculation of the beta function and then the fact that $\Gamma(3/4)\Gamma(1/4) = \frac{\pi}{\sin(\pi/4)} = \pi\sqrt{2}$ and $\Gamma(1/2) = \sqrt{\pi}$. We need only to calculate the length of one of the perpendicular sides which can be done similarly. □

21. (a) It follows immediately from Proposition 4.1. □

(b)(c) It will be a broken line of three pieces since the integral is not convergent at ∞ . □

(d) we have the following similar calculation

$$\begin{aligned} \int_0^1 z^{-\beta_1} (1-z)^{-\beta_2} dz &= \frac{\Gamma(1-\beta_1)\Gamma(1-\beta_2)}{\Gamma(2-\beta_1-\beta_2)} = \frac{\Gamma(\alpha_1)\Gamma(\alpha_2)}{\Gamma(\beta_3)} \\ &= \frac{\sin(\alpha_3\pi)}{\pi} \Gamma(\alpha_1)\Gamma(\alpha_2)\Gamma(\alpha_3). \end{aligned}$$

Where we use $\Gamma(\beta_3)\Gamma(1-\beta_3) = \frac{\pi}{\sin((1-\beta_3)\pi)}$. A similar calculation with the change of variables $u = \frac{1}{z}$ can be used to compute the integral

$$\left| \int_1^\infty z^{-\beta_1} (1-z)^{-\beta_2} dz \right| = \frac{\Gamma(\alpha_2)\Gamma(\alpha_3)}{\Gamma(\beta_1)}$$

Similar calculations finish the problem. □

22. We give a conformal mapping $g = \frac{i-z}{i+z} : \mathbb{H} \rightarrow \mathbb{D}$. WLOG we assume $B_i \neq \pm 1$. Hence no

vertices is the image of the infy. Then we know $F \circ g : \mathbb{H} \rightarrow P$ can be written as

$$F \circ g = c'_1 \int_0^z \frac{dt}{\prod_{i=1}^n (t - A_i)^{\beta_i}} + c'_2$$

With $\sum_{i=1}^n \beta_i = 2$. Let $t = g^{-1}(\zeta) \Rightarrow$

$$\begin{aligned} F(g(z)) &= c_1 \int_1^{g(z)} \frac{d\zeta}{(1 + \zeta)^2 \prod_{i=1}^n (i \frac{1-\zeta}{1+\zeta} - A_i)^{\beta_i}} + c_2 = \\ &= c_1 \int_1^{g(z)} \frac{d\zeta}{(1 + \zeta)^2 \prod_{i=1}^n (\zeta - B_i)^{\beta_i}} + c_2 \end{aligned}$$

With $|B_i| = 1$.

If some $B_i = \pm 1$, then we follow the proof of theorem 4.7 and similarly we draw the conclusion. $\square$

23. Since $\beta_1 = \beta_2 = \dots \beta_n$ and by ex 22, we know the F maps the unit disc conformally onto the interior of a regular polygon. Let $\zeta = e^{i\theta}$, we have

$$\begin{aligned} \left| \int_1^{e^{\frac{2\pi}{n}}} (1 - \zeta^n)^{\frac{-2}{n}} d\zeta \right| &= \left| \int_1^{e^{\frac{2\pi}{n}}} (1 - e^{in\theta})^{\frac{-2}{n}} de^{i\theta} \right| = \\ \left| \int_1^{e^{\frac{2\pi}{n}}} (e^{-\frac{1}{2}in\theta} - e^{\frac{1}{2}in\theta})^{\frac{-2}{n}} d\theta \right| &= 2^{-\frac{2}{n}} \int_0^{\frac{2\pi}{n}} \sin \frac{n}{2} \theta d\theta = \\ &= \frac{1}{n} 2^{\frac{n-2}{n}} \int_0^{\pi} (\sin \theta)^{\frac{-2}{n}} d\theta \end{aligned}$$

The computation of the other side is same. Hence the perimeter is $2^{\frac{n-2}{n}} \int_0^{\pi} (\sin \theta)^{\frac{-2}{n}} d\theta$. $\square$

24. (a) We change the variables $x = (1 - \tilde{k}^2 y^2)^{-\frac{1}{2}}$ and we have

$$K'(k) = \int_0^1 \frac{(1 - \tilde{k}^2 y^2)}{\sqrt{\tilde{k}^2 y^2 (1 - k^2 - \tilde{k}^2 y^2)}} \cdot \frac{1}{2} (1 - \tilde{k}^2 y^2)^{\frac{-3}{2}} y \tilde{k} dy = K(\tilde{k})$$

$\square$

(b) Just follow the hint and calculate it like (a). $\square$

(c) By ex 9 of chapter 6, let $t = x^2$ and we have

$$\frac{\pi}{2} F\left(\frac{1}{2}, \frac{1}{2}, 1; k^2\right) = \frac{1}{2} \int_0^1 \frac{dt}{\sqrt{t(1-t)(1-k^2 t)}} = K(k)$$

$\square$

10.2 Problem

2. (a) Observe that

$$\begin{aligned} \frac{([f(\gamma(t_0))]', [f(\eta(t_0))]')}{|[f(\gamma(t_0))]'[f(\eta(t_0))]'|} &= \frac{(f'(z_0)\gamma'(t_0), f'(z_0)\eta'(t_0))}{|[f(\gamma(t_0))]'[f(\eta(t_0))]'|} = \frac{|f'(z_0)|^2(\gamma'(t_0), \eta'(t_0))}{|f'(z_0)|^2|\gamma'(t_0)\eta'(t_0)|} = \frac{(\gamma'(t_0), \eta'(t_0))}{|\gamma'(t_0)\eta'(t_0)|}. \\ \frac{([f(\gamma(t_0))]', -i[f(\eta(t_0))]')}{|[f(\gamma(t_0))]'[f(\eta(t_0))]'|} &= \frac{(f'(z_0)\gamma'(t_0), -if'(z_0)\eta'(t_0))}{|[f(\gamma(t_0))]'[f(\eta(t_0))]'|} = \frac{|f'(z_0)|^2(\gamma'(t_0), -i\eta'(t_0))}{|f'(z_0)|^2|\gamma'(t_0)\eta'(t_0)|} = \frac{(\gamma'(t_0), -i\eta'(t_0))}{|\gamma'(t_0)\eta'(t_0)|}. \square \end{aligned}$$

(b) Choose $\gamma(t) = z_0 + t$, $\eta(t) = z_0 + it$. Since $f = u + iv$ preserve angles we have

$$\begin{aligned} f(\gamma(t)) &= u(x_0 + t, y_0) + iv(x_0 + t, y_0), f(\eta(t)) = u(x_0, y_0 + t) + iv(x_0, y_0 + t) \Rightarrow \\ &< \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}, \frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y} > = 0, \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right) / \left(\frac{\partial v}{\partial y}, -\frac{\partial u}{\partial y} \right) \Leftrightarrow \left(\frac{\partial u}{\partial x}, \frac{\partial v}{\partial x} \right) = k \left(\frac{\partial v}{\partial y}, -\frac{\partial u}{\partial y} \right) (k > 0) \end{aligned}$$

(here we use $<, >$ to denote inner product)

And by problem 1 we know

$$\lim_{t \rightarrow 0} \frac{f(\gamma(t)) - f(z_0)}{t} = \lim_{t \rightarrow 0} \frac{f(\eta(t)) - f(z_0)}{t} \Rightarrow \left| \left(\frac{\partial u}{\partial x}, \frac{\partial v}{\partial x} \right) \right| = \left| \left(\frac{\partial v}{\partial y}, -\frac{\partial u}{\partial y} \right) \right| \Rightarrow k = 1$$

Hence f satisfies C.-R. equation at z_0 . Then we draw the conclusion. $\square$

3. (a) By Schwarz-Pick lemma,

$$\begin{aligned} d(f(z_1), f(z_2)) &= \inf_{\eta: f(z_1) \rightarrow f(z_2)} \int_0^1 \frac{|\eta'(t)|}{1 - |\eta(t)|^2} dt \leq \inf_{f \circ \gamma: \gamma: z_1 \rightarrow z_2} \int_0^1 \frac{|[f(\gamma(t))]'|}{1 - |f(\gamma(t))|^2} dt \leq \\ &\inf_{\gamma: z_1 \rightarrow z_2} \int_0^1 \frac{|\gamma'(t)|}{1 - |\gamma(t)|^2} dt = d(z_1, z_2), z_1, z_2 \in \mathbb{D} \end{aligned}$$

$\square$

(b) By ex 13 we know in Schwarz-Pick lemma the equality occurs when $f \in \text{Aut}(\mathbb{D})$. Hence the second inequality above is equality. Since the f^{-1} is conformal map, we know $\{\eta : f(z_1) \rightarrow f(z_2)\} = \{f \circ \gamma : \gamma : z_1 \rightarrow z_2\} \Rightarrow$ the first inequality is equality. $\square$

(c) We know there exists $f(z) \in \text{Aut}(\mathbb{D})$ s.t. $f(z_1) = 0$, $f(0) = z_1$. Assume $f(z_2) = se^{i\theta}$. Then we take $\varphi(z) = e^{-i\theta} f(z)$. $\square$

(d) Let $\gamma : 0 \rightarrow s$, $\gamma(t) = \gamma_1(t) + i\gamma_2(t) \Rightarrow \gamma_1(\mathbb{R} \rightarrow \mathbb{R}) : 0 \rightarrow s$. Notice that

$$\begin{aligned} |\gamma'(t)| &\geq |\gamma_1'(t)|, |\gamma(t)| \geq |\gamma_1(t)| \Rightarrow \\ \int_0^1 \frac{|\gamma'(t)|}{1 - |\gamma(t)|^2} dt &\geq \int_0^1 \frac{|\gamma_1'(t)|}{1 - |\gamma_1(t)|^2} dt = \int_0^1 \frac{|d\gamma_1(t)|}{1 - |\gamma_1(t)|^2} \geq \int_0^s \frac{t}{1 - t^2} dt = \frac{1}{2} \log \frac{1+s}{1-s} \end{aligned}$$

Let $\gamma(t) = st$, we know the equality occurs. Hence

$$d(0, s) = \frac{1}{2} \log \frac{1+s}{1-s}$$

□

(e) By (c) we know $s = \left| \frac{z_1 - z_2}{1 - \bar{z}_1 z_2} \right| \Rightarrow$

$$d(z_1, z_2) = d(0, s) = \frac{1}{2} \log \frac{|1 - \bar{z}_1 z_2| + |z_1 - z_2|}{|1 - \bar{z}_1 z_2| - |z_1 - z_2|}$$

□

4. (a) From the definition we know that

$$\begin{aligned} |b|^2 - |d|^2 &= -1, |a|^2 - |c|^2 = 1, a\bar{b} = c\bar{d}, ad - bc = 1 \Rightarrow \\ a &= \frac{bc + 1}{d} \Rightarrow \frac{bc + 1}{d} \bar{b} = c\bar{d} \Rightarrow c = \bar{b} \end{aligned}$$

Hence all matrices in $SU(1, 1)$ are of the form $\begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix}$, $|a|^2 - |b|^2 = 1$

□

(b) Let f_M be the associated fractional linear transform of $M \in SU(1, 1)$.

From direct calculation, we can obtain $f_M \circ f'_M = f_{MM'}$. Observe that every $\varphi = e^{i\theta} \frac{z - \alpha}{1 - \bar{\alpha}z} \in \text{Aut}(\mathbb{D})$ can be written as f_M where

$$M = \begin{pmatrix} \frac{e^{\frac{1}{2}i\theta}}{\sqrt{1-|\alpha|^2}} & -\frac{\alpha e^{\frac{1}{2}i\theta}}{\sqrt{1-|\alpha|^2}} \\ -\frac{e^{\frac{1}{2}i\theta}}{\sqrt{1-|\alpha|^2}} & \frac{e^{\frac{1}{2}i\theta}}{\sqrt{1-|\alpha|^2}} \end{pmatrix}$$

By (a), it is not difficult to know that every $M \in SU(1, 1)$ can be written as the form above (let $|a| = \frac{1}{\sqrt{1-|\alpha|^2}}$ and compute α). Hence by the homomorphism $M \rightarrow f_M$ we get

$$SU(1, 1)/\{\pm 1\} \cong \text{Aut}(\mathbb{D})$$

□

5. (a) Obviously $\{r_n\}$ have a convergent subsequence $\{r_{k_n}\}$. If $r_{k_n} \rightarrow c$ and $c \neq 0$, then $F = 0$ is trivial. If any subsequences converge to 0, then we have $r_n \rightarrow 0$. If $F \neq 0$, we know $\exists t \in \mathbb{R}_+$, $F(it) \neq 0$. WLOG we assume $F(i) \neq 0$ (or we can choose such t and consider $F(tz)$). Let Consider $g(z) = i \frac{1-z}{1+z}$ which is a conformal map from $\mathbb{D}$ to $\mathbb{H}$. Let $f = F \circ g : \mathbb{D} \rightarrow \mathbb{C}$. Since $F(ir_k) = 0 \Rightarrow f(z_k) = 0$, $z_k = 1 - \frac{2r_k}{1+r_k}$. Denote $\{t_n\}$ the zero point of f . Since F is bounded, we have $\log |f(0)| = \log |F(i)| \neq \infty$ and $|f| \leq M$. Choose $R < 1$ and f vanishes nowhere in C_R . We

use Jensen's formula.

$$\log |f(0)| = \sum_{|t_k| < R} \log \frac{|t_k|}{R} + \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta$$

It is easy to prove that $x - \log x > 1, x < 1 \Rightarrow$

$$\begin{aligned} C \sum_{n=1}^{\infty} r_n &\leq \sum_{n=1}^{\infty} \frac{2r_n}{1+r_n} = \sum_{n=1}^{\infty} (1 - |z_n|) < -\lim_{R \rightarrow 1} \sum_{|z_n| < R} \log |z_n| \leq -\lim_{R \rightarrow 1} \sum_{|t_n| < R} \log |t_n| = \\ &\lim_{R \rightarrow 1} \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta - \log |f(0)| \leq M - \log |f(0)| < +\infty \end{aligned}$$

Contradiction! □

(b) Use the notation of (a). Consider

$$f(z) = \prod_{n=1}^{\infty} \frac{z_n - z}{1 - \bar{z}_n z} \frac{|z_n|}{z_n}$$

By Problem 2 of chapter 5, we know $f(z)$ defines a holomorphic function on the unit disc having precisely the zeros z_n and f is bounded. Hence $F = f \circ g^{-1}$ is the desired function. □

6. Actually it is Caratheodory's Theorem. □

7. (a) Denote D_k the $D(0, r_k)$. Since $D_k \subset f(D_k)$ and they are all simply connected, obviously $D_k \subset D_{r(k)}$. Write $f(z) = zg(z) \Rightarrow f'(z) = g(z) + zg'(z), f'(0) = g(0)$. Since f is injective, it has no other zero point except 0. Take $r < r_k \Rightarrow$

$$\left| \frac{z}{f(z)} \right| < \frac{r}{r} = 1, |z| = r$$

Hence $|\frac{1}{g(z)}| < 1, |z| \leq r$ by maximal modulus principle. So $|f'(0)| > 1$ □

(b) Obviously, $|F_n(z)| > |F_{n-1}(z)| > \dots > |z|$ and $F_n(0) = 0$. So it is an expansion. By chain rule $F'_n(0) = \prod_{k=0}^n f'_k(0)$. Since $F_n(r_{k_0}z) : \mathbb{D} \rightarrow \mathbb{D}$, by Schwarz lemma we have $|F_n(z)| \leq |\frac{z}{r_{k_0}}|, |z| \leq r_{k_0} \Rightarrow |F'_n(0)| \leq \frac{1}{r_{k_0}}$. So $\{|F_n(0)|\}$ is bounded and monotone increasing. So $\lim_{n \rightarrow +\infty} |f'_n(0)| = 1$. □

(d)

$$|f'(0)| = \left| \frac{|\beta|^2 - 1}{(1 - \bar{\beta}z)^2} \right|_{z=\beta} \frac{1}{2\sqrt{x}} \Big|_{x=\alpha} \frac{|\alpha|^2 - 1}{(1 - \bar{\alpha}z)^2} \Big|_{z=0} = \frac{1+r_k}{2\sqrt{r_k}}$$

8. For convenience we denote $a_1 = 1$.

(a) $v(r, \theta) = \sum_{i=1}^{\infty} a_i r^i \sin i\theta$. Observe that

$$\frac{2}{\pi} \int_0^{\pi} \sin m\theta \sin n\theta d\theta = \begin{cases} 1, m = n \\ 0, m \neq n \end{cases}$$

Hence

$$a_n r^n = \frac{2}{\pi} \int_0^\pi v(r, \theta) \sin n\theta d\theta$$

□

(b) Since $|\sin \theta| \leq \sin \theta$, $\theta \in [0, \pi]$, we assume $|\sin n\theta| \leq n \sin \theta$. Then $|\sin(n+1)\theta| = |\sin \theta \cos n\theta + \sin n\theta \cos \theta| \leq |\sin \theta| + |\sin n\theta| \leq (n+1) \sin \theta$. By induction we draw the conclusion. □

(c) Since $a_n \in \mathbb{R}$, $f(\bar{z}) = \sum_{n=1}^\infty a_n \bar{z}^n = \overline{f(z)}$, $f(\mathbb{D})$ is symmetric with respect to the real axis. Let $z_0 \in UHP \cap \mathbb{D}$. Assume $f(z_0) \in UHP$. If there exists z_1 s.t. $f(z_1) \notin UHP$, then we can choose a smooth curve $\gamma \subset UHP$ from z_0 to z_1 . Since f is continuous, there exists $z' \in \gamma$ s.t. $f(z') \in \mathbb{R} \Rightarrow f(\bar{z}') = f(z')$. However, f is injective, contradiction! □

(d) Assume the radius of convergence of $f'(z)$ is R , then we know $\overline{\lim}_{n \rightarrow +\infty} \frac{1}{n^{-1} \sqrt[n]{|na_n|}} = \frac{1}{R}$. Hence we can choose M, N which satisfy $\max_{n \in [1, N]} |na_n| \leq M$ and $\forall n > N, |na_n| \leq C \frac{1}{R^{n-1}}$ for some

C . Choose $r < R$. By (b) we have $|v(r, \theta)| \leq \sum_{n=1}^\infty |na_n| r^n \sin \theta \Rightarrow$

$$\frac{|v(r, \theta)|}{r \sin \theta} \leq 1 + \sum_{n=2}^\infty |na_n| r^{n-1} \leq 1 + M \sum_{n=2}^N r^{n-1} + C \sum_{n=N+1}^\infty \left(\frac{r}{R}\right)^n = 1 + O(r).$$

So

$$v(r, \theta) = r \sin \theta [1 + O(r)]$$

Hence if r is small, we know $v(r, \theta) > 0$, $\theta \in (0, \pi)$. We choose such $z = re^{i\theta} \in UHP$ and we know $f(z) \in UHP$. By (c) we have $f(\mathbb{D} \cap UHP) \subset UHP \Rightarrow v(r, \theta) \sin \theta \geq 0$, $\theta \in [0, \pi]$, $r \in (0, 1)$. □

(e) By (a), (b), (d) we know that

$$|a_n r^n| \leq \frac{2n}{\pi} \int_0^\pi |v(r, \theta) \sin \theta| d\theta = n \frac{2}{\pi} \int_0^\pi v(r, \theta) \sin \theta d\theta = nr, r \in (0, 1), \theta \in [0, \pi].$$

Let $r \rightarrow 1$, we have $|a_n| \leq n$. □

(f)

$$\begin{aligned} \frac{1}{1-z} &= 1 + z + z^2 + \dots, |z| < 1 \Rightarrow \\ \left(\frac{1}{1-z}\right)' &= \frac{1}{(1-z)^2} = 1 + 2z + 3z^2 + \dots \Rightarrow \\ f(z) &= z + 2z^2 + 3z^3 + \dots, |z| < 1 \end{aligned}$$

Obviously, $f(0) = 0$, $f'(0) = 1$. Let $\omega \neq 0$, then $f(z) = \omega \Leftrightarrow$

$$z + \frac{1}{z} - 2 = \frac{1}{\omega}$$

So two solutions z_1, z_2 of the equation above must satisfy $|z_1 z_2| = 1$. Since f is defined in $\mathbb{D}$, f is injective holomorphic function. $\square$

9.(a) Obviously, $a_n \geq b_n$. Since

$$\begin{aligned} a_{n+1} &= \frac{a_n + b_n}{2} \leq \frac{a_n + a_n}{2} = a_n \\ b_{n+1} &= \sqrt{a_n b_n} \geq b_n \end{aligned}$$

and

$$\begin{aligned} \frac{a_{n+1} + b_{n+1}}{2} - \sqrt{a_{n+1} b_{n+1}} &= \frac{1}{2} \left(\frac{a_n + b_n}{2} - \sqrt{a_n b_n} \right) + \sqrt{a_n b_n} - \sqrt{\frac{a_n + b_n}{2} \sqrt{a_n b_n}} \leq \\ &\frac{1}{2} \left(\frac{a_n + b_n}{2} - \sqrt{a_n b_n} \right) \leq \dots \leq \frac{a - b}{2^{n+2}}, \end{aligned}$$

We have $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n$. $\square$

(b) Let $x = \sin \theta$, then

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}} &= \frac{1}{a} \int_0^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{\cos^2 \theta + \frac{b^2}{a^2} \sin^2 \theta}} = \\ \frac{1}{a} \int_0^1 \frac{dx}{\sqrt{(1 - x^2 + \frac{b^2}{a^2} x^2)(1 - x^2)}} &= \frac{1}{a} \int_0^1 \frac{dx}{\sqrt{(1 - k^2 x^2)(1 - x^2)}} \end{aligned}$$

Use the notation of ex 24, we have $\tilde{k} = \frac{b}{a}$. Since

$$I\left(\frac{a+b}{2}, \sqrt{ab}\right) = \frac{2}{a+b} \int_0^1 \frac{dx}{\sqrt{(1 - \frac{(a-b)^2}{(a+b)^2} x^2)(1 - x^2)}}$$

and by ex 24

$$\begin{aligned} K(k) &= \frac{2}{1 + \tilde{k}} K\left(\frac{1 - \tilde{k}}{1 + \tilde{k}}\right) = \frac{2a}{a+b} K\left(\frac{a-b}{a+b}\right) \Rightarrow \\ I(a, b) &= \frac{1}{a} K(k) = I\left(\frac{a+b}{2}, \sqrt{ab}\right) \end{aligned}$$

Hence by (a) we know

$$\frac{2}{\pi} I(a, b) = \frac{2}{\pi} \lim_{n \rightarrow +\infty} I(a_n, b_n) = \frac{1}{M(a, b)}$$

11 Chapter 9

11.1 Exercise

1. With the notation given in the problem,

$$n\omega_2 = \frac{np}{q}\omega_1 = \frac{1-mq}{q}\omega_1 = \frac{1}{q}\omega_1 - m\omega_1.$$

This gives us that

$$f\left(z + \frac{1}{q}\omega_1\right) = f\left(z + \frac{1}{q}\omega_1 - m\omega_1\right) = f(z + n\omega_2) = f(z)$$

This shows that $\frac{1}{q}\omega_1$ is a period of f and it is smaller than ω_2 and ω_1 in size, since $\frac{1}{q}\omega_1 = \frac{1}{p}\omega_2$. $\square$

(b) By Stein *Fourier Analysis* page 107, we know $\{m + n\tau\}_{m,n \in \mathbb{Z}}$ is dense in $\mathbb{R}$ when $\tau \in \mathbb{R} - \mathbb{Q}$. Since we have $f(x + (n\tau + m)\omega_2) = f(x)$ and f is continuous, we can easily verify that $f = \text{constant}$. $\square$

2. The argument of Theorem 4.1 in Chapter 3 shows that

$$\int_{\partial P} z \frac{f'(z)}{f(z)} dz = 2\pi i (a_1 + \cdots + a_r - b_1 - \cdots - b_r)$$

We observe that

$$\begin{aligned} \frac{1}{2\pi i} \int_{\partial P} z \frac{f'(z)}{f(z)} dz &= \frac{1}{2\pi i} \int_{z_0}^{z_0+\omega_1} z \frac{f'(z)}{f(z)} dz + \frac{1}{2\pi i} \int_{z_0+\omega_1}^{z_0+\omega_1+\omega_2} z \frac{f'(z)}{f(z)} dz + \\ &\quad \frac{1}{2\pi i} \int_{z_0+\omega_1+\omega_2}^{z_0+\omega_2} z \frac{f'(z)}{f(z)} dz + \frac{1}{2\pi i} \int_{z_0+\omega_2}^{z_0} z \frac{f'(z)}{f(z)} dz = \\ &= \frac{1}{2\pi i} \int_{z_0}^{z_0+\omega_1} z \frac{f'(z)}{f(z)} dz + \frac{1}{2\pi i} \int_{z_0}^{z_0+\omega_2} (z + \omega_1) \frac{f'(z)}{f(z)} dz + \\ &\quad \frac{1}{2\pi i} \int_{z_0+\omega_1}^{z_0} (z + \omega_2) \frac{f'(z)}{f(z)} dz + \frac{1}{2\pi i} \int_{z_0+\omega_2}^{z_0} z \frac{f'(z)}{f(z)} dz = \\ &= \frac{1}{2\pi i} \log f(z)|_{z_0}^{z_0+\omega_2} \cdot \omega_1 + \frac{1}{2\pi i} \log f(z)|_{z_0+\omega_1}^{z_0} \cdot \omega_2 = \\ &\quad n\omega_1 + m\omega_2 \end{aligned}$$

Hence we draw the conclusion. $\square$

This follows from the residue theorem.

3. Denote $K = \{(x, y) | 1 \leq x^2 + y^2 \leq R^2, x, y \geq 0\}$

$$\iint_K \frac{1}{x^2 + y^2} dx dy = \int_1^R \int_0^{\frac{\pi}{2}} \frac{1}{r} d\theta dr = \frac{\pi}{2} \log R$$

Hence

$$\begin{aligned} \left| \sum_K \frac{1}{m^2 + n^2} - \frac{\pi}{2} \log R \right| &= \left| \iint_K \frac{1}{[x]^2 + [y]^2} - \frac{1}{x^2 + y^2} dx dy \right| \leq \\ &4 \iint_K \frac{[x] + [y]}{([x]^2 + [y]^2)^4} dx dy \rightarrow C, R \rightarrow +\infty \end{aligned}$$

Hence we have

$$\sum_{(m,n) \neq (0,0)} \frac{1}{m^2 + n^2} = 2\pi \log R + O(1)$$

4. Since

$$\begin{aligned} \wp(z) &= \frac{1}{z^2} + \sum_{|\omega| \leq R} \left[\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right] + \sum_{|\omega| > R} O\left(\frac{1}{\omega^3}\right) = \\ &\frac{1}{z^2} + \sum_{|\omega| \leq R} \left[\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right] + O\left(\frac{1}{R}\right) = \wp^R(z) + O\left(\frac{1}{R}\right) \end{aligned}$$

and

$$\wp^R(z+1) - \wp^R(z) = O\left(\sum_{R-c < |\omega| < R+c} \frac{1}{\omega^2}\right) = O\left(\frac{1}{R}\right)$$

Hence let $R \rightarrow +\infty$, we have $\wp(z+1) = \wp(z)$. Similarly we have $\wp(z) = \wp(z+\tau)$. $\square$

5. (a) If we establish convergence then since it is an infinite product it can only vanish at a point where one of the terms vanishes, thus it has simple zeros at all the periods and does not vanish anywhere else. $\square$

(b) Notice that

$$\begin{aligned} \frac{\sigma'}{\sigma} &= d(\log \sigma) = \frac{1}{z} + \sum_j \left[\frac{z}{\tau_j} + \frac{z^2}{2\tau_j^2} + \log\left(1 - \frac{1}{\tau_j}\right) \right]' = \\ &\frac{1}{z} + \sum_{(m,n) \neq (0,0)} \left[\frac{1}{z - n - m\tau} + \frac{1}{n + m\tau} + \frac{z}{(n + m\tau)^2} \right] = \\ &\frac{1}{z} + \sum_{\omega \in \Lambda^*} \frac{z^2}{(z - \omega)\omega^2} \end{aligned}$$

Hence like the $\wp(z)$, this series is convergent whenever z is not a lattice point. $\square$

(c) Just derivate it term by term. $\square$

6. We have $(\wp')^2 = 4\wp^3 - a\wp - b$, hence differentiating both sides and dividing by $\wp'$ gives $2\wp'' = 12\wp^2 - a$. $\square$

7. We see that $8 \sum_{m \text{ odd}} \frac{1}{m^2} = \frac{\pi^2}{\sin(\pi/2)^2} = \pi^2$. Use the fact that

$$\frac{1}{4} \sum_m \frac{1}{m^2} = \sum_{m \text{ even}} \frac{1}{m^2} = \left(\sum_m - \sum_{m \text{ odd}} \right) \frac{1}{m^2}$$

which results in $\sum_m = \frac{4}{3} \sum_{m \text{ odd}}$. Similar analysis works for $\zeta(4)$. □

8.(a) Trivial by theorem 2.5. □

(b) By theorem 2.5 we have

$$|E_4(\tau) - 2\zeta(4)| \leq C \sum_{r=1}^{\infty} \frac{r^4}{e^{2\pi r t}} \leq c e^{-2\pi t}$$

($t \geq 1$ guarantees the series is convergent.) □

(c) Since $E_4(\tau) = \tau^{-4} E(\frac{1}{\tau})$, we draw the conclusion by (b) □

11.2 Problem

1. (a) We notice that

$$S_1(z) - \wp(z) = \lim_{R \rightarrow +\infty} \sum_{1 \leq n^2 + m^2 \leq R^2} \frac{1}{(n + m\tau)^2}$$

and

$$\begin{aligned} \iint_{1 \leq x^2 + y^2 \leq R^2} \frac{1}{(x + y\tau)^2} dx dy &= \left(\int_{-R}^{-1} + \int_1^R \right) \left(\int_{-\sqrt{R^2 - y^2}}^{-1} + \int_1^{\sqrt{R^2 - y^2}} \right) \frac{1}{(x + y\tau)^2} dx dy = \\ &= - \left(\int_{-R}^{-1} + \int_1^R \right) \left(\frac{2\sqrt{R^2 - y^2}}{y^2\tau^2 - (R^2 - y^2)} + \frac{2}{y^2\tau^2 - 1} \right) dy = 0 \end{aligned}$$

Since

$$\begin{aligned} \left| \sum_{\substack{1 \leq m^2 + n^2 \leq R \\ m, n \geq 0}} \frac{1}{(n + m\tau)^2} - \iint_{\substack{1 \leq x^2 + y^2 \leq R^2 \\ x, y \geq 0}} \frac{1}{(x + y\tau)^2} dx dy \right| &= \left| \iint_{\substack{1 \leq x^2 + y^2 \leq R^2 \\ x, y \geq 0}} \frac{1}{([x] + [y]\tau)^2} - \frac{1}{(x + y\tau)^2} dx dy \right| \leq \\ &\leq \iint_{\substack{1 \leq x^2 + y^2 \leq R^2 \\ x, y \geq 0}} \frac{|(\{x\} + \{y\}\tau)(x + [x] + (y + [y])\tau)|}{|([x] + [y]\tau)^4|} \leq \iint_{\substack{1 \leq x^2 + y^2 \leq R^2 \\ x, y \geq 0}} \frac{4|(1 + \tau)| |([x] + [y]\tau)|}{|([x] + [y]\tau)^4|} \leq \iint_{\substack{1 \leq x^2 + y^2 \leq R^2 \\ x, y \geq 0}} \frac{4|(1 + \tau)|}{|([x] + [y]\tau)^3|} \end{aligned}$$

We have

$$\lim_{R \rightarrow \infty} \left| \sum_{\substack{1 \leq m^2 + n^2 \leq R \\ m, n \geq 0}} \frac{1}{(n + m\tau)^2} - \iint_{\substack{1 \leq x^2 + y^2 \leq R^2 \\ x, y \geq 0}} \frac{1}{(x + y\tau)^2} dx dy \right| \leq \lim_{R \rightarrow \infty} \iint_{\substack{1 \leq x^2 + y^2 \leq R^2 \\ x, y \geq 0}} \frac{4|(1 + \tau)|}{|([x] + [y]\tau)^3|} < +\infty$$

Hence

$$\begin{aligned} \lim_{R \rightarrow \infty} \left| \sum_{1 \leq m^2 + n^2 \leq R^2} \frac{1}{(n + m\tau)^2} \right| &= \lim_{R \rightarrow \infty} \left| \sum_{1 \leq m^2 + n^2 \leq R^2} \frac{1}{(n + m\tau)^2} - \iint_{1 \leq x^2 + y^2 \leq R^2} \frac{1}{(x + y\tau)^2} dx dy \right| \leq \\ &\lim_{R \rightarrow \infty} 2 \left| \iint_{\substack{1 \leq x^2 + y^2 \leq R^2 \\ x, y \geq 0}} \right| + 2 \left| \iint_{\substack{1 \leq x^2 + y^2 \leq R^2 \\ x \leq 0, y \geq 0}} \right| < +\infty \end{aligned}$$

Then we draw the conclusion. $\square$

(b) Notice that

$$S_2(z) - \wp(z) = \sum_m \sum_n \frac{1}{(n + m\tau)^2}$$

(c) The same as (b).

2. By lemma 2.4 we have

$$\sum_{n=-\infty}^{+\infty} \frac{1}{(n + z + m\tau)^2} = \frac{\pi^2}{\sin^2[(z + m\tau)\pi]} \Rightarrow \wp(z) - \sum_{m=-\infty}^{+\infty} \frac{\pi^2}{\sin^2(z + m\tau)\pi} = \wp(z) - S_2(z) = F(\tau)$$

3. We restrict our sight to the standard parallelogram $\{a + b\tau | 0 \leq a < 1, 0 \leq b < 1\}$. Consider the formula $\wp'^2 = \wp^3 - g_2\wp - g_3$. $\forall x, y \in \mathbb{C}$ satisfy $y^2 = x^3 - g_2x - g_3, y \neq 0$, we claim that there exists exact one $u \in \{a + b\tau | 0 \leq a < 1, 0 \leq b < 1\}$ satisfies $\wp(u) = x, \wp'(u) = y$. Clearly $\wp(u) - x = 0$ have two distinct roots in the parallelogram or $y = \wp' = 0$ which has been excluded. We denote them u_1, u_2 and we have

$$u_2 = \begin{cases} -u_1 + 1 + \tau, u_1 = a + b\tau, a, b \neq 0 \\ -u_1 + 1, u = a \\ -u_1 + \tau, u = b\tau \end{cases}$$

since $\wp$ is even and double period. Hence $\wp'(u_1) = -\wp'(u_2)$. However we have $\wp'(u_i)^2 = y^2$ so there exists only one u satisfy this equation. We observe that

$$[I(\wp(z))]' = \frac{\wp'(z)}{\sqrt{\wp^3(z) - g_2\wp(z) - g_3}} = \frac{\wp'(z)}{\sqrt{\wp'^2(z)}} = \pm 1$$

So choose proper z we know $I(\wp(z)) = z - \alpha \Rightarrow I(\wp(z + \alpha)) = z$. $\square$

4. (a) We only compute $\wp$ in vertical lines, we have

$$\begin{aligned} S = \wp(-\frac{n}{2} + cti) &= \wp(\frac{n}{2} + cti) \Rightarrow 2S = \wp(-\frac{n}{2} + cti) + \wp(\frac{n}{2} + cti) = \frac{1}{(-\frac{n}{2} + cti)^2} + \frac{1}{(\frac{n}{2} + cti)^2} + \\ &\sum_{(a,b) \neq (0,0)} \left[\frac{1}{(-\frac{n}{2} + cti - a + bti)^2} - \frac{1}{(-a + bti)^2} + \frac{1}{(\frac{n}{2} + cti + a + bti)^2} - \frac{1}{(a + bti)^2} \right] \in \mathbb{R} \end{aligned}$$

Hence $S \in \mathbb{R}$. □

(b) Consider the rectangle $Q = (0, \frac{1}{2}) \times (0, \frac{1}{2}\tau)$

Assume $\exists z \in Q, \wp(z) \in UHP$. By (a) we know $\wp(Q) \in UHP \Leftarrow \wp(Q) \cap \mathbb{R} = \emptyset$. Consider $\wp(z) - \wp(\frac{1+\tau}{2})$. Obviously we have

$$\begin{aligned}\lim_{t \rightarrow 0+} \wp(it) &= -\infty \\ \lim_{t \rightarrow 0+} \wp(t) &= +\infty\end{aligned}$$

Hence we have $\wp(\partial Q) - \wp(\frac{1+\tau}{2}) = \mathbb{R} \Rightarrow \wp(\partial Q) = \mathbb{R}$. If $\exists a \in Q, \wp(a) \in \mathbb{R}$, we have $\wp(z) - \wp(a)$ have at least three zero points in $[0, 1] \times [0, \tau]$, contradiction to $ord\wp = 2$. Hence $\wp(Q) \in UHP$. Then choose $u \in UHP$, we know $\wp(z) - u = 0$ have zero point $m + ni \in [0, 1] \times [0, \tau]$ since $ord\wp = 2$. If $m + ni \notin Q$, we can easily find a zero point in Q since $\wp$ is double period and even. If $m + ni \in Q$, similarly we can find a zero point in $(0, 1) \times (0, \tau)$ but not in Q . Since $ord\wp = 2$, these two points are all zero points, so there is only one zero point in Q . Hence $\wp$ is a conformal map. □

12 Chapter 10

12.1 Exercise

2. Multiply the recursive relation by x^n and sum from $n = 2$ up to infinity. This gives

$$F(x) - F_0 - F_1x = x(F(x) - F_0) + x^2F(x).$$

This proves the first part in a formal sense, to get convergence in a ball around 0 we will show that $F_n \sim c\alpha^n$ for some $\alpha > 0$, which is enough to get convergence in a ball around 0.

For (c) and (d) follow the partial fraction decomposition, then use the geometric series, but absolute convergence in a small disc we can combine the two sums. Then matching up the appropriate powers of x^n gives the result.

3. Do the same things as in (2) but in more generality. In the case $\alpha = \beta$ we just note that we have $U(x) = \frac{u_0}{(1-\alpha x)^2} + (u_1 - au_0) \frac{x}{(1-\alpha x)^2}$. Now notice that $\frac{1}{(1-x)^2} = \sum_{n \geq 1} nx^{n-1}$ to finish off the calculations.

4. We have $\prod_{n \geq 1} (1 - x^n) = \sum_{k=-\infty}^{\infty} (-1)^k x^{k(3k+1)/2}$. So we see that

$$\left(\sum_{n \geq 0} p(n)x^n \right) \left(\sum_{k=-\infty}^{\infty} (-1)^k x^{k(3k+1)/2} \right) = 1.$$

Multiplying out and comparing powers of x^n on each side gives the desired formula

$$\sum_k (-1)^k p \left(n - \frac{k(3k+1)}{2} \right) = 0$$

for $n \geq 1$

- 5.

$$\begin{aligned} \log F(x) &= \sum_{n=1}^{\infty} \log \frac{1}{1-x^n} \\ &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{1}{m} x^{nm} \\ &= \sum_{m=1}^{\infty} \frac{1}{m} \sum_{n=1}^{\infty} x^{nm} \\ &= \sum_{m=1}^{\infty} \frac{1}{m} \frac{x^m}{1-x^m} \end{aligned}$$

Notice $mx^{m-1}(1-x) < 1 - m^x < m(1-x)$ when $0 < x < 1$, then

$$\sum_{m=1}^{\infty} \frac{1}{m^2} \frac{x^m}{1-x} < \log F(x) < \sum_{m=1}^{\infty} \frac{1}{m^2} \frac{x}{1-x}$$

Then $\log F(x) \sim \frac{\pi^2}{6} \frac{1}{1-x}$.

6.

$$\log F(e^{-y}) \sim \frac{\pi^2}{6} \frac{1}{1-e^{-y}} \sim \frac{\pi^2}{6} \frac{1}{y}$$

as $y \rightarrow 0$, then $F(e^{-y}) = \sum_{n=0}^{\infty} p(n)e^{-ny} \leq C_1 e^{\frac{c_1}{y}}$. So $p(n)e^{-ny} \leq C_1 e^{\frac{c_1}{y}}$, let $y = \frac{1}{\sqrt{n}}$ then $p(n) \leq e^{c' n^{\frac{1}{2}}}$. In the opposite direction, $F(e^{-y}) = \sum_{n=0}^{\infty} p(n)e^{-ny} \geq C_2 e^{\frac{c_2}{y}}$. Then

$$\sum_{n=0}^m p(n)e^{-ny} \geq C_2 e^{\frac{c_2}{y}} - \sum_{n=m+1}^{\infty} p(n)e^{-ny} \geq C_2 e^{\frac{c_2}{y}} - \sum_{n=m+1}^{\infty} e^{cn^{\frac{1}{2}}} e^{-ny}$$

Take $y = \frac{A}{\sqrt{m}}$, then $mp(m) \geq \sum_{n=0}^m p(n)e^{-n\frac{A}{\sqrt{m}}} \geq C_2 e^{\frac{c_2\sqrt{m}}{A}} - \sum_{n=m+1}^{\infty} e^{cn^{\frac{1}{2}}} e^{-n\frac{A}{\sqrt{m}}}$. i.e. $p(m) \geq e^{c'' n^{\frac{1}{2}}}$.

7.

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z} = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1} e^{2\pi i z})(1 + q^{2n-1} e^{-2\pi i z})$$

where $q = e^{\pi i \tau}$. (a) Let $z = \frac{\tau}{2}$

$$\begin{aligned} \sum_{n=-\infty}^{\infty} e^{\pi i (n^2+n)\tau} &= \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n})(1 + q^{2n-2}) \\ \sum_{n=-\infty}^{\infty} q^{n^2+n} &= \prod_{n=1}^{\infty} (1 - q^{4n})(1 + q^{2n-2}) \\ \sum_{n=-\infty}^{\infty} x^{\frac{n^2+n}{2}} &= \prod_{n=1}^{\infty} (1 - x^{2n})(1 + x^{n-1}) \\ &= \prod_{n=0}^{\infty} (1 - x^{2n+2})(1 + x^n) \end{aligned}$$

(b) Let $z = \frac{3\tau}{10} + \frac{1}{2}$

$$\begin{aligned} \sum_{n=-\infty}^{\infty} (-1)^n e^{\pi i (n^2 + \frac{3}{5}n)\tau} &= \prod_{n=1}^{\infty} (1 - q^{2n})(1 - q^{2n+\frac{2}{5}})(1 - q^{2n-\frac{8}{5}}) \\ \sum_{n=-\infty}^{\infty} (-1)^n q^{n^2 + \frac{3}{5}n} &= \prod_{n=1}^{\infty} (1 - q^{2n})(1 - q^{2n+\frac{2}{5}})(1 - q^{2n-\frac{8}{5}}) \\ \sum_{n=-\infty}^{\infty} (-1)^n x^{\frac{n^2 + \frac{3}{5}n}{2}} &= \prod_{n=1}^{\infty} (1 - x^{5n})(1 - x^{5n+1})(1 - x^{5n+4}) \end{aligned}$$

8. Trivial number theory exercise.

9. (a) $r_2(p) = 4(d_1(p) - d_3(p)) = 4 \times 2 = 8$.

(b) When $a = 2t$, $r_2(p) = 4(d_1(p) - d_3(p)) = 4(t + 1 - t) = 4$; $a = 2t - 1$, $r_2(p) = 4(d_1(p) - d_3(p)) = 4(t - t) = 0$.

(c) By a combination method we can calculate d_1 and d_3 .

10. (a) $r_2(p) = 0$ when p is a prime of the form $4k + 3$ and $r_2(5^k) = 4k$.

(b) $r_4(2^k) = 24$ and $r_4(q^k) = 8(1 + q + q^2 + \cdots + q^k)$ when q is odd prime.

11.

$$\sum_{n=1}^{\infty} d(n)z^n = \sum_{n=1}^{\infty} (z^n + z^{2n} + \cdots) = \sum_{n=1}^{\infty} \frac{z^n}{1 - z^n}$$

$$\sum_{n=1}^{\infty} \sigma_l(n)z^n = \sum_{n=1}^{\infty} n^l (z^n + z^{2n} + \cdots) = \sum_{n=1}^{\infty} \frac{n^l z^n}{1 - z^n}$$

12. (a)

$$\sum_{n=1}^{\infty} \frac{q^n}{(1 - q^n)^2} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} m q^{mn} = \sum_{m=1}^{\infty} m \sum_{n=1}^{\infty} q^{mn} = \sum_{m=1}^{\infty} \frac{mq^m}{1 - q^m}$$

(b)

$$\begin{aligned} & \sum_{n=1}^{\infty} \sigma_1^*(n)q^n \\ &= \sum_{n=1}^{\infty} n(q^n + q^{2n} + \cdots) - \sum_{n=1}^{\infty} 4n(q^{4n} + q^{8n} + \cdots) \\ &= \sum_{n=1}^{\infty} \frac{nq^n}{1 - q^n} - \sum_{n=1}^{\infty} \frac{4nq^{4n}}{1 - q^{4n}} \\ &= \sum_{n=1}^{\infty} \frac{q^n}{(1 - q^n)^2} - 4 \sum_{n=1}^{\infty} \frac{q^{4n}}{(1 - q^{4n})^2} \end{aligned}$$

(c)

$$\begin{aligned} & \sum_{n=1}^{\infty} \frac{q^n}{(1 - q^n)^2} - \sum_{n=1}^{\infty} \frac{4q^{4n}}{(1 - q^{4n})^2} \\ &= \sum_{n=1}^{\infty} \frac{q^n}{(1 - q^n)^2} - \sum_{n=1}^{\infty} \left(\frac{q^{2n}}{(1 - q^{2n})^2} - \frac{q^{2n}}{(1 + q^{2n})^2} \right) \\ &= \sum_{n=1}^{\infty} \frac{q^n}{(1 - (-1)^n q^n)^2} \end{aligned}$$

12.2 Problem

1. If $4^a(8k + 7) = x^2 + y^2 + z^2$, we have $x^2 + y^2 + z^2 \equiv 0 \pmod{4} \Rightarrow 4|x, y, z$. Hence we reduce the condition to $8k + 7 = x^2 + y^2 + z^2 \Rightarrow x^2 + y^2 + z^2 \equiv 3 \pmod{4} \Rightarrow x = 2t_1 + 1, y = 2t_2 + 1, z = 2t_3 + 1 \Rightarrow 8k + 7 = 4 \sum_{i=1}^3 (t_i^2 + t_i) + 3 \Leftrightarrow 2k + 1 = \sum_{i=1}^3 t_i(t_i + 1)$. However, the left is odd and the right is even. Contradiction! $\square$

2. (a)

$$\Im(g(\tau)) = \frac{\Im(\tau)}{|c\tau + d|^2}$$

Choose a $g_0 \in G$ s.t. $\Im(g_0(\tau))$ is maximal, then apply finitely many of $T_1(\tau) = \tau + 1$ to make $|\Re(g_0(\tau))| \leq \frac{1}{2}$. Then we conclude $|g_0(\tau)| \geq 1$. Otherwise

$$\Im(Sg_0(\tau)) = \Im\left(-\frac{1}{g_0(\tau)}\right) = -\frac{\Im(\overline{g_0(\tau)})}{|g_0(\tau)|^2} > \Im(g_0(\tau))$$

and this contradicts the maximality of $\Im(g_0(\tau))$.

(b) Suppose $\tau, \tau' \in \mathcal{F}$ are congruent. WLOG, $\tau' = g(\tau)$ and $\Im(\tau') \geq \Im(\tau)$ (otherwise $\tau = g^{-1}(\tau')$ and $\Im(\tau') \leq \Im(\tau)$). Let $g = \frac{a\tau+b}{c\tau+d}$, then $\Im(\tau') \geq \Im(\tau)$ gives $|c\tau + d| \leq 1$. Then c can only be 0 or ± 1 . If $c = 0$, then $g(\tau) = \tau + b$, that is $g(\tau) = \tau \pm 1$ when both $\tau, \tau' \in \mathcal{F}$. If $c = \pm 1$, then $g(\tau) = \frac{a\tau+1}{\tau} = a + \frac{1}{\tau}$, that is $g(\tau) = \frac{1}{\tau}$.

(c) For given g , $2i$ is in $\mathcal{F}$, consider $g(2i)$. By (a), we have a composition of S, T_1 and there inverse, say h , s.t. $hg(2i) \in \mathcal{F}$. Now both $2i$ and $hg(2i)$ are in $\mathcal{F}$, by (b), $Shg(2i) = 2i, T_1^{\pm 1}hg(2i) = 2i$ or $hg(2i) = 2i$. It suffices to prove that if a transformation fixes $2i$ then it is an identity, which is trivial:

$$\frac{2ia + b}{2ic + d} = 2i$$

gives $a = d = 1, b = c = 0$ when a, b, c, d are integers and $ad - bc = 1$.

3. Almost the same as Problem 2.

4. (a) Use the conformal mapping from $\mathbb{H}$ to $\mathbb{D}$

$$F(\tau) = \frac{i - \tau}{i + \tau}$$

Let $z_n = F(\tau_n)$

$$\begin{aligned} \sum_n (1 - |z_n|) &= \sum_n (1 - |F(\tau_n)|) \\ &= \sum_n \left(1 - \left|\frac{i - x_n - iy_n}{i + x_n + iy_n}\right|\right) \\ &= \sum_n \left(1 - \sqrt{\frac{x_n^2 + (1 - y_n)^2}{x_n^2 + (1 + y_n)^2}}\right) \\ &< \sum_n \left(1 - \frac{1 - y_n}{1 + y_n}\right) \\ &= \sum_n \frac{2y_n}{1 + y_n} \\ &< \sum_n 2y = \infty \end{aligned}$$

By Problem 1,2 Chapter 5, there exists no holomorphic (not identically zero) function in unit disc with zeros in z_n . Then there exists no holomorphic (not identically zero) function in upper

half plane with zeros in τ_n .

(b) Bezout's theorem.

(c)

$$\begin{aligned}
& \frac{3}{2} \sum_{(c,d)=1, c \not\equiv d \pmod{2}} \frac{1}{c^2 + d^2} \\
&= \sum_{(c,d)=1, c \not\equiv d \pmod{2}} \frac{1}{c^2 + d^2} + \frac{1}{(c+d)^2 + (c-d)^2} \\
&= \sum_{(a,b)=1} \frac{1}{a^2 + b^2} \\
&= \frac{6}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sum_{(a,b)=1} \frac{1}{a^2 + b^2} \\
&= \frac{6}{\pi^2} \sum_{k,l=1}^{\infty} \frac{1}{k^2 + l^2} = \infty
\end{aligned}$$

(d) WLOG, $F(i) = 0$, otherwise replace $F(\tau)$ by $F(\tau) - F(i)$. For each relative prime c, d with opposite parity, choose $g_{c,d}(\tau) = \frac{a\tau+b}{c\tau+d} \in G$ s.t. $\Im(g_{c,d}(i)) = \frac{1}{c^2+d^2}$. Then F also vanishes at all $g_{c,d}(i)$, and by (a) F is identically zero.

13 Appendix:Tests of Taishan College

13.1 The Mid-Term Test 2021

Time:100 minutes

- 1.(10 pt) Let $a, b, c \in \mathbb{R}$, $f(z) = ay^3 + bx^2y + i(x^3 + cxy^2)$ is entire. Compute a, b, c .
- 2.(15 pt) Show that

$$\frac{z}{1-z^2} + \frac{z^2}{1-z^4} + \cdots + \frac{z^{2^n}}{1-z^{2^{n+1}}} + \cdots = \frac{z}{1-z}, |z| < 1.$$

- 3.(10 pt) Show that $\mathbb{N}$ cannot be partitioned into a finite number union of disjoint arithmetic progressions.
- 4.(10 pt) If f is entire and it is not constant, show that $\forall a \in \mathbb{R}, \exists z \in \mathbb{C}$ s.t. $\operatorname{Re}(f(z)) = a$
- 5.(15 pt) Suppose f is non-vanishing continuous function on $\overline{\mathbb{D}}$ that is analytic in $\mathbb{D}$. Prove that if $|f(z)| = 1, \forall |z| = 1$, then f is constant.
- 6.(20 pt) Suppose $f(z)$ is holomorphic in $\overline{\mathbb{D}}$. Let $f(0) = i, f(\frac{1}{2}) = \frac{1}{2}$. Let

$$F(z) = \frac{1}{2\pi i} \int_{|\xi|=1} \frac{f(\xi)}{\xi - z} d\xi \quad \text{and} \quad G(z) = \frac{1}{2\pi i} \int_{|\xi|=1} \frac{\overline{f(\xi)}}{\xi - z} d\xi.$$

Compute $F(0), G(\frac{1}{2})$.

- 7.(10 pt) Show that there are n points $\{z_k\}_{k=1}^n$ in the unit closed disc, prove that there exists a point z on the unit circle s.t. the product of their distance $\prod_{k=1}^n |z_k - z|$ is not less than 1.
- 8.(10 pt) Let $t > 0, A \in \mathbb{R}$. Compute

$$\lim_{t \rightarrow +\infty} \frac{1}{2\pi i} \int_{1-it}^{1+it} \frac{A^z}{z} dz.$$

[Hint: the result relies on A .]

13.2 The Mid-Term Test 2022

Time:120 minutes

1.(20 pt)Find all closed curves C satisfy

$$\int_C \frac{1}{z^2 + z + 1} dz = 0$$

2.(20 pt) Compute

$$\int_{|z|=\frac{7}{2}} \frac{(2(z+1)^3 + z+1)^4}{(z^4-1)^4(z^3-2)^3(z^2-3)^2(z-4)} dz.$$

3.(20 pt) Let $a > 1$. Compute the number of solutions of the equation $z + e^{-z} = a$, $\text{Re}(z) > 0$.

4.(20 pt) Let $f(z)$ be analytic at z_0 and $f'(z_0) \neq 0$. Show that there exists $\delta > 0$ s.t. f is injective in the neighbourhood $\{z : |z - z_0| < \delta\}$

5.(8 pt) Let Ω be bounded reigon of $\mathbb{C}$ and $f : \Omega \rightarrow \Omega$ be holomorphic. Suppose $f(z_0) = z_0$ and $f'(z_0) = 1$. Show that $f(z) = z$.

6.(6 pt) Assume two non-zero complex number $\omega_1, \omega_2 \in \mathbb{C}$ satisfy $\omega_1/\omega_2 \notin \mathbb{R}$. Let $\Lambda = \{m\omega_1 + n\omega_2 | m, n \in \mathbb{Z}\}$. Let Γ be the boundary of $\{a\omega_1 + b\omega_2 | a, b \in [0, 1]\}$. Suppose f is meromorphic and satisfy

(i) The poles of f is not in Γ . (ii) $\forall z \in \mathbb{C}, f(z) = f(z + \omega_1) = f(z + \omega_2)$.

Show that

$$\frac{1}{2\pi i} \int_{\Gamma} \frac{zf'(z)}{f(z)} dz \in \Lambda.$$

7.(6 pt) Let $g(z)$ be a polynomial on z , consider a cubic equation $w^3 - 3zw + g(z) = 0$ on w .

(i) Find one solution when $g = -z^3 - 1$.

(ii) Find all polynomials $g(z)$ such that the cubic equation has three distinct entire functions as solutions.

solution to 7. Claim: if g is polynomial, then the solutions to this equation is polynomials.

Let f be one of its solutions. If g is not a constant then its infinite point is an essential singularity. By great Picard's theorem, g takes all value infinitely times with only one exception. Then there exists A and distinct z_n such that $f(z_n) = A$.

$$f^3(z_n) - 3z_nf(z_n) + g(z_n) = 0$$

$$A^3 - 3z_nA + g(z_n) = 0$$

there is a contradiction since $A^3 - 3Az + g(z)$ is a non-zero polynomial (non-zero can be achieved by selecting A) which cannot have infinitely many zeros. so the solutions f 's are always polynomials.

By Vieta's theorem

$$\begin{cases} f_1 + f_2 + f_3 = 0 \\ f_1f_2 + f_1f_3 + f_2f_3 = -3z \\ f_1f_2f_3 = -g(z) \end{cases}$$

Then we have $f_1^2 + f_1f_2 + f_2^2 = 3z$ i.e. $(f_1 - \xi f_2)(f_1 - \xi^2 f_2) = 3z$ where ξ is the primitive cubic root. Then we have $\begin{cases} f_1 - \xi f_2 = c \\ f_1 - \xi^2 f_2 = \frac{3z}{c} \end{cases}$ or $\begin{cases} f_1 - \xi f_2 = \frac{3z}{c} \\ f_1 - \xi^2 f_2 = c \end{cases}$ Then we can solve f_1, f_2 respectively, then solve f_3 and g which we omit here.

13.3 The Final Exam 2021

Time:120 minutes

1.(10 pt) Let $a, b, c \in \mathbb{R}$,

$$f(z) = e^{2x} (2 \cos^2 y - a) + be^x (x \cos y - y \sin y) + x^2 - y^2 + ic (e^x \cos y - x) (e^x \sin y - y)$$

is analytic. Compute a, b, c .

2.(10 pt) $\forall r > 0$, denote γ_r the upper half circle of radius r (counterclockwise). Show that

(i) $\left| \int_{\gamma} \frac{e^x}{z} dz \right| \leq \pi e$. (ii) $\left| \int_{\gamma_r} e^{iz} dz \right| < \pi$.

3. (20 pt) Suppose $f(z)$ is analytic in $\{z \in \mathbb{C} : |z| \leq 3\}$ and $f(0) = i, f(\frac{1}{2}) = \frac{1}{2}, f(2) = 2$. Let

$$F(z) = \frac{1}{2\pi i} \int_{|\xi|=1} \frac{f(\xi)}{\xi - z} d\xi, G(z) = \frac{1}{2\pi i} \int_{|\xi|=1} \frac{\overline{f(\xi)}}{\xi - z} d\xi,$$

Compute $F(0), F(2), G(\frac{1}{2}), G(2)$.

4.(10 pt) Compute

$$\int_{|z|=2} \frac{1}{z^2 + 2} dz$$

5.(10 pt) Let $f(z)$ be analytic in $|z| \leq 3$. Suppose f have only one zero point $z_0 (|z_0| \leq 1)$ and $f'(z_0) \neq 0$. Compute

$$\int_{|z|=2} \frac{zf'(z)}{f(z)} dz.$$

6.(10 pt) Let $a \in \mathbb{C}$. Let f be entire function and $\forall z \in \mathbb{C}, f(z) \neq a$. Show that there exists entire function g s.t. $f(z) = a + e^{g(z)}$.

7.(20 pt) Let $g(z)$ be a polynomial on z , consider a cubic equation $w^3 - 3zw + g(z) = 0$ on w .

(i) Find one solution when $g = -z^3 - 1$.

(ii) Find all polynomials $g(z)$ such that the cubic equation has three distinct entire functions as solutions.

8.(10 pt) Let $f : \mathbb{H} \rightarrow \mathbb{H}$ be analytic function. Suppose f have two distinct fixed point in $\mathbb{H}$. Show that $f(z) = z, z \in \mathbb{H}$.

13.4 The Final Exam 2022

Time:120 minutes

1.(15 pt)Let $a, b, c \in \mathbb{R}$. $f = ay^3 + bx^2y + i(x^3 + cxy^2)$ is entire. Compute a, b, c .

2.(30 pt)Compute

$$\frac{1}{2\pi i} \int_{|z|=1} \frac{1}{z(3z+1)} dz$$

$$\frac{1}{2\pi i} \int_{|z|=1} \frac{z \sin z}{(1-e^z)^3} dz$$

3.(15 pt) Suppose f is non-vanishing continuous function on $\overline{\mathbb{D}}$ that is analytic in $\mathbb{D}$. Prove that if $|f(z)| = 1, \forall |z| = 1$, then f is constant.

4.(10 pt)Let $a \in \mathbb{C}$. Let f be entire function and $\forall z \in \mathbb{C}, f(z) \neq a$. Show that there exists entire function g s.t. $f(z) = a + e^{g(z)}$.

5.(10 pt)Let f be holomorphic and injective in $\operatorname{Re}(z) > 0$. Suppose $\operatorname{Re}(f) > 0$ and $a > 0$ is a fixed point of f . Show that $|f'(a)| \leq 1$.

6.(6 pt)Denote $A(r) = \sup_{|z|=r} |f(z)|$. If f is a transcendental entire function, show that

$$\lim_{r \rightarrow \infty} \frac{\ln A(r)}{\ln r} = \infty$$

7.(4 pt)Let f be analytic in $|z| \leq R$ and $|f| \leq M, f(0) \neq 0$. Show that the number of zero point in $|z| \leq \frac{R}{3}$ is no more than $\frac{1}{\ln 2} \ln \frac{M}{|f(0)|}$.

8.(5 pt)(i) Show that for harmonic function u in a simply connected region Ω , there exists a harmonic function v s.t. $f = u + iv$ is holomorphic.

(ii) Prove that u cannot reach the extreme value in Ω .

9.(5 pt)Let $f : \mathbb{D} \rightarrow \mathbb{C}$ be holomorphic and injective function satisfies $f(0) = 0, f'(0) = 1$. Suppose

$$\operatorname{Re}\left(z \frac{f''(z)}{f'(z)}\right) \leq \frac{4|z| - 2|z|^2}{1 - |z|^2}$$

Show that

(i)

$$|f'(z)| \leq \frac{1 + |z|}{(1 - |z|)^3}$$

(ii)

$$|f(z)| \leq \frac{|z|}{(1 - |z|)^2}$$

[Hint:use (i)]

14 Reference

8.31

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